

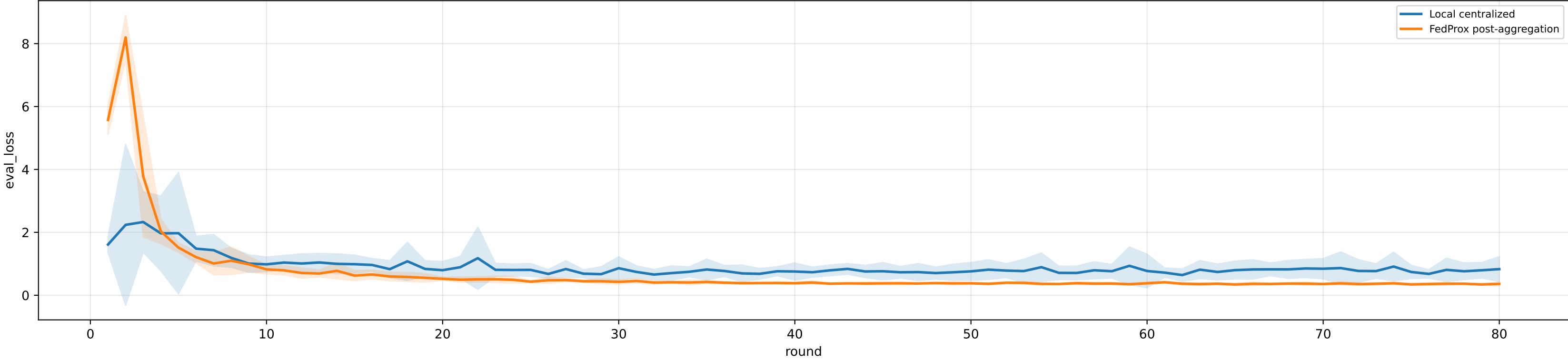
Controlled q-label heterogeneity: iid | FedProx diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

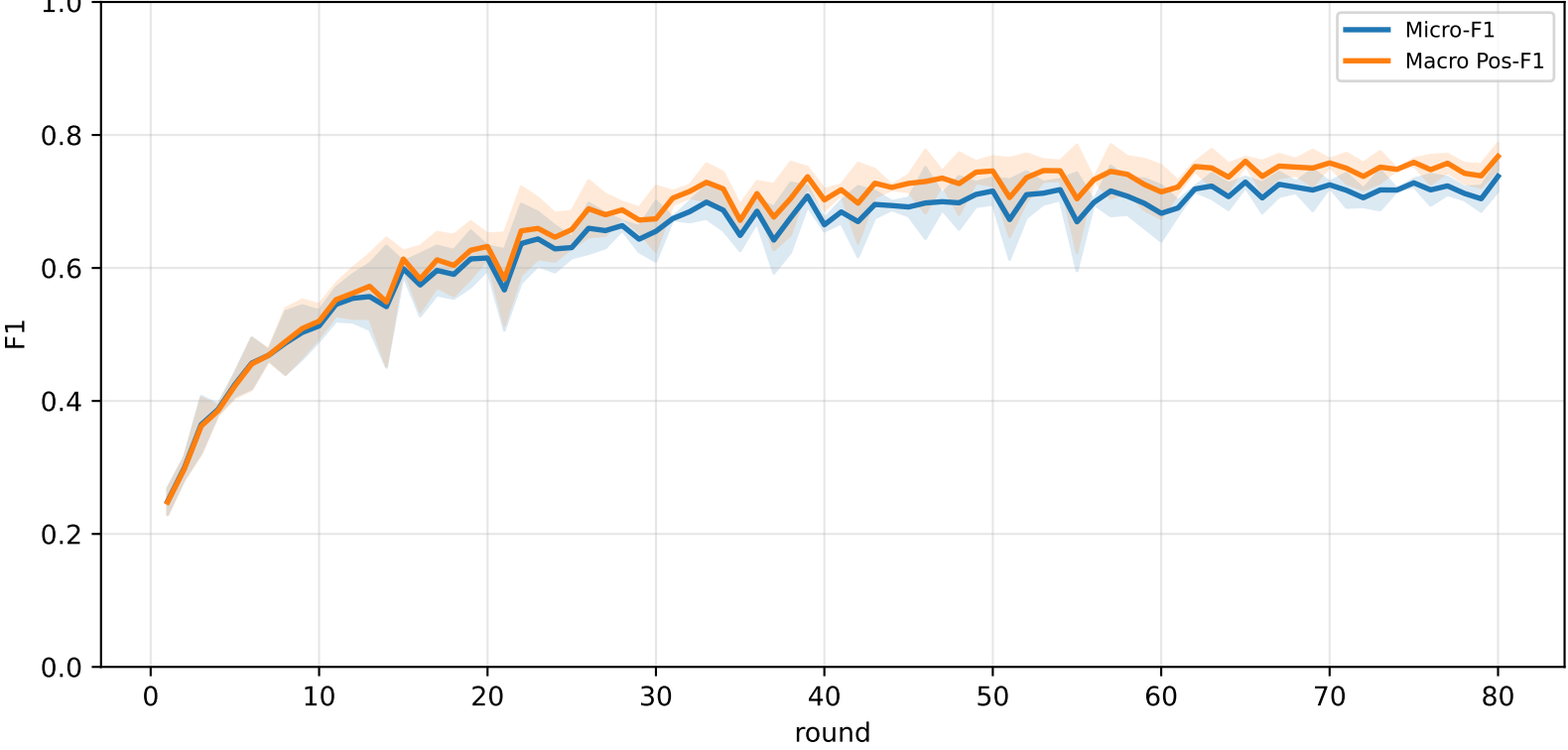
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	64.2% ± 0.6%	67.8% ± 0.9%	79.6% ± 4.0%	78.3% ± 5.1%	66.8% ± 1.9%	61.9% ± 2.0%	48.9% ± 0.2%
FedProx	65.8% ± 1.0%	68.8% ± 0.6%	83.6% ± 2.3%	77.3% ± 1.3%	67.3% ± 3.8%	63.0% ± 0.4%	52.7% ± 1.5%

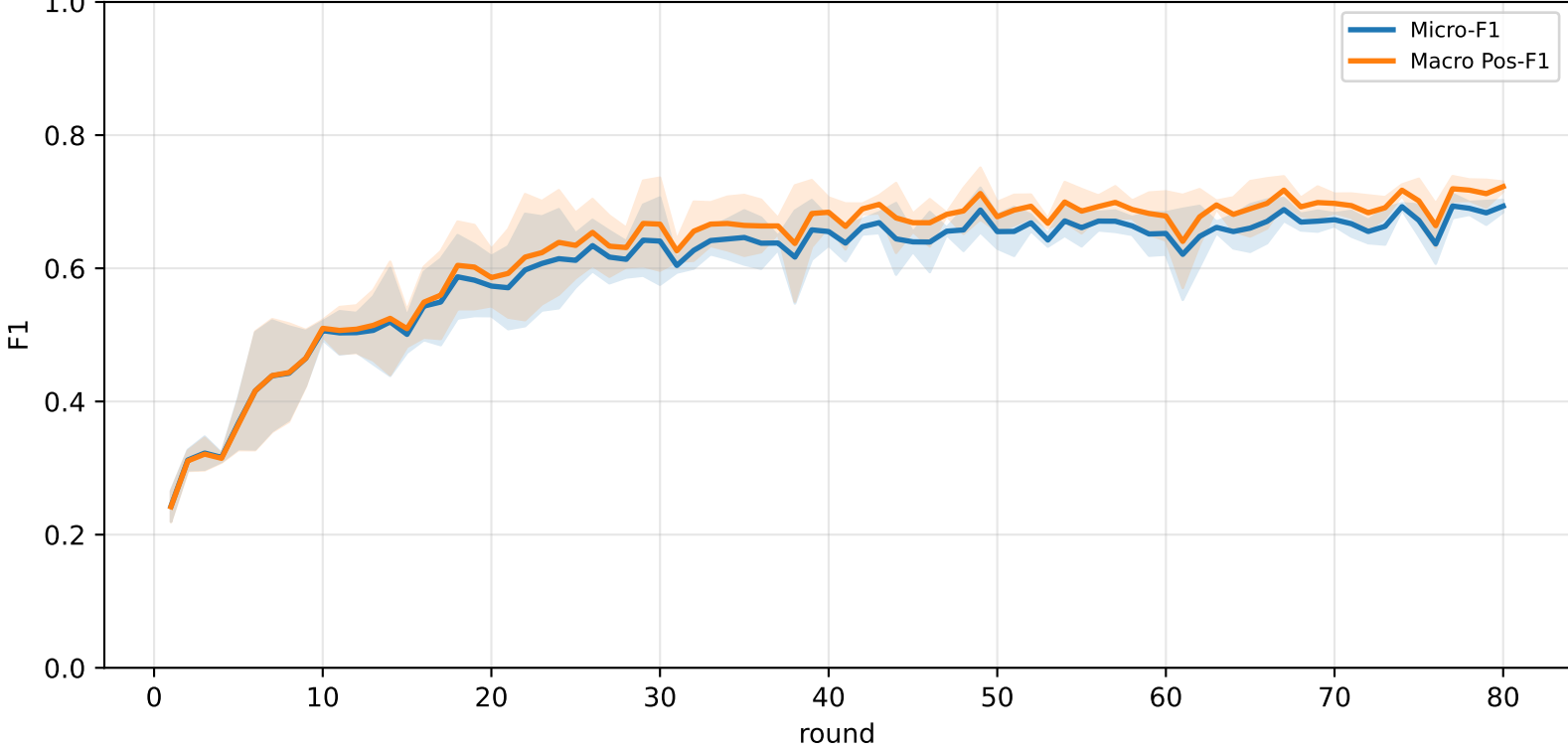
Validation loss comparison



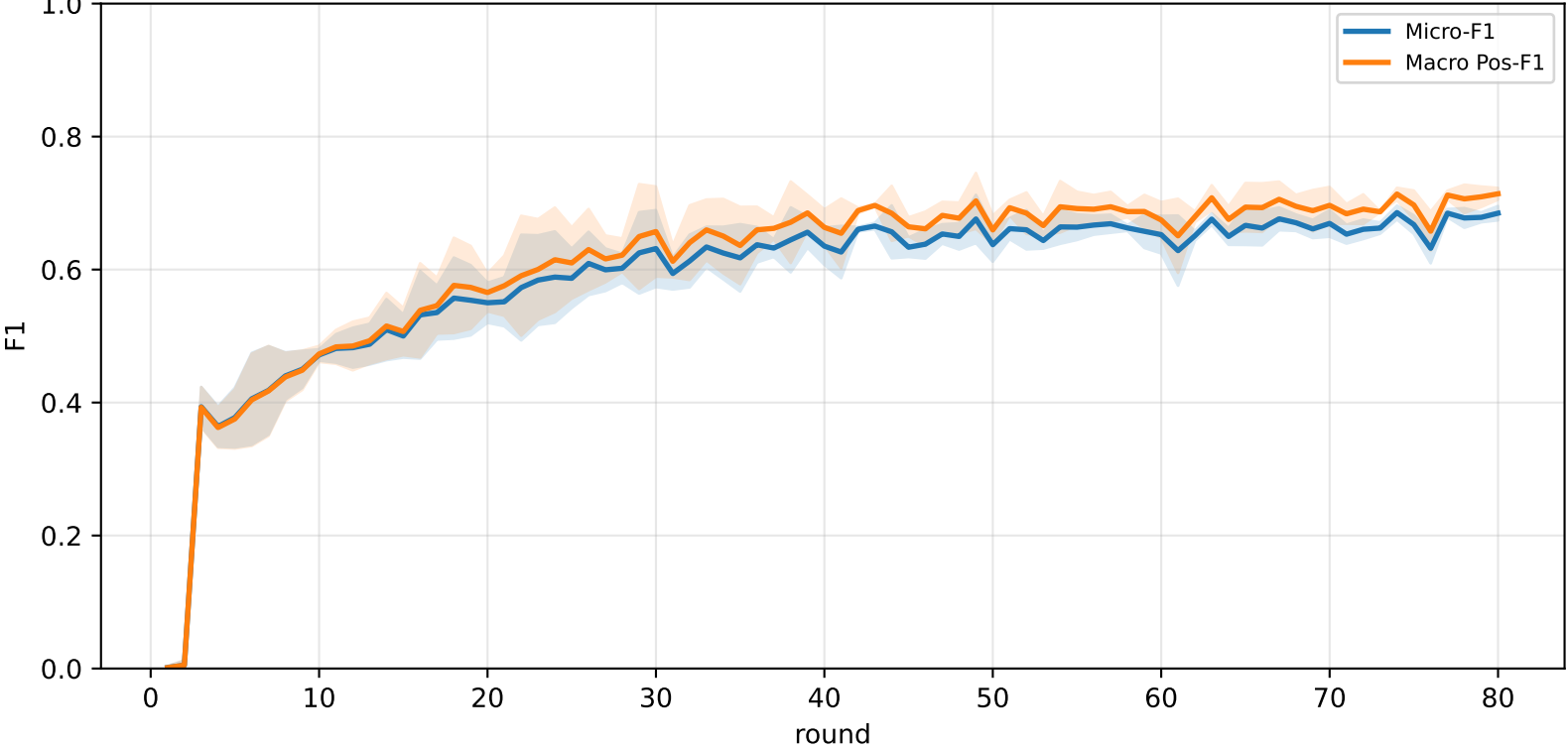
Pooled local centralized micro / macro F1



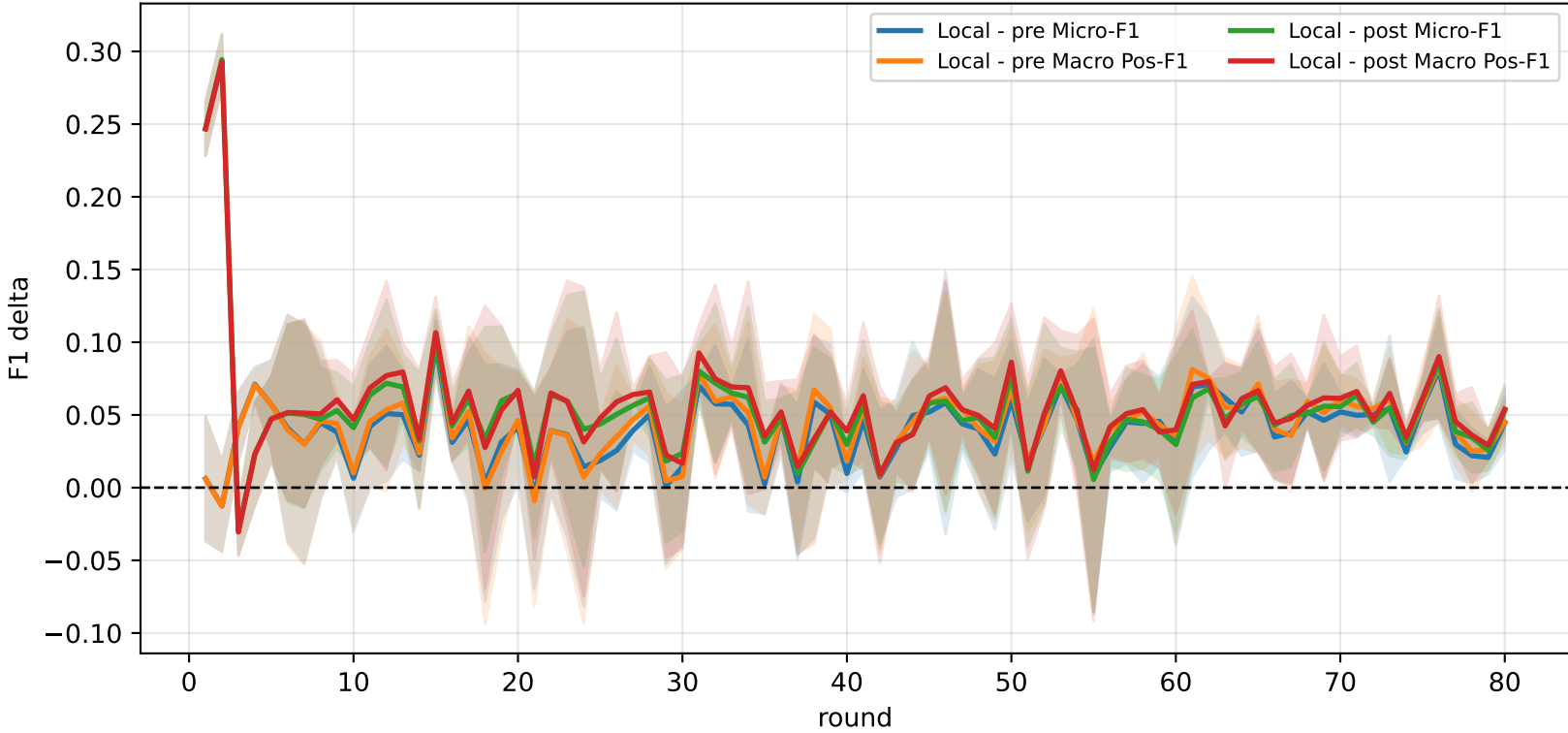
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

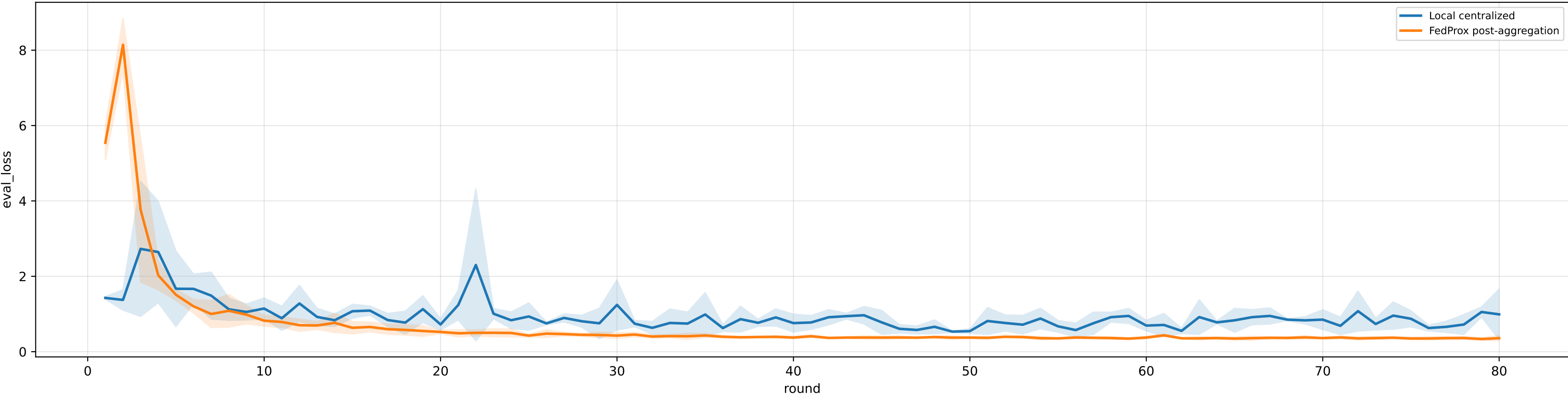


family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

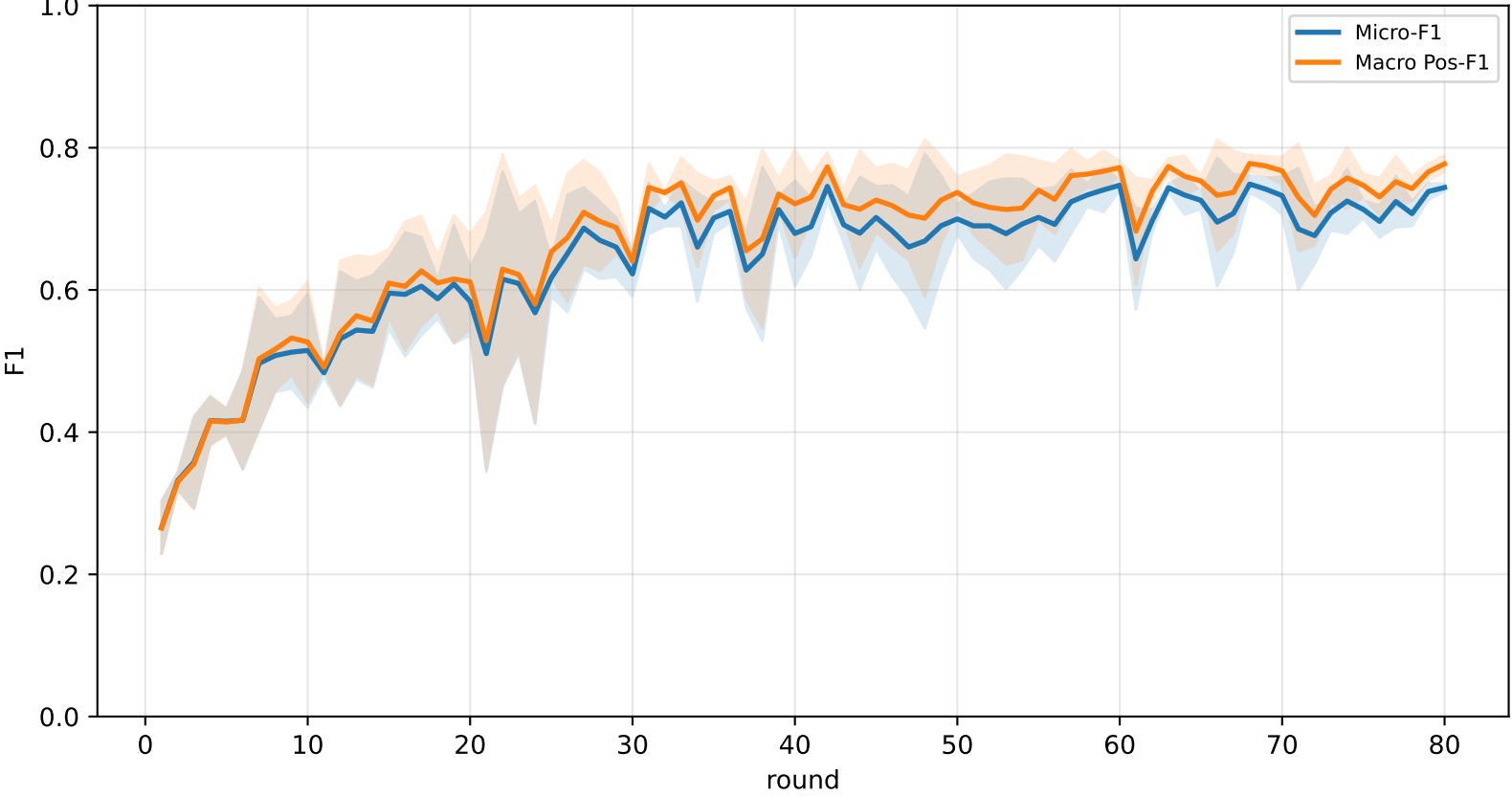
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	61.7% ± 2.9%	65.8% ± 2.5%	83.4% ± 5.2%	77.4% ± 2.3%	64.8% ± 2.9%	56.1% ± 5.8%	47.3% ± 2.8%
FedProx	66.1% ± 1.0%	69.0% ± 0.5%	84.1% ± 2.9%	77.5% ± 1.7%	67.1% ± 3.2%	63.5% ± 0.1%	52.9% ± 1.7%

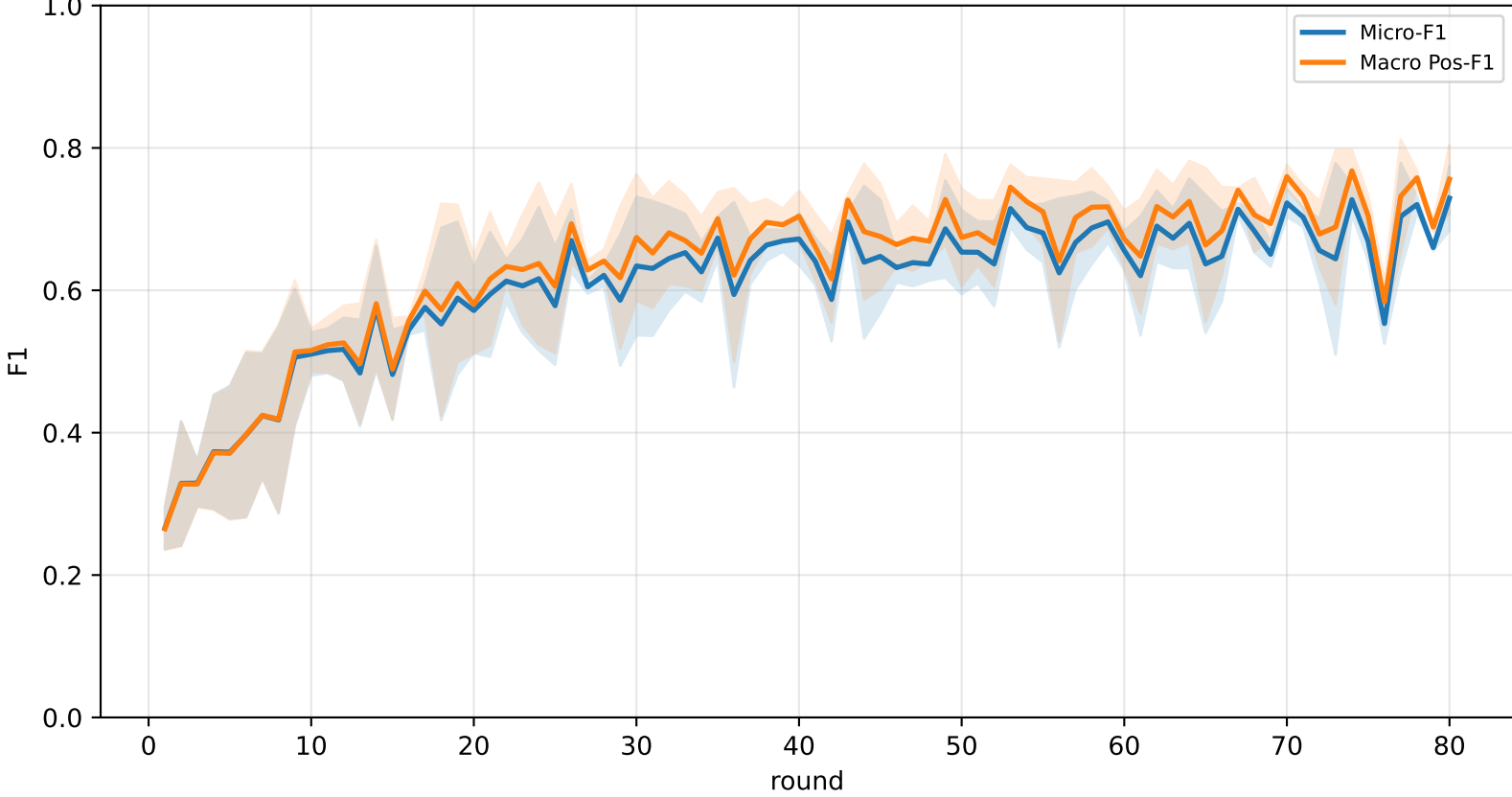
Validation loss comparison



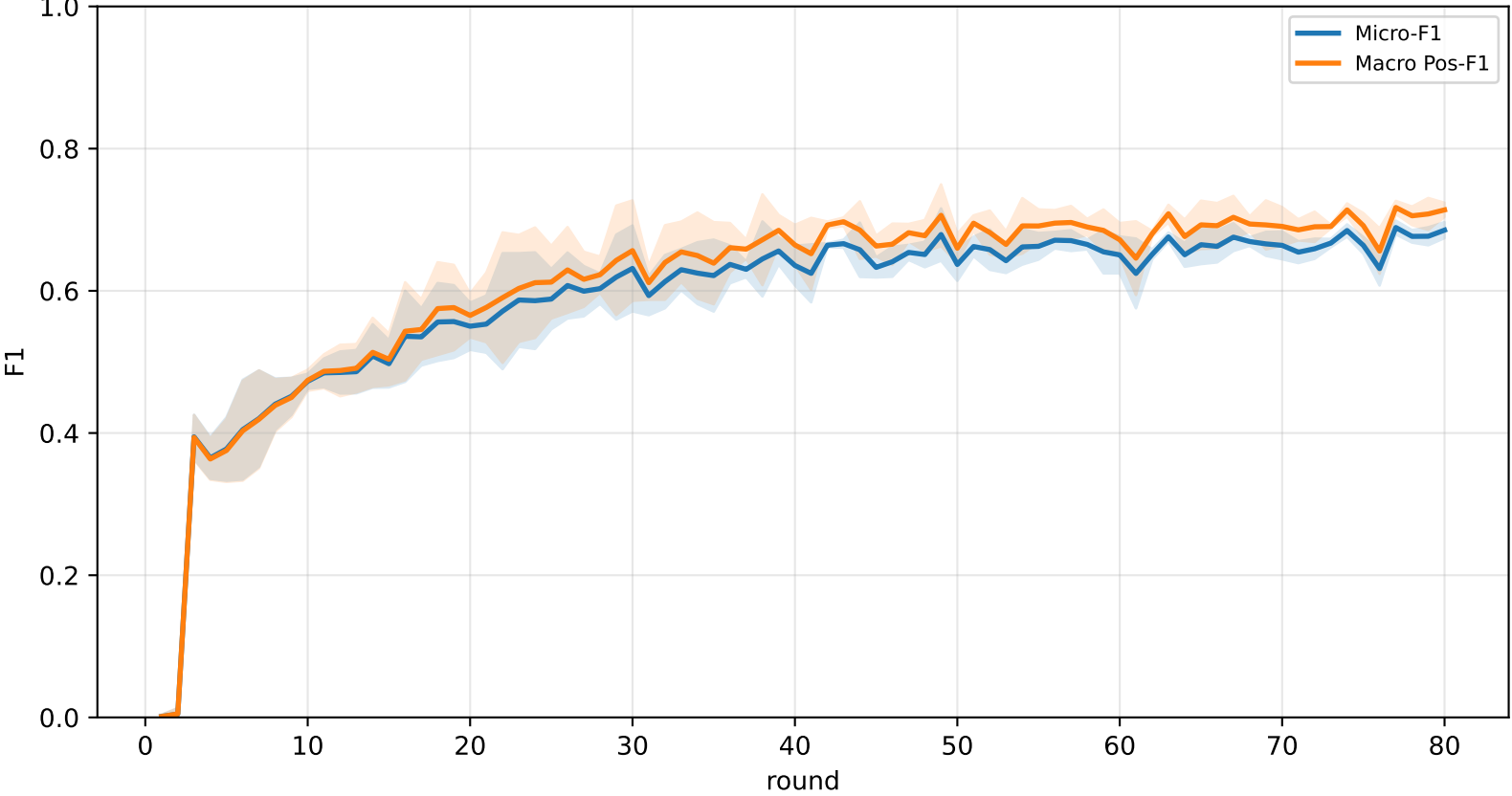
Pooled local centralized micro / macro F1



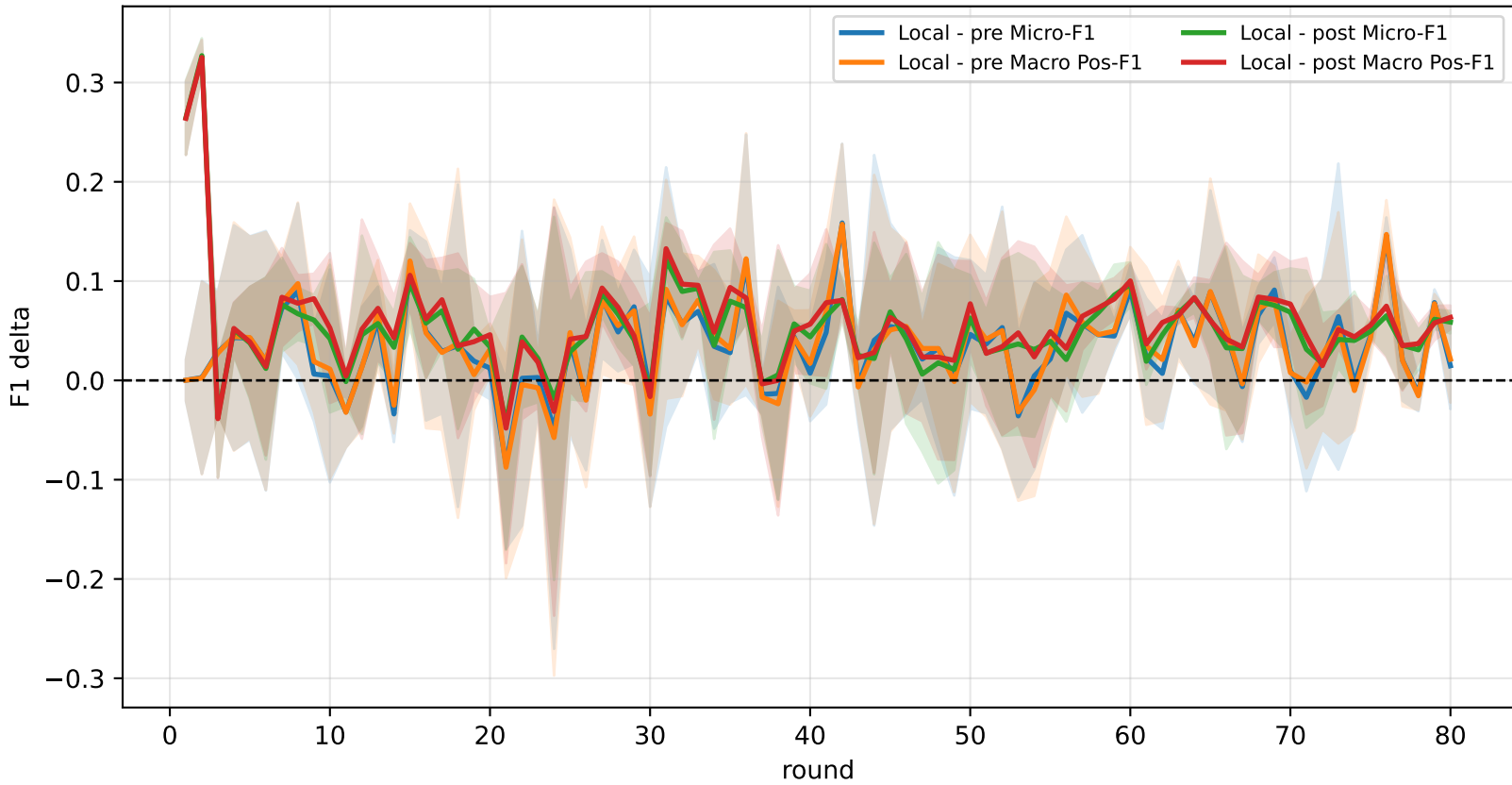
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

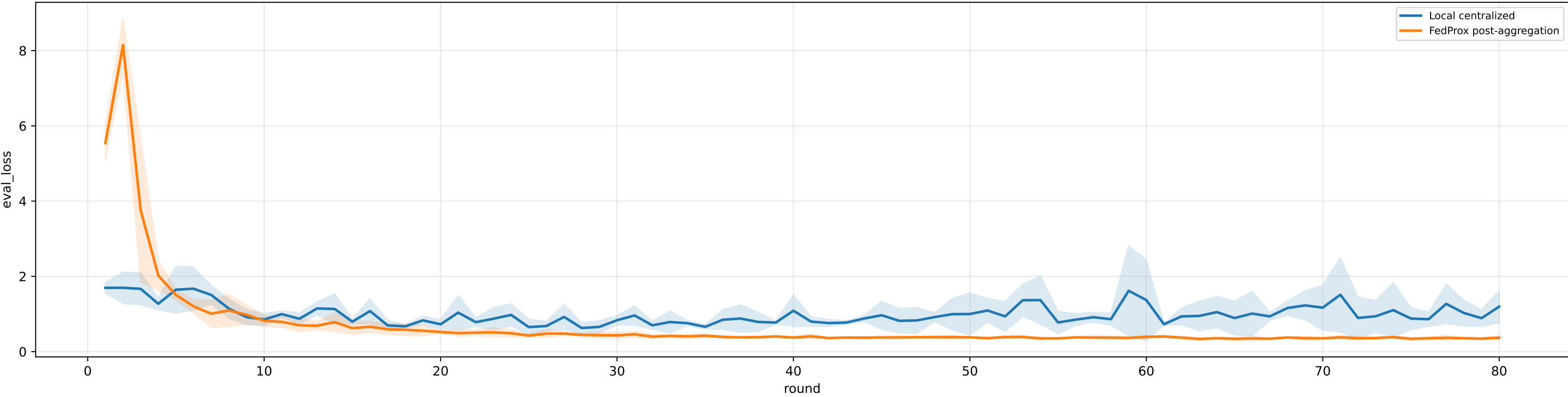


family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

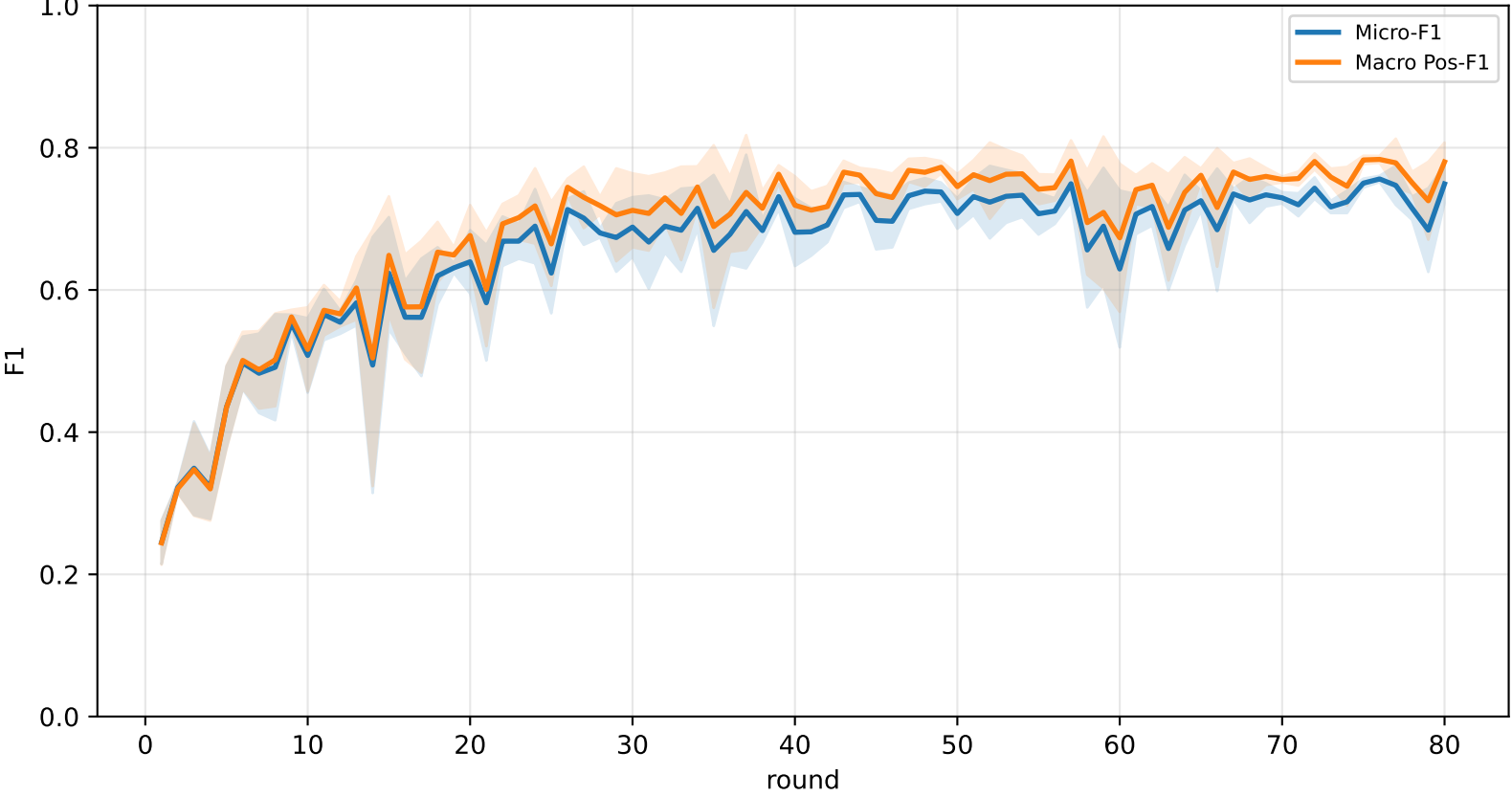
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	59.3% ± 1.8%	62.3% ± 1.2%	70.6% ± 12.2%	70.7% ± 7.8%	63.8% ± 2.1%	59.4% ± 6.7%	46.7% ± 4.1%
FedProx	66.0% ± 0.6%	69.1% ± 0.3%	84.0% ± 2.3%	78.2% ± 1.5%	67.6% ± 3.3%	62.9% ± 0.9%	52.6% ± 1.5%

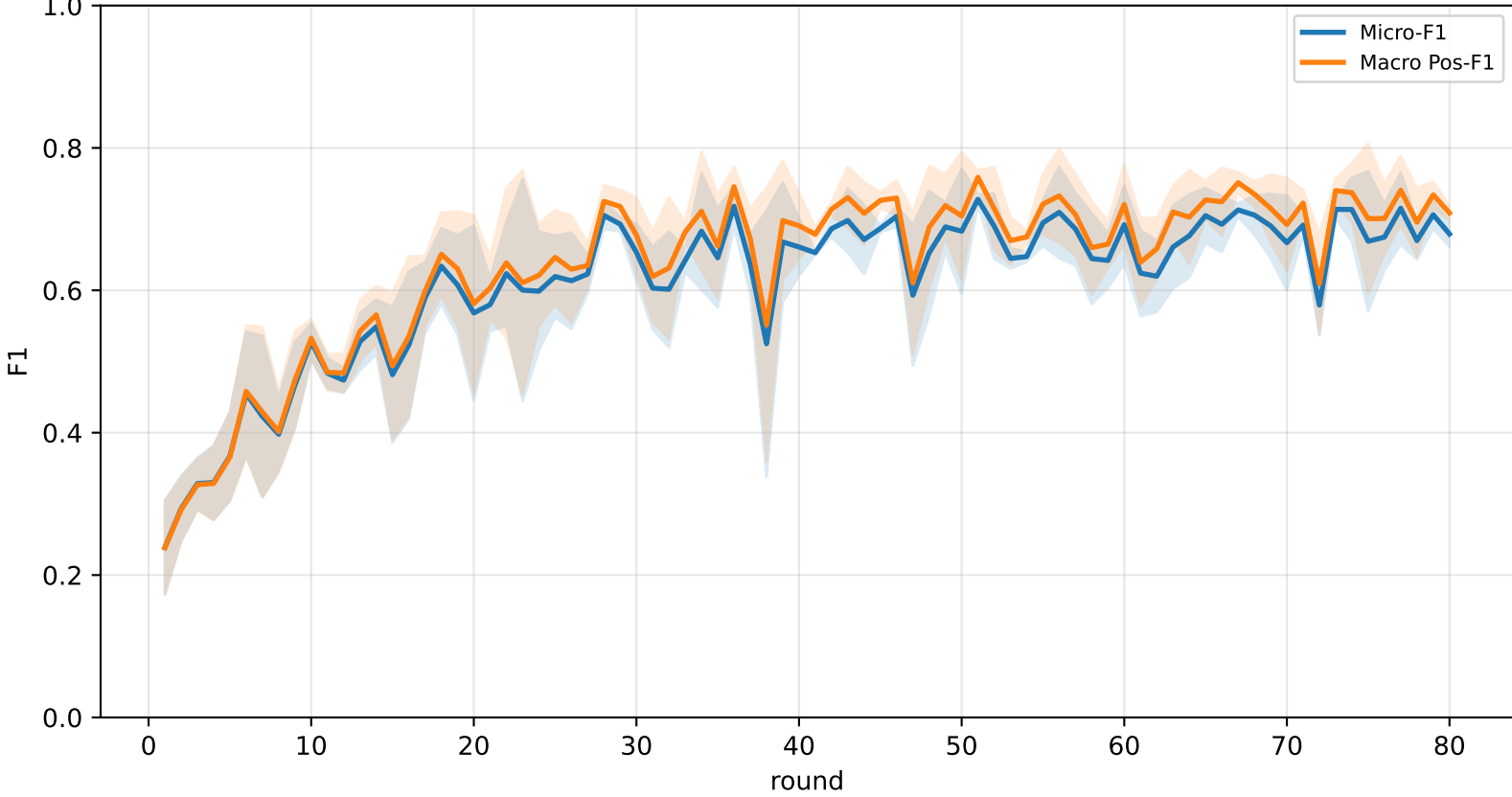
Validation loss comparison



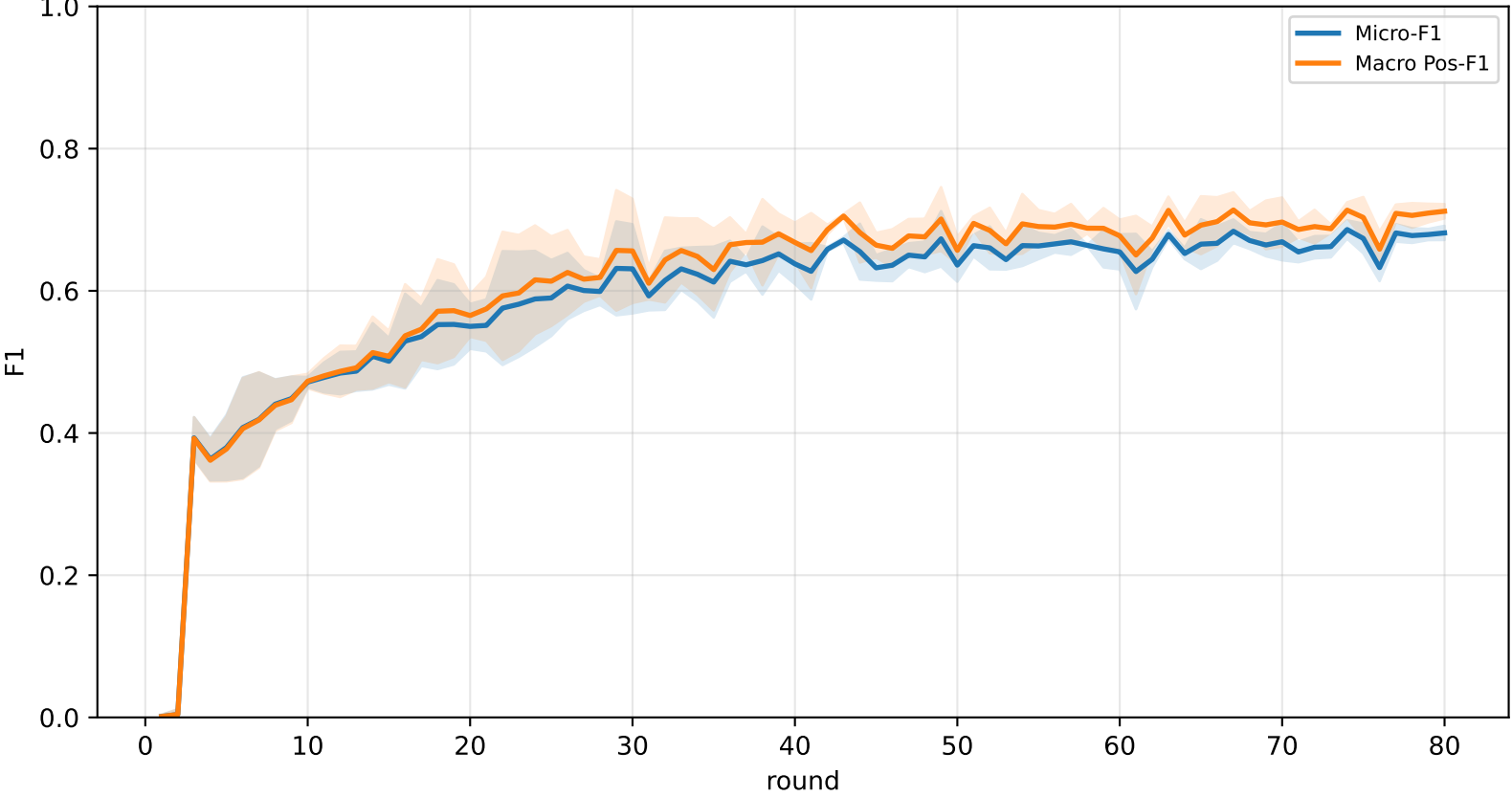
Pooled local centralized micro / macro F1



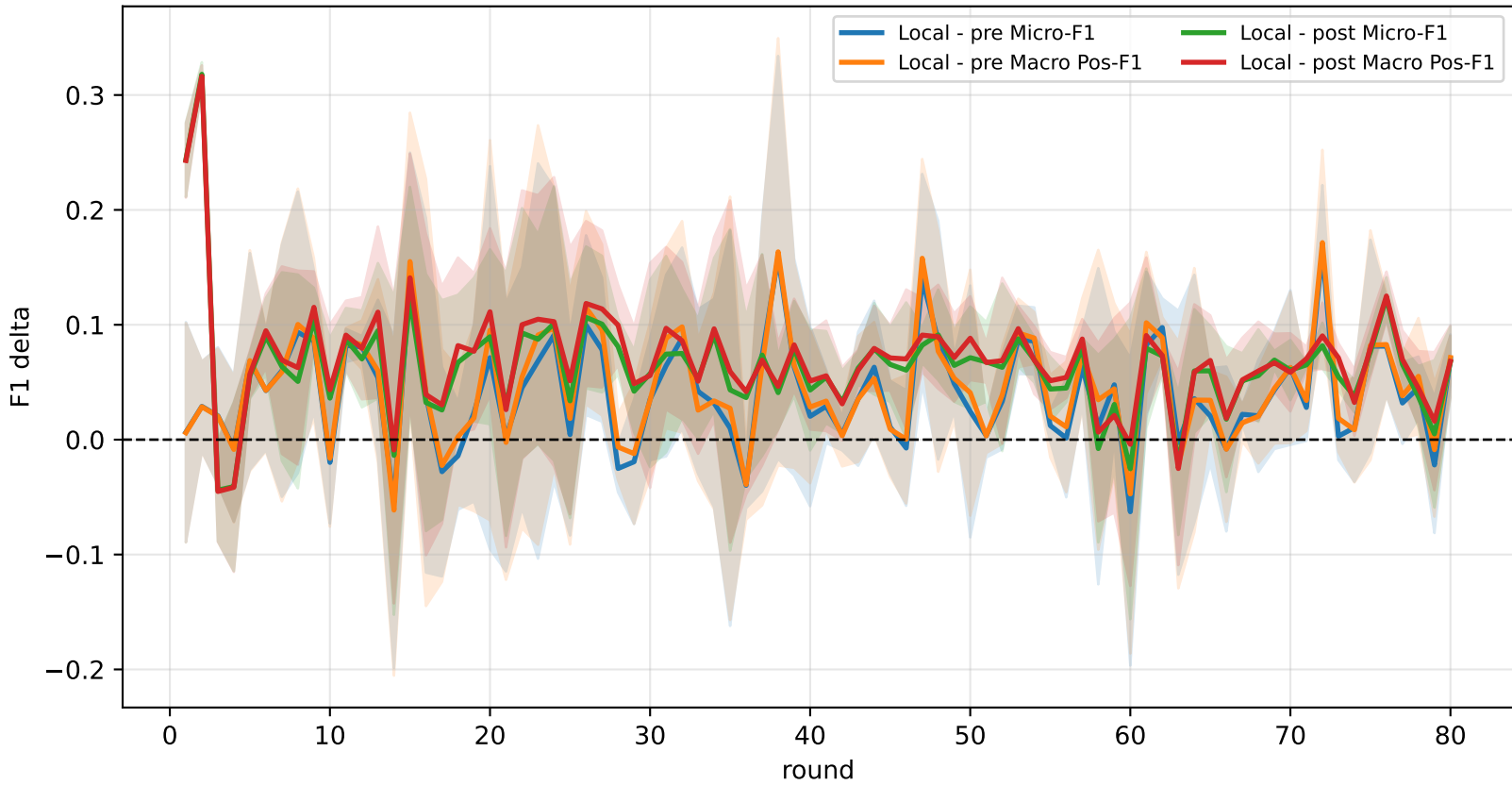
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

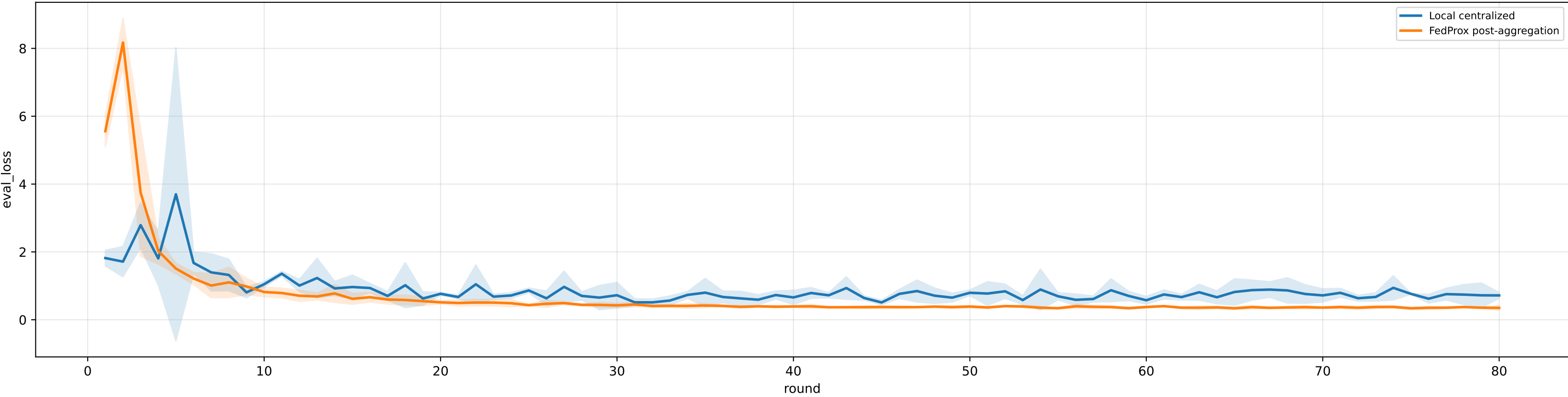


family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

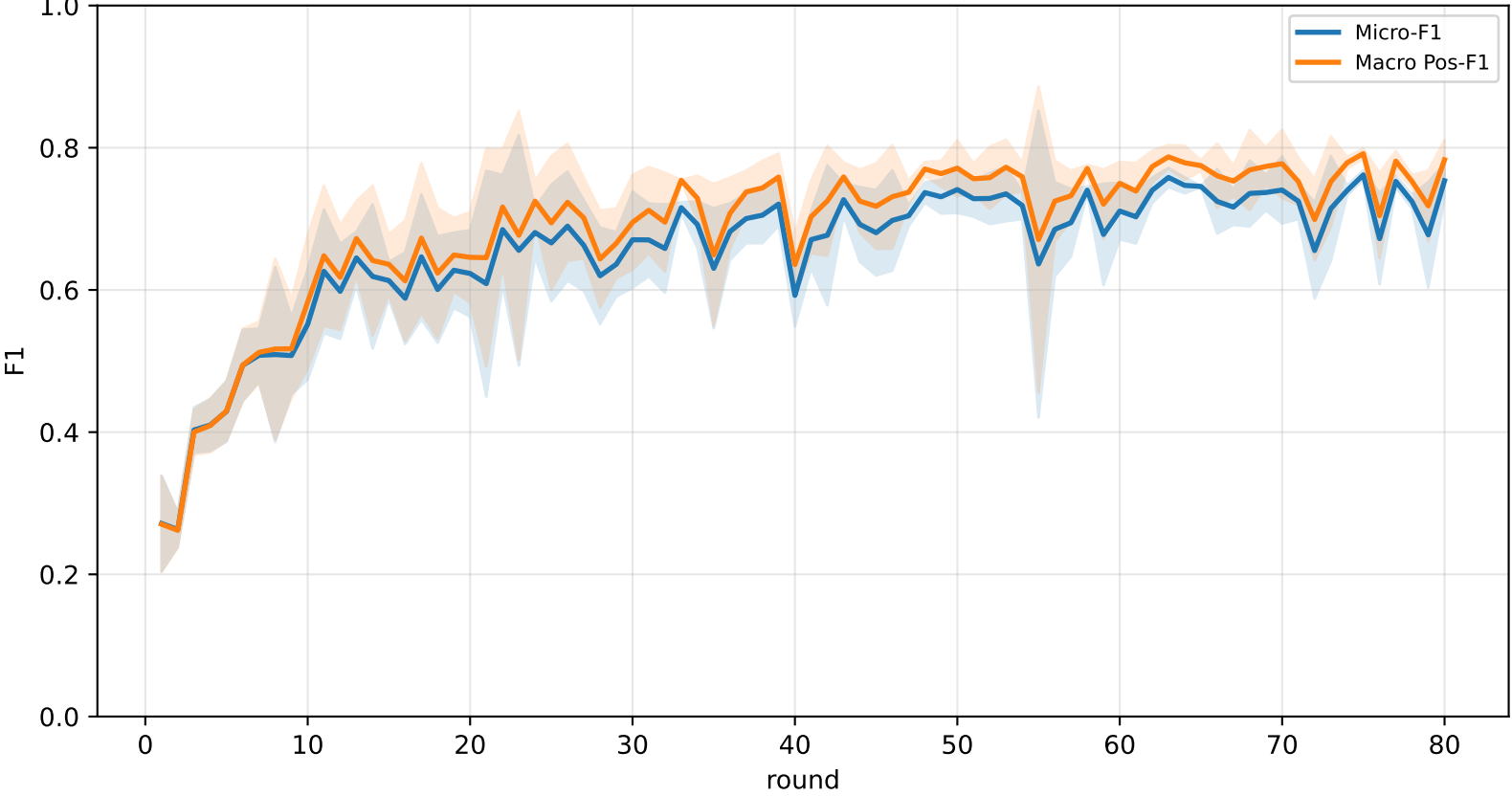
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	65.1% ± 6.6%	68.5% ± 6.7%	78.1% ± 11.0%	82.9% ± 5.3%	69.7% ± 5.5%	62.2% ± 8.5%	49.8% ± 5.3%
FedProx	65.7% ± 0.7%	68.6% ± 0.4%	83.2% ± 2.8%	77.5% ± 1.3%	66.9% ± 3.3%	62.7% ± 0.6%	52.8% ± 1.3%

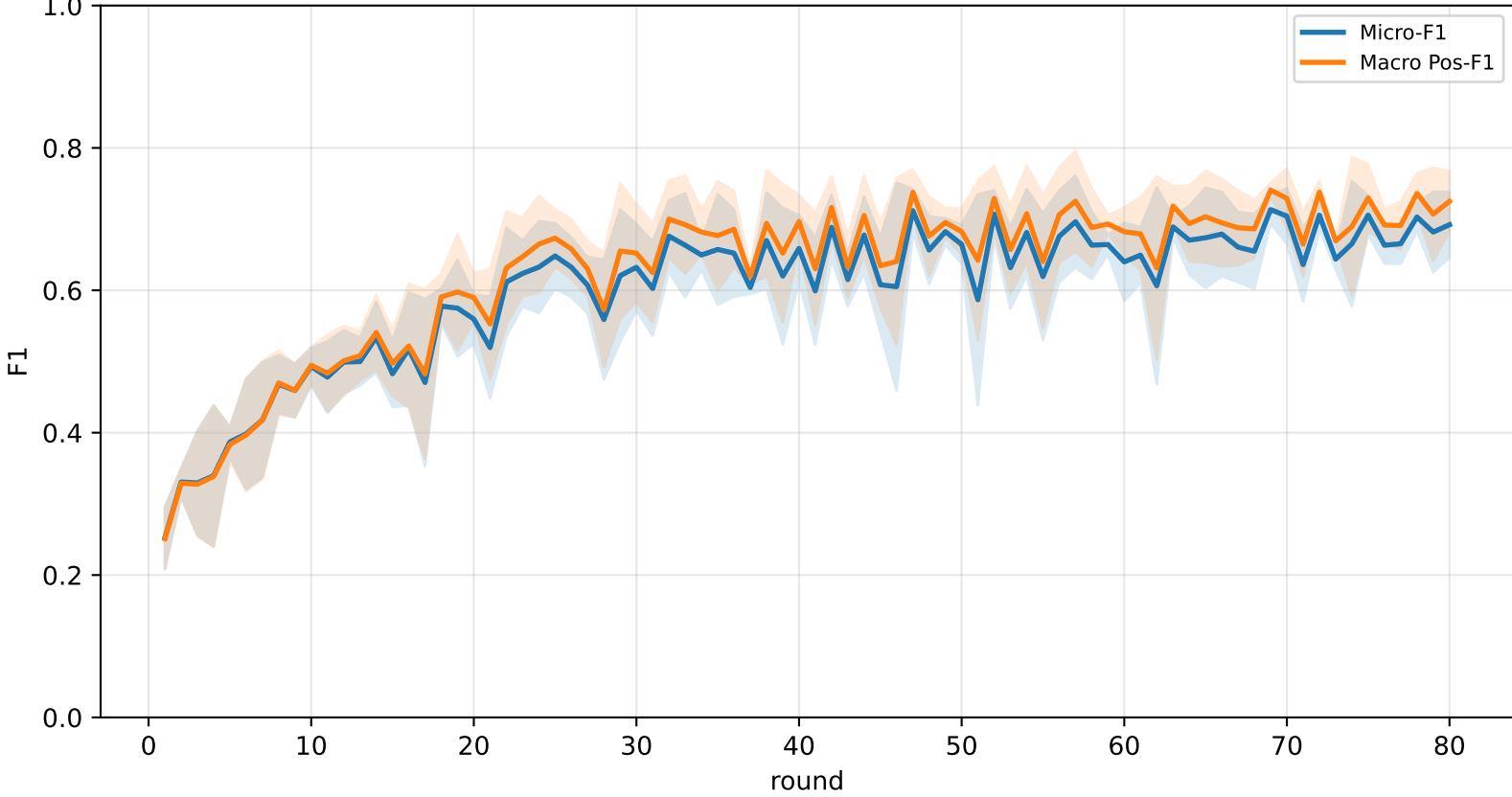
Validation loss comparison



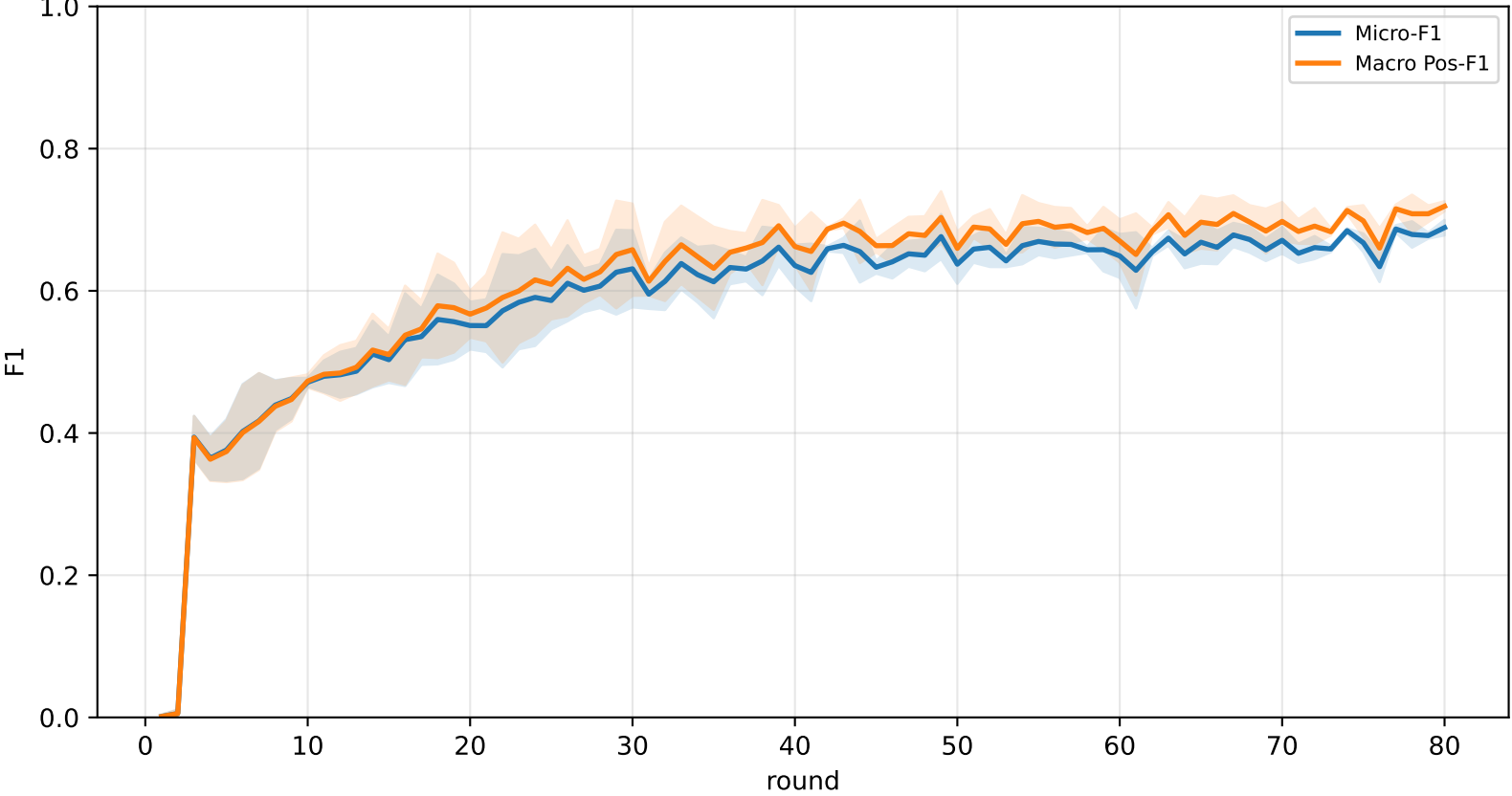
Pooled local centralized micro / macro F1



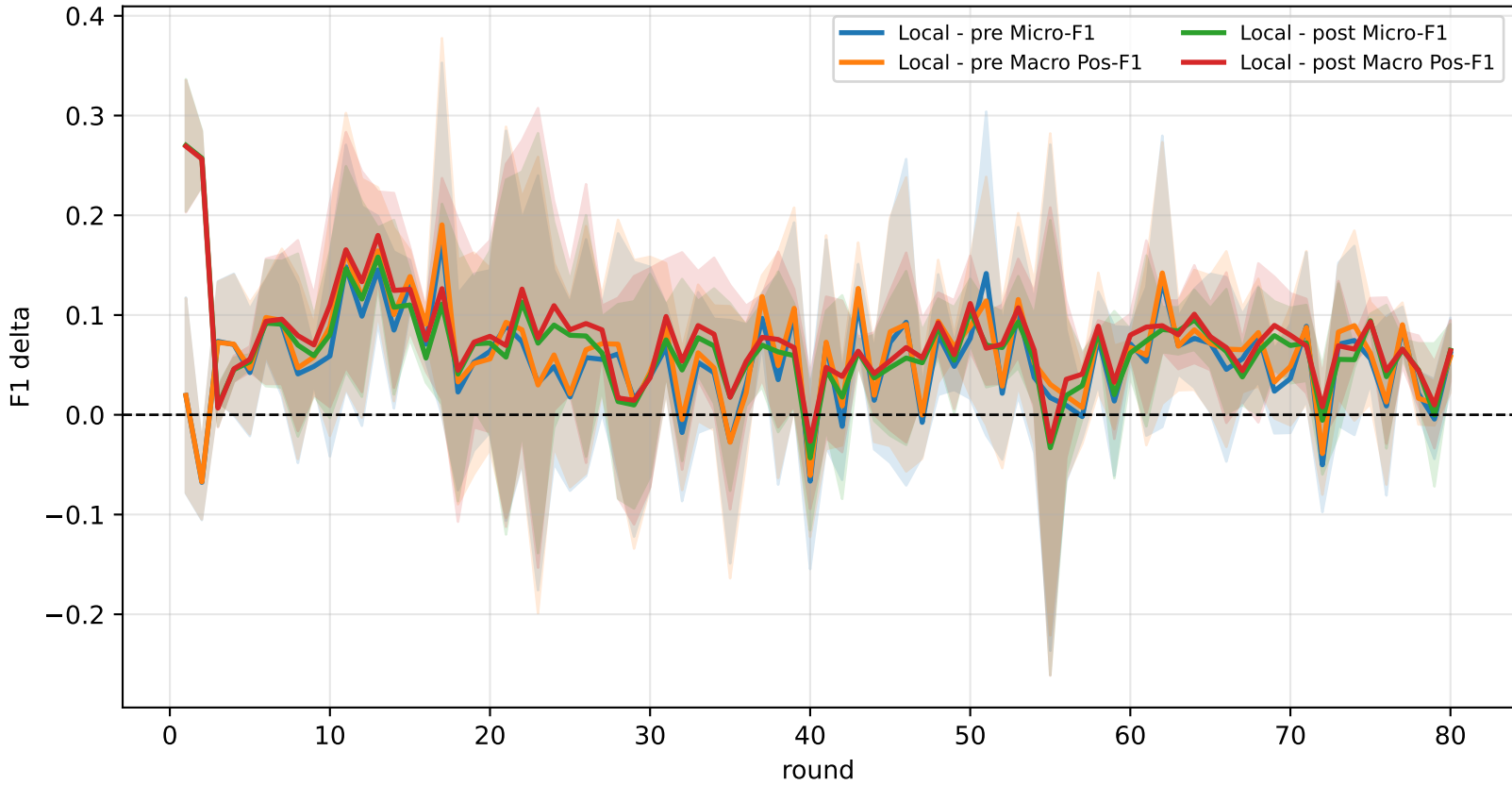
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

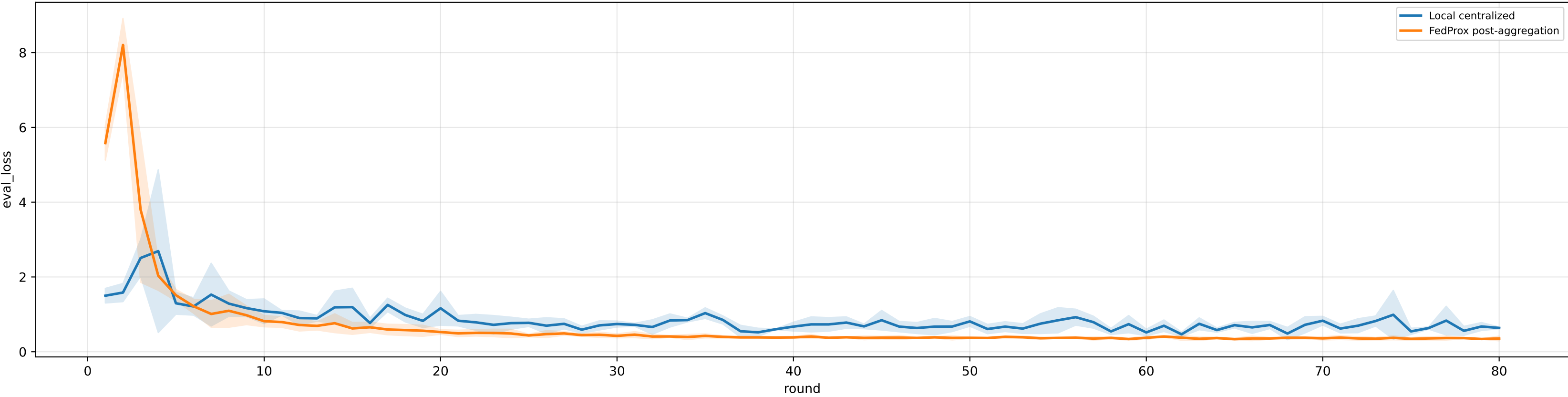


family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

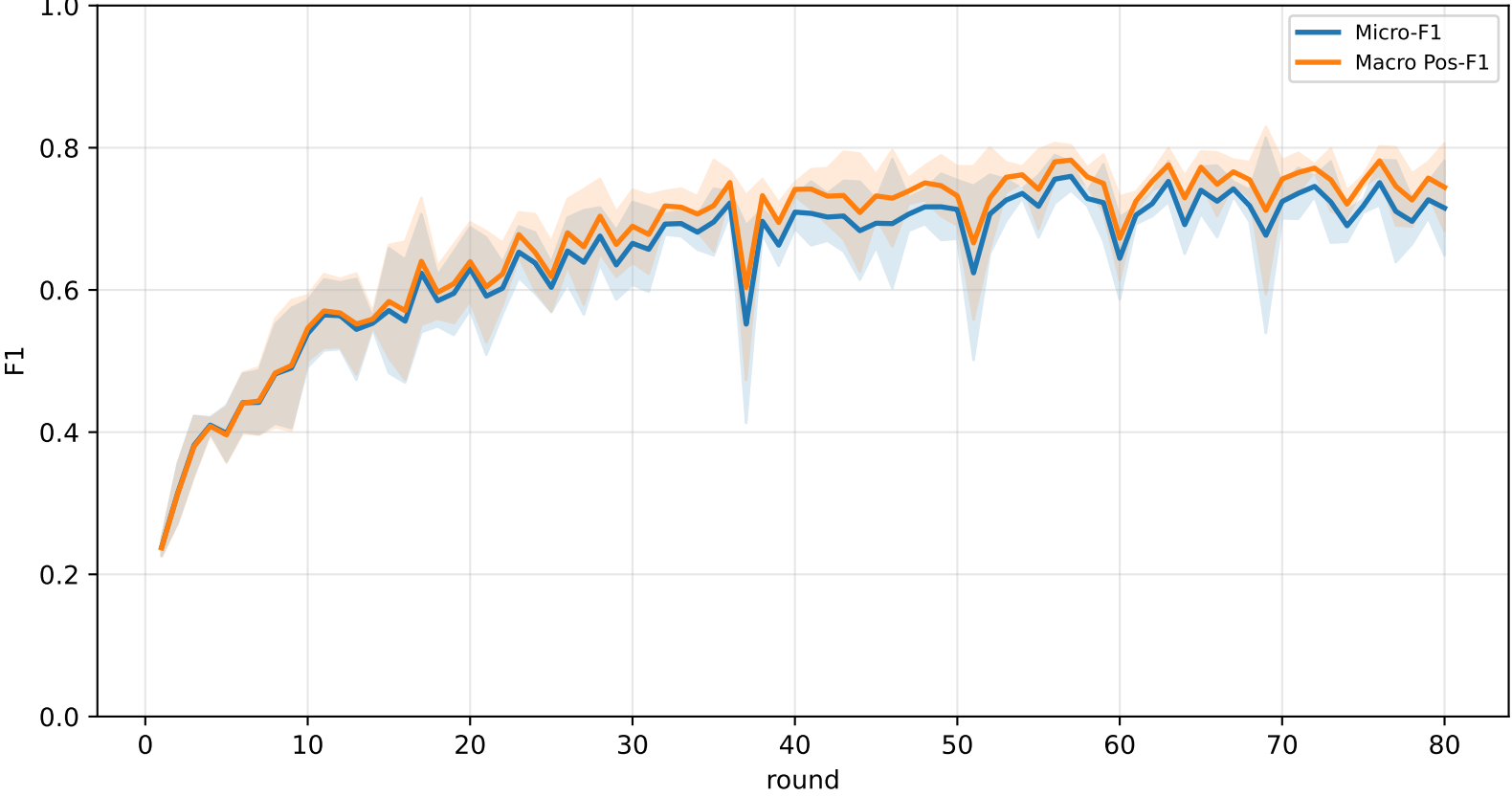
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	66.7% ± 3.2%	69.8% ± 3.5%	83.1% ± 3.5%	76.9% ± 8.5%	67.2% ± 1.6%	69.7% ± 4.1%	52.1% ± 4.2%
FedProx	65.4% ± 1.1%	68.3% ± 0.7%	82.8% ± 2.3%	76.6% ± 1.2%	67.0% ± 4.5%	62.5% ± 0.3%	52.5% ± 1.4%

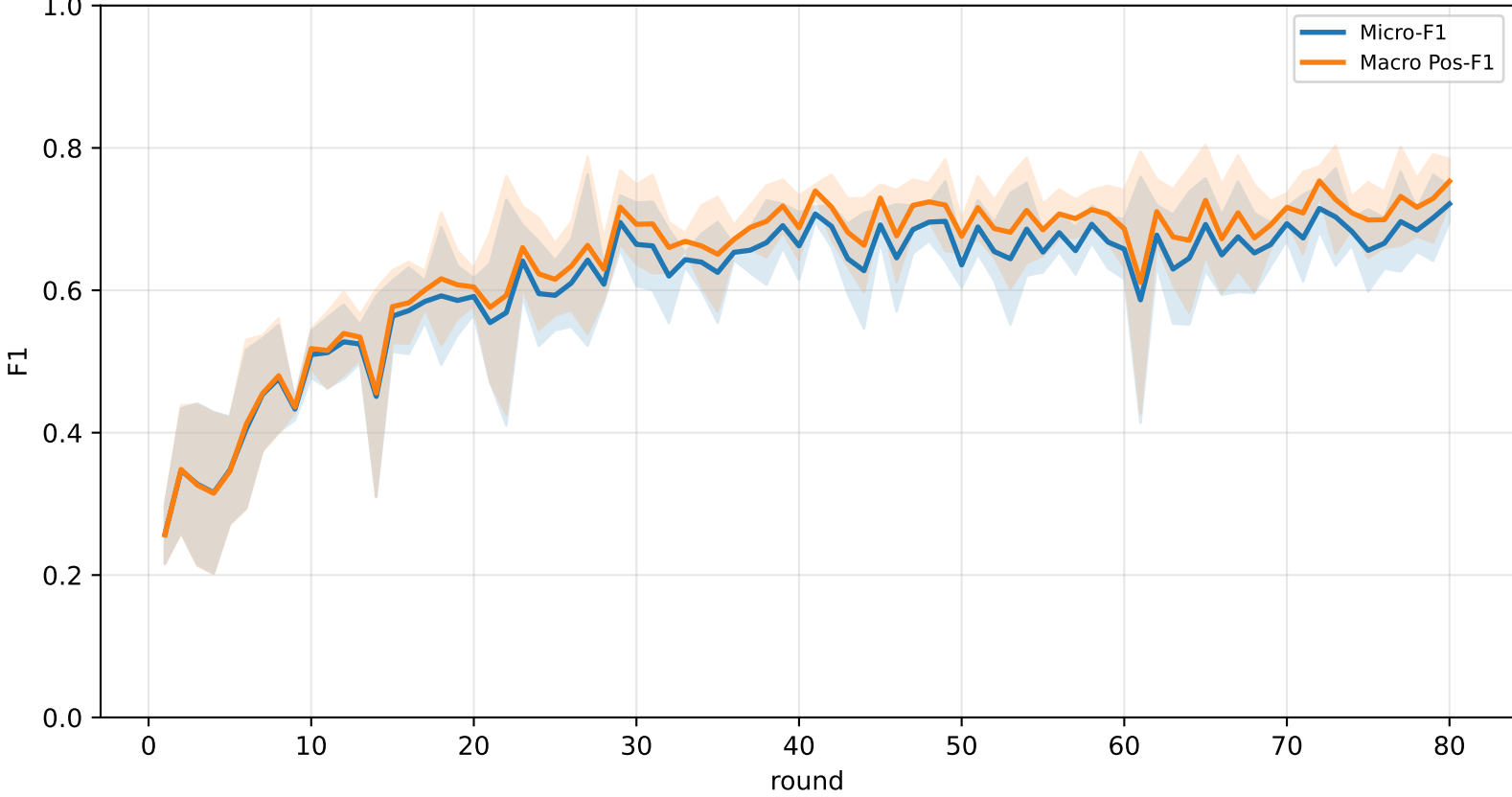
Validation loss comparison



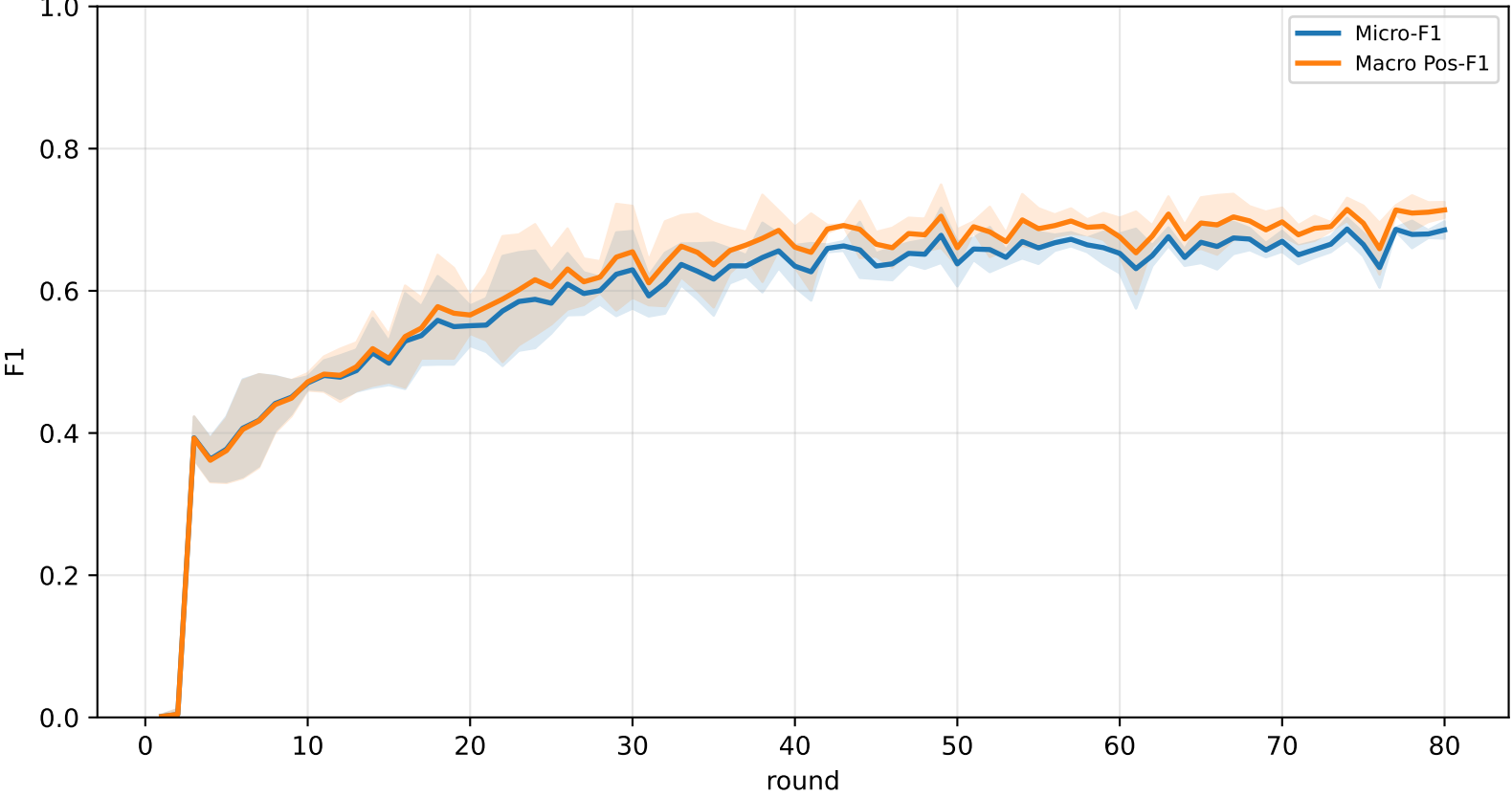
Pooled local centralized micro / macro F1



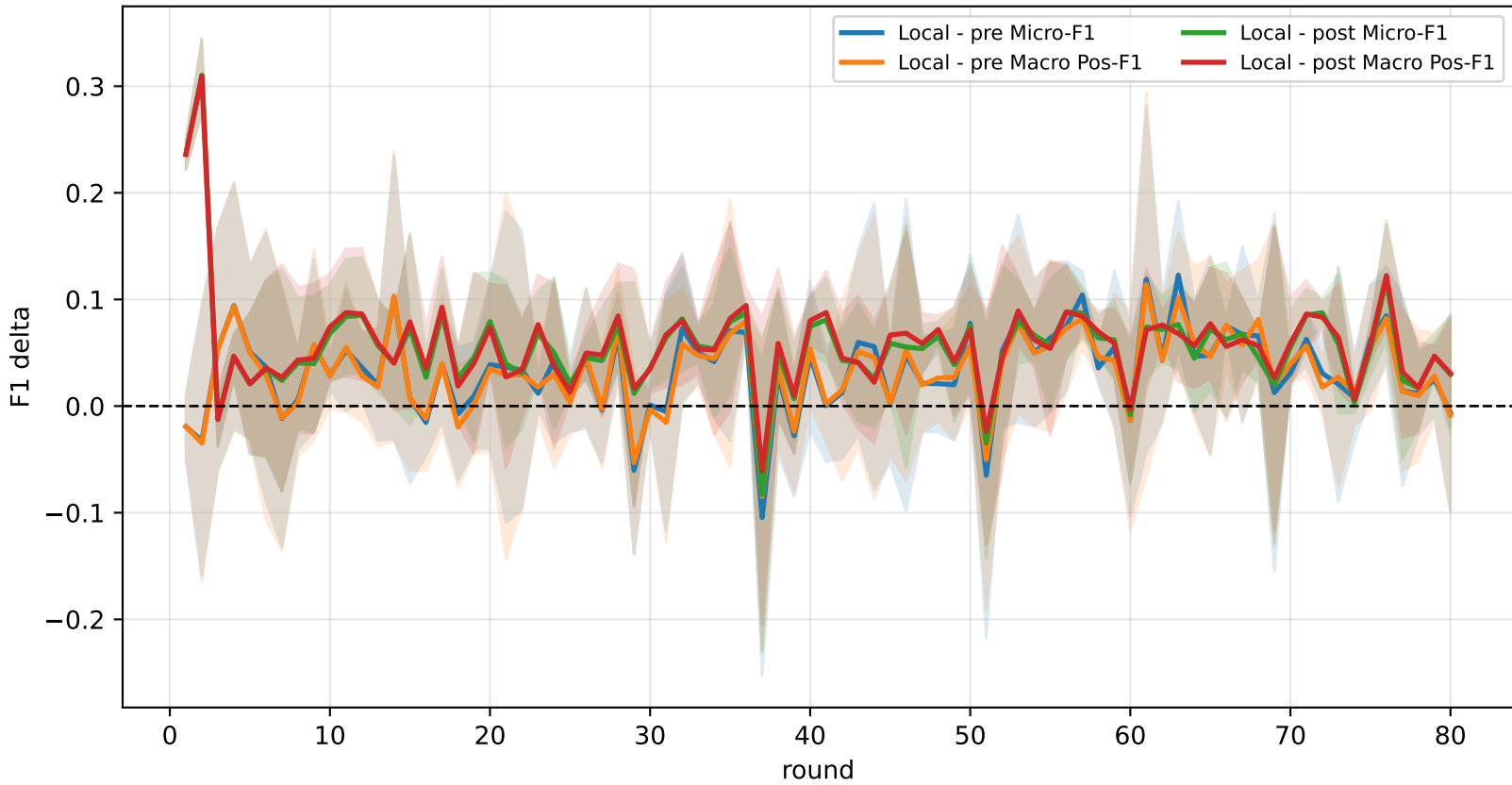
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

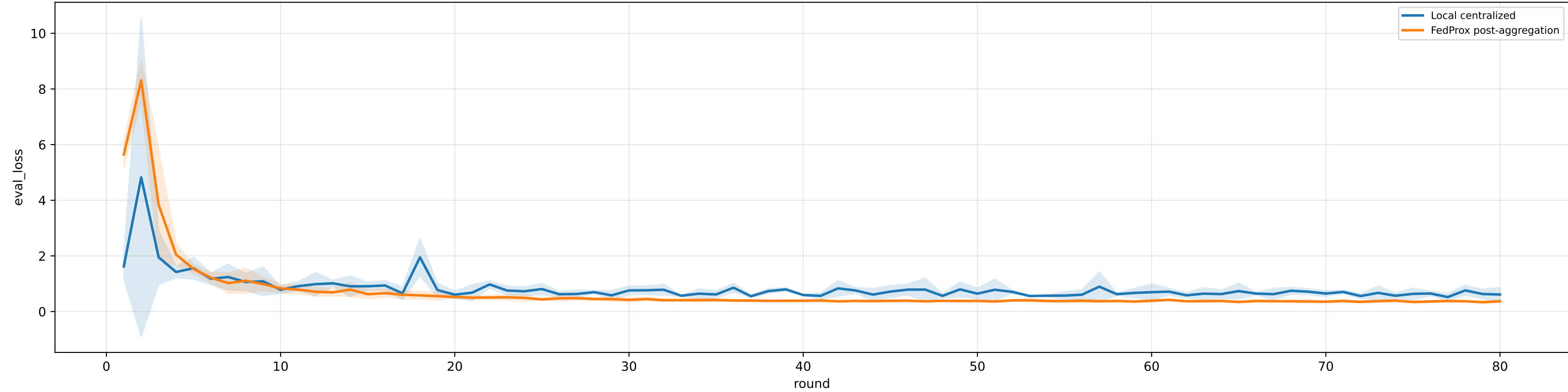


family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

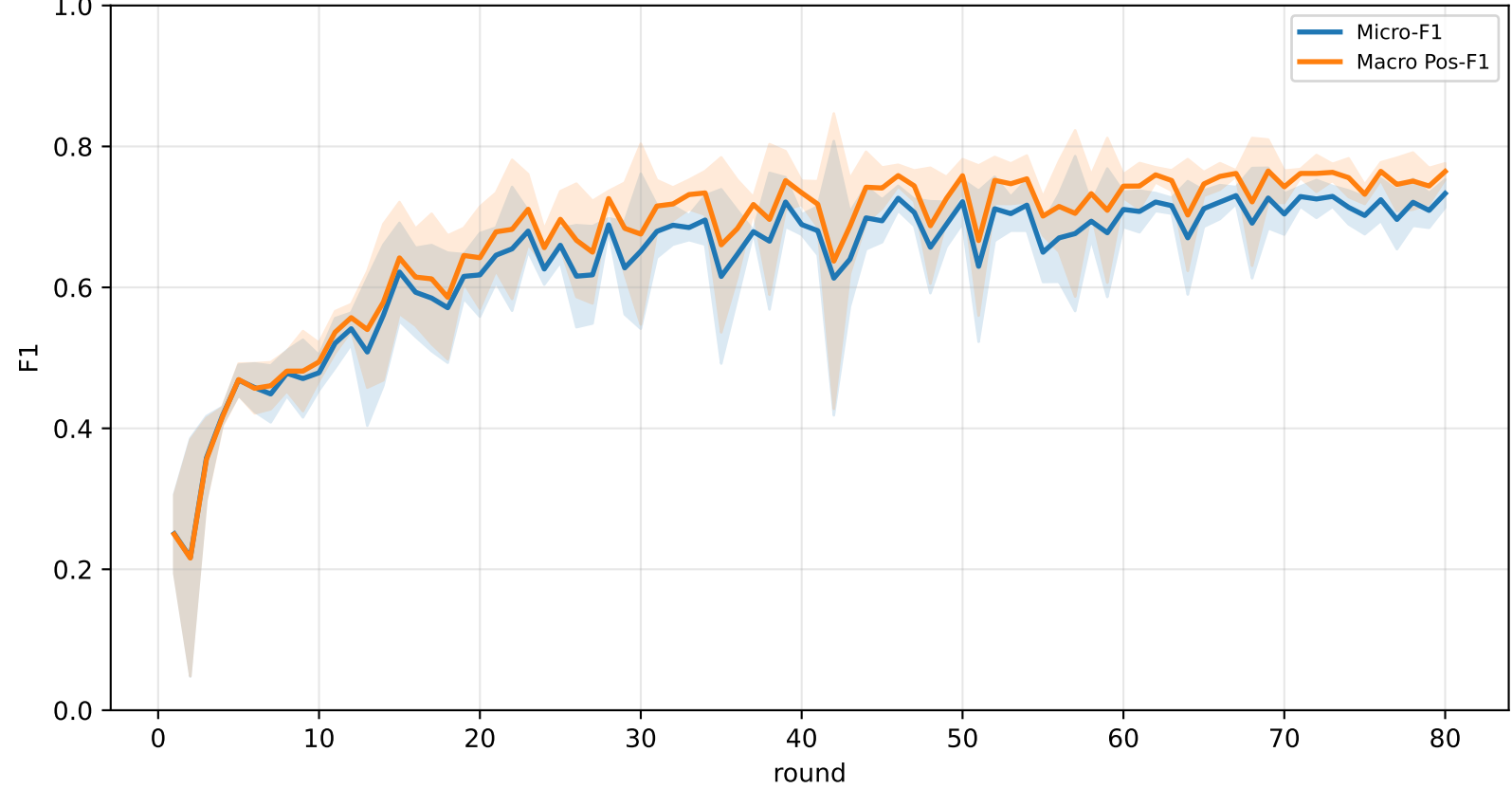
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	68.1% ± 1.5%	72.6% ± 1.2%	89.4% ± 1.7%	86.4% ± 0.7%	69.7% ± 3.6%	67.1% ± 2.2%	50.4% ± 1.9%
FedProx	65.9% ± 1.5%	68.9% ± 1.2%	83.9% ± 1.5%	76.8% ± 1.4%	67.9% ± 4.5%	63.4% ± 0.7%	52.6% ± 1.9%

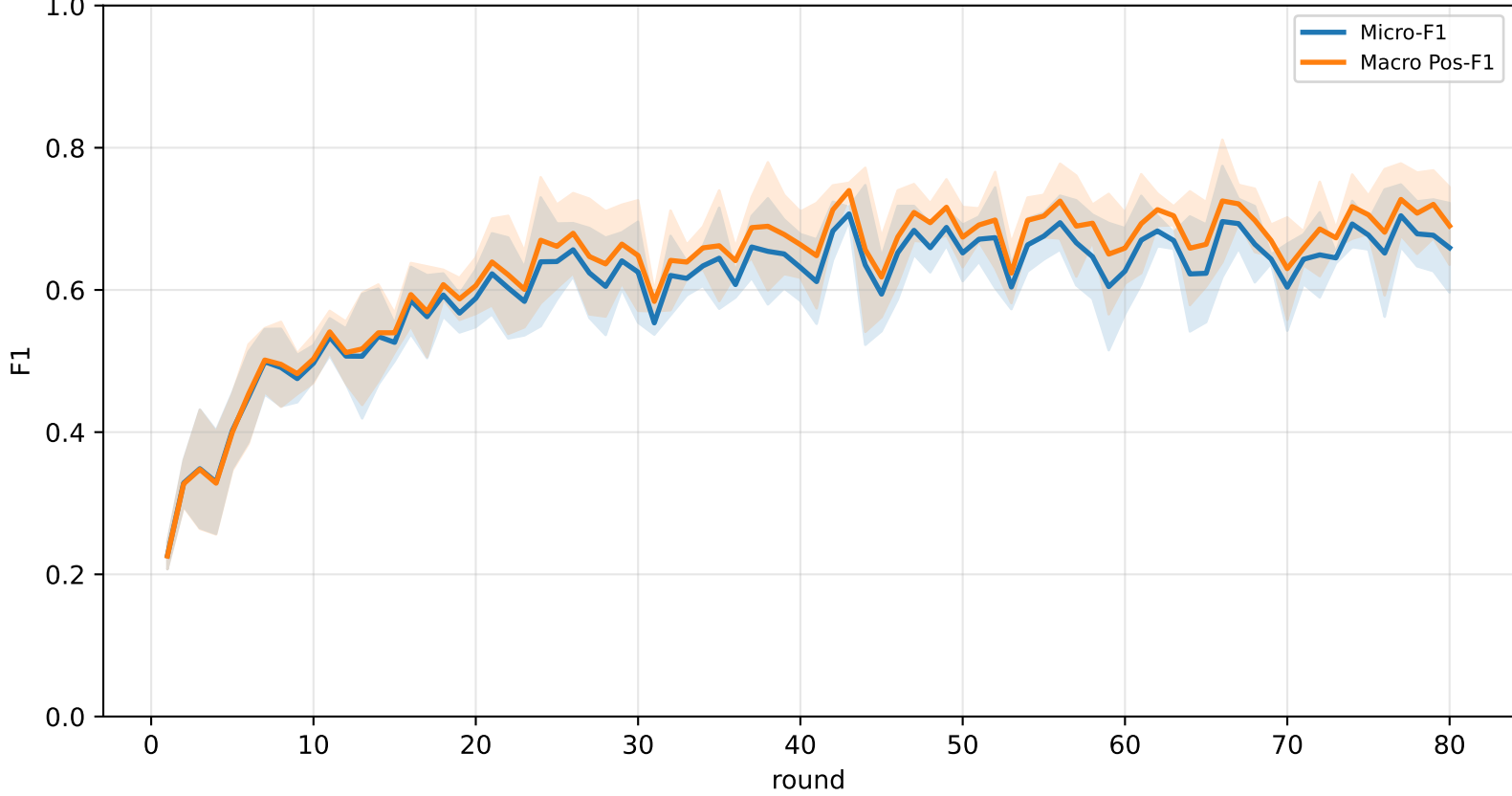
Validation loss comparison



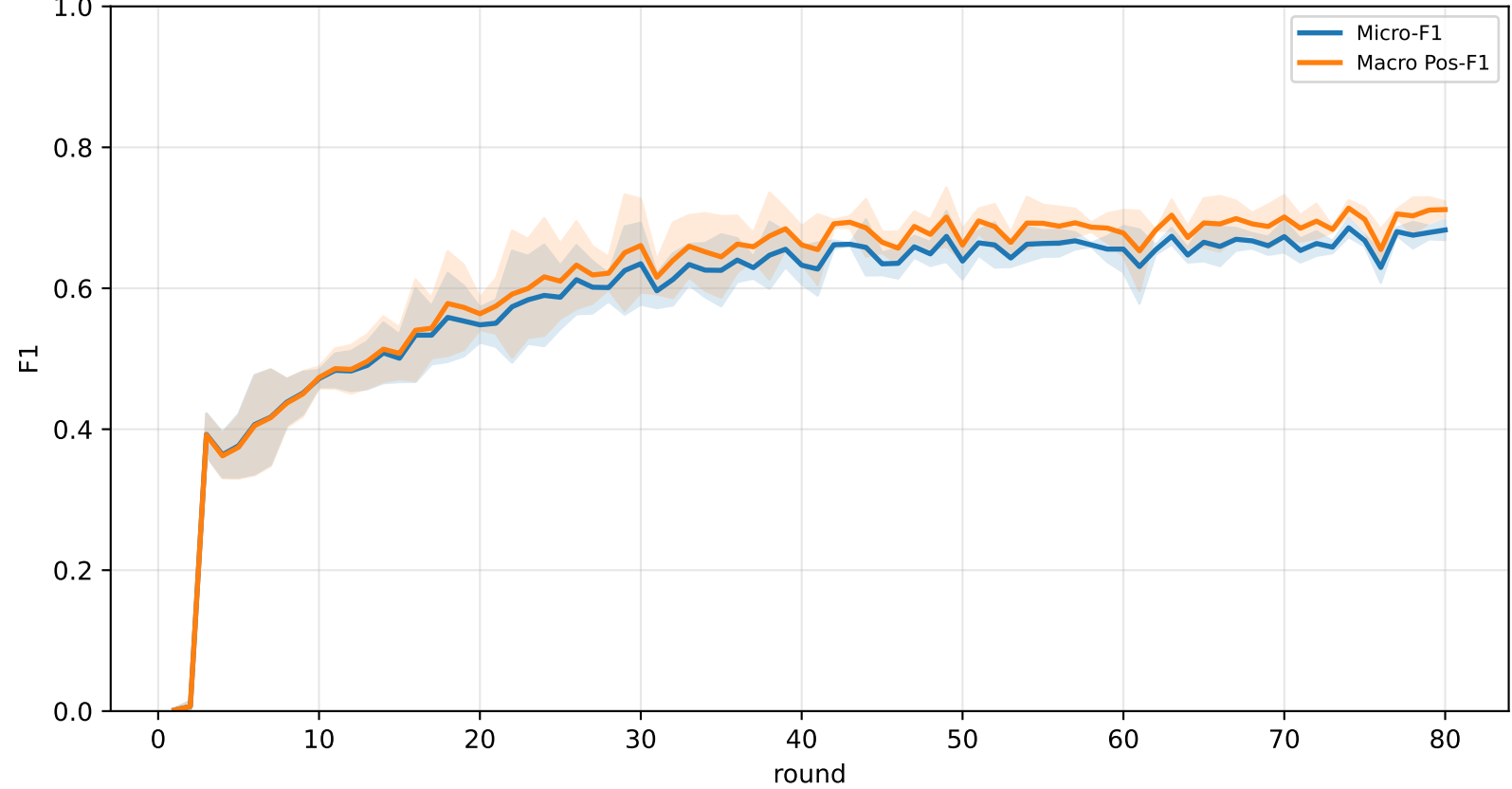
Pooled local centralized micro / macro F1



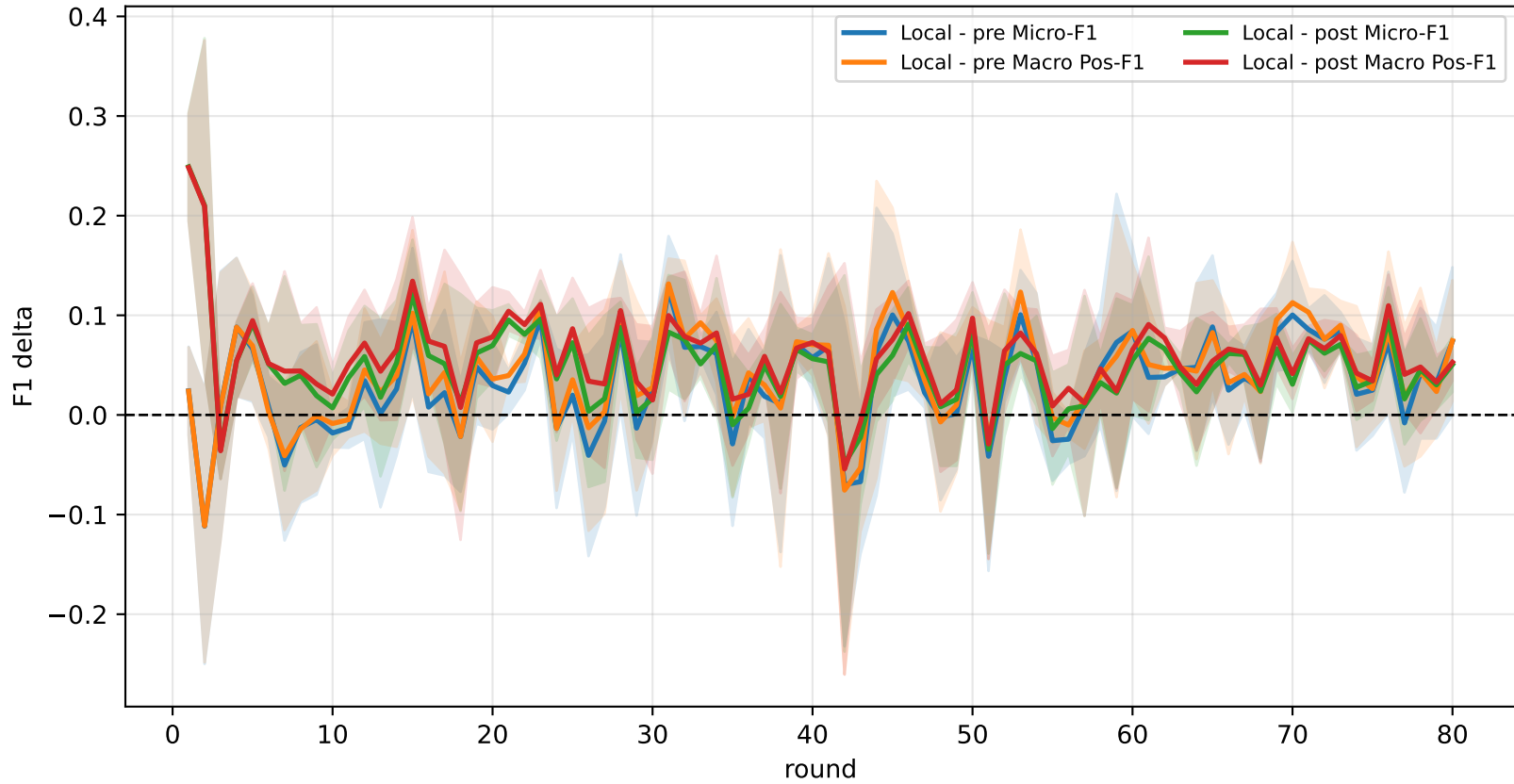
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)



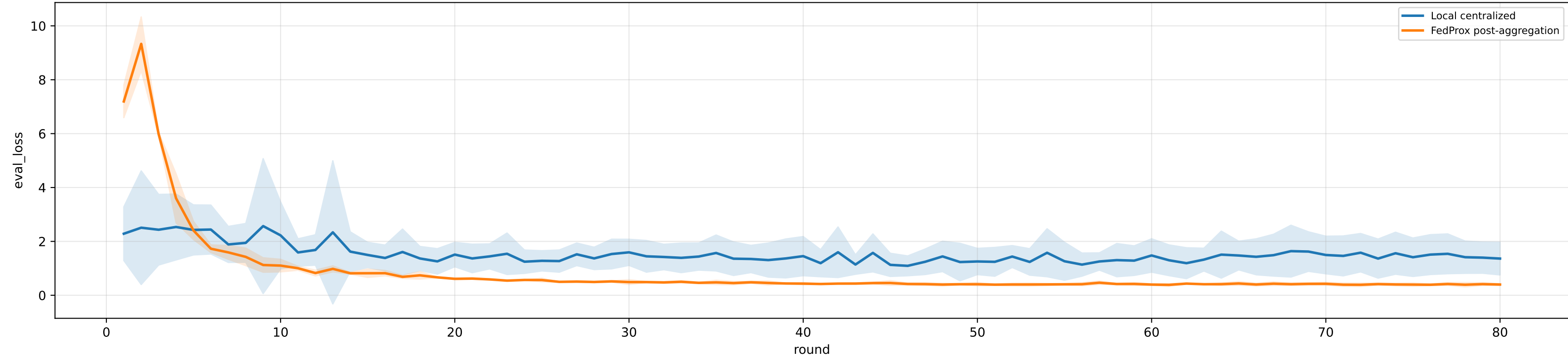
Controlled q-label heterogeneity: Mid | FedProx diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.8%±0.0% / true 6.4% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

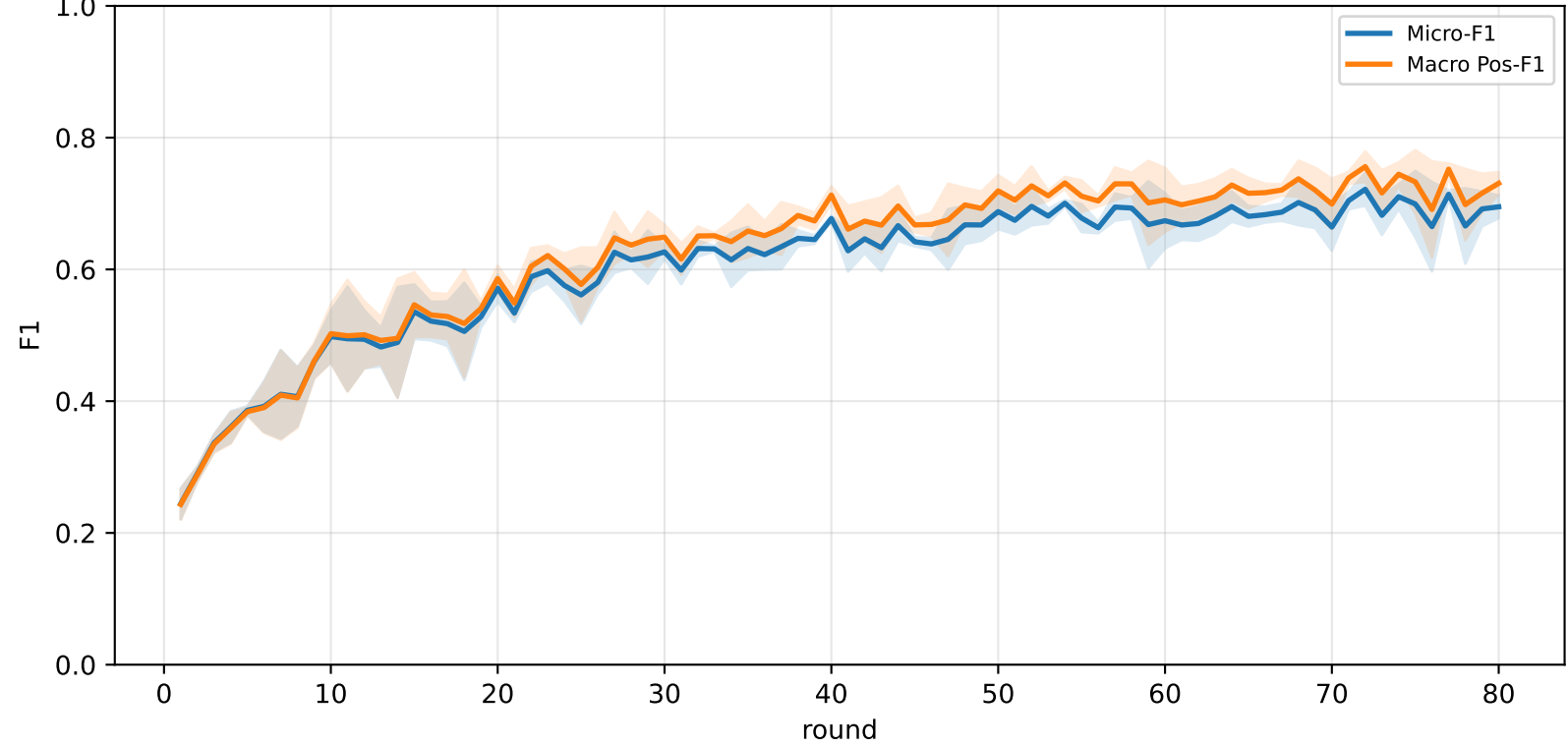
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	60.3% ± 0.4%	63.4% ± 0.4%	74.7% ± 4.2%	70.7% ± 0.1%	60.9% ± 2.0%	57.0% ± 1.1%	47.0% ± 0.4%
FedProx	66.4% ± 1.2%	68.8% ± 0.9%	84.0% ± 1.6%	74.1% ± 4.4%	66.5% ± 0.9%	64.2% ± 0.5%	55.2% ± 2.4%

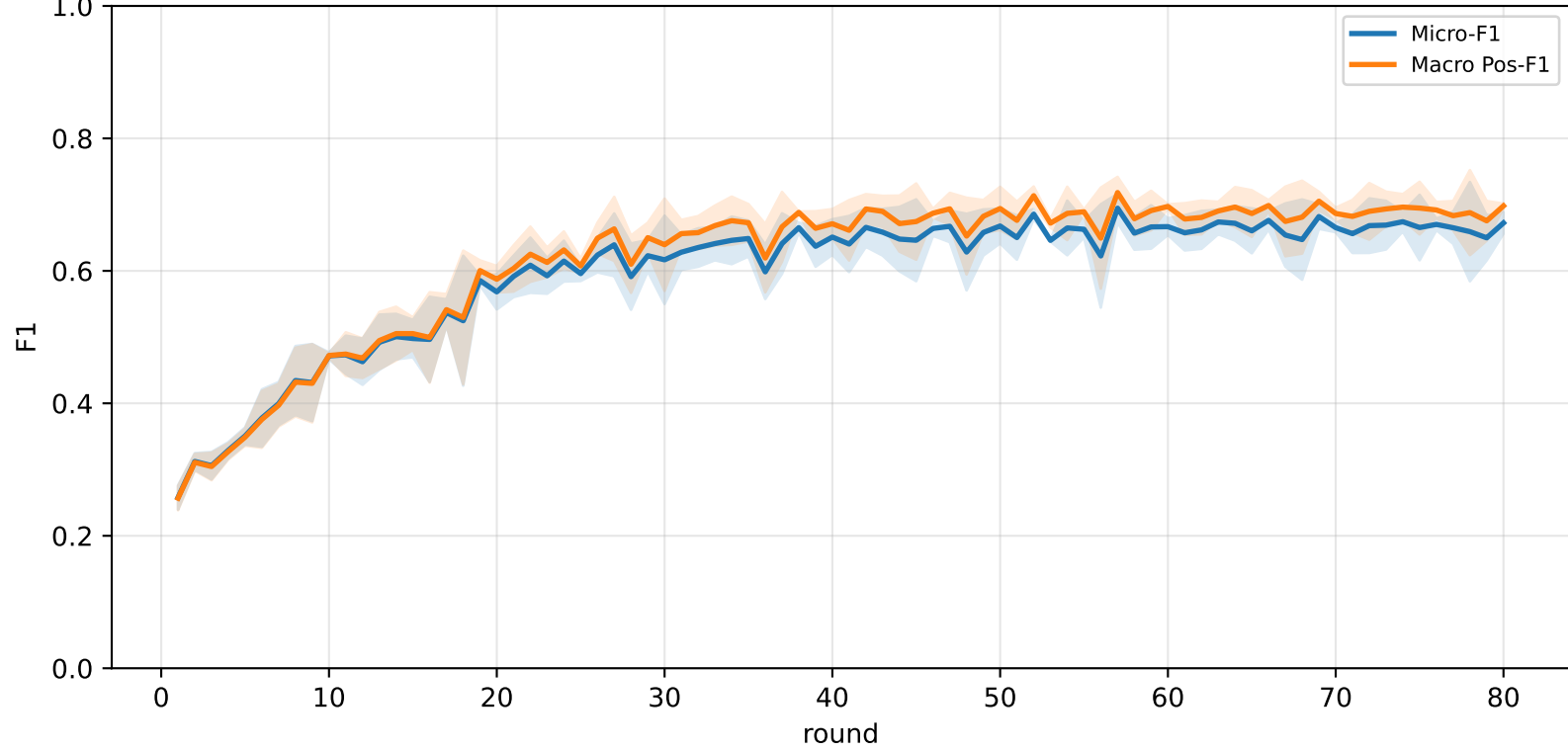
Validation loss comparison



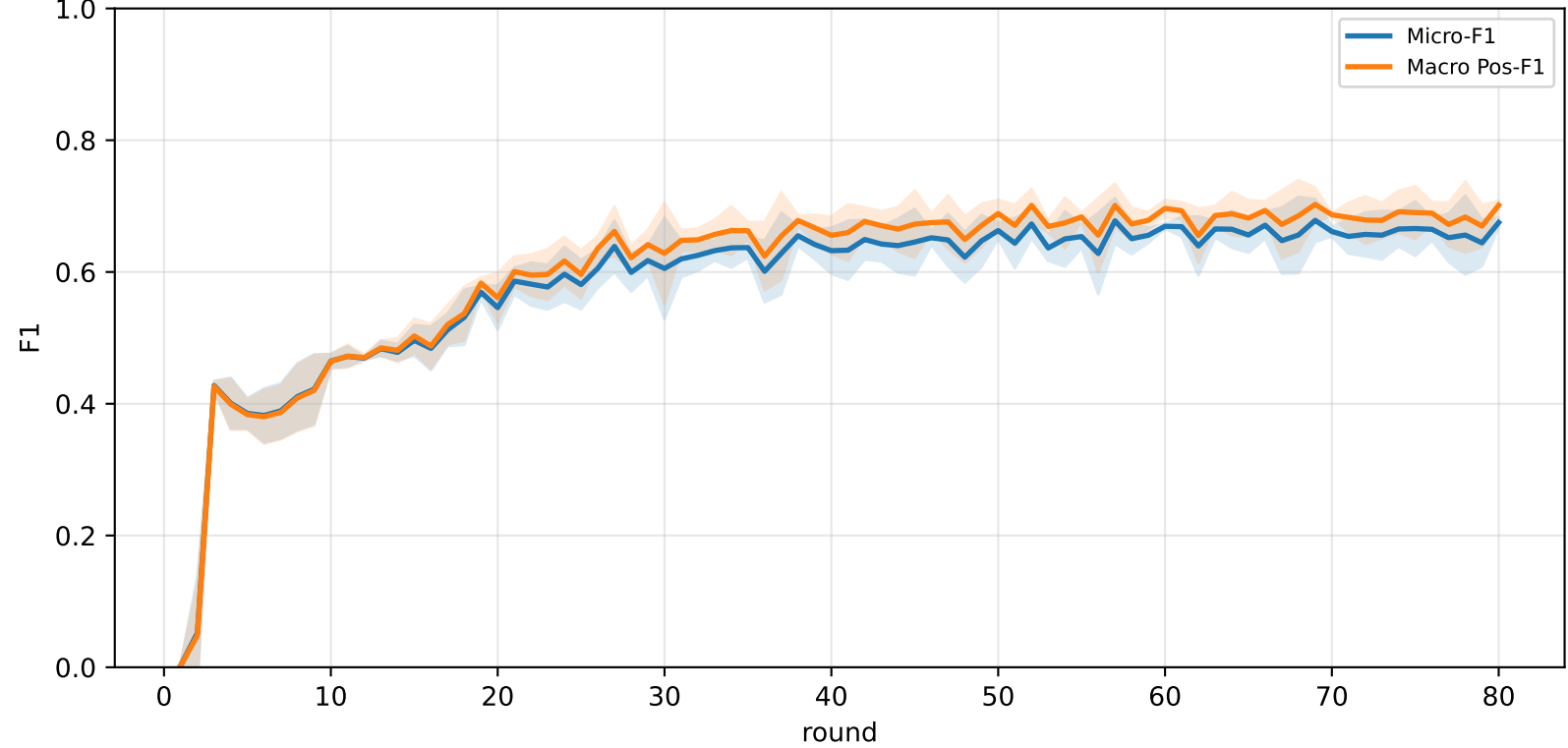
Pooled local centralized micro / macro F1



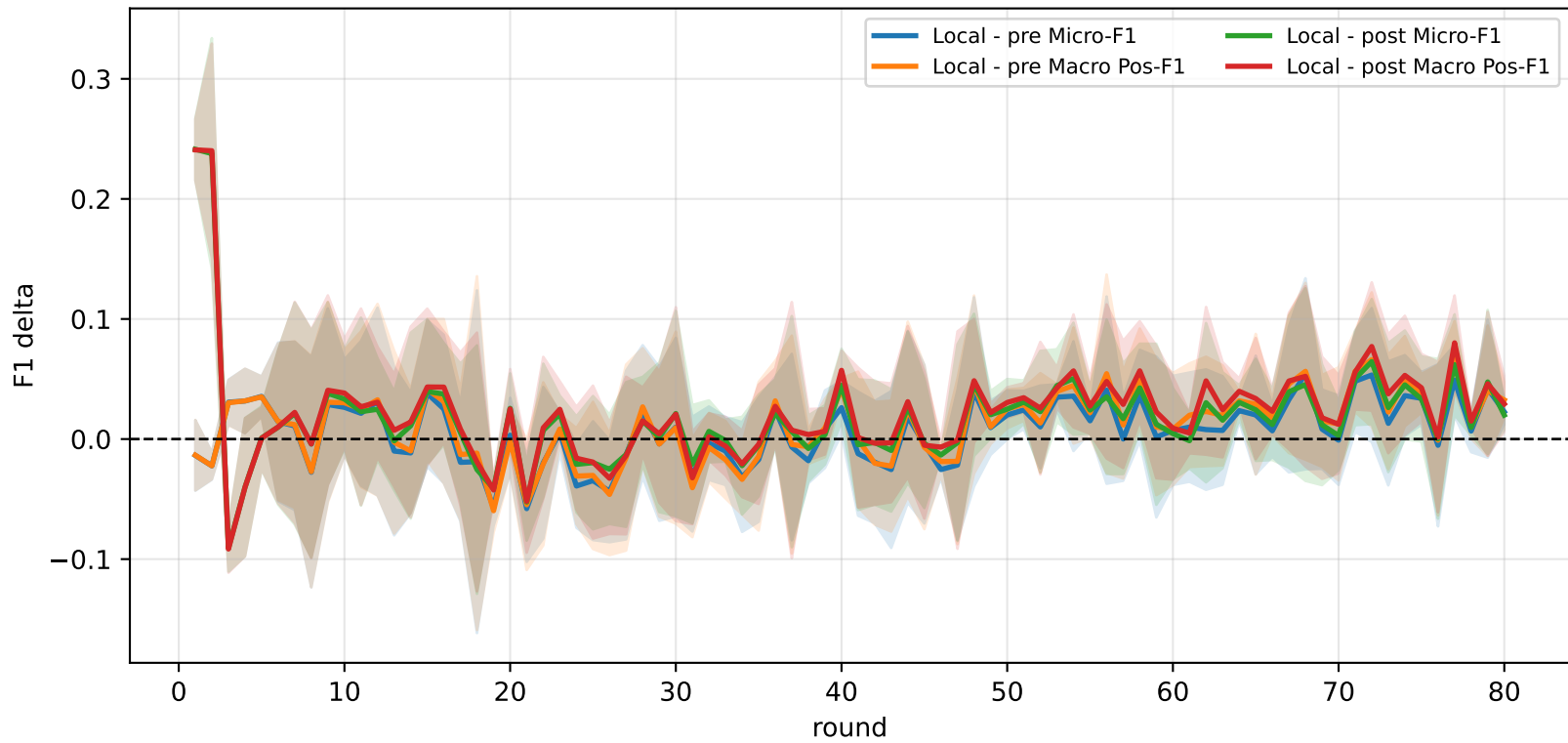
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

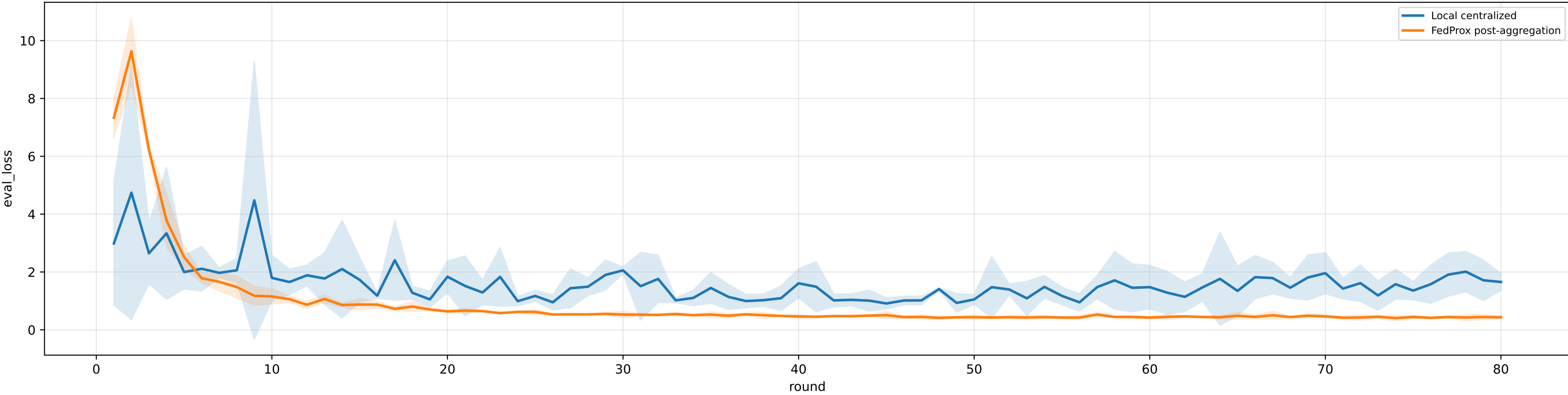


family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

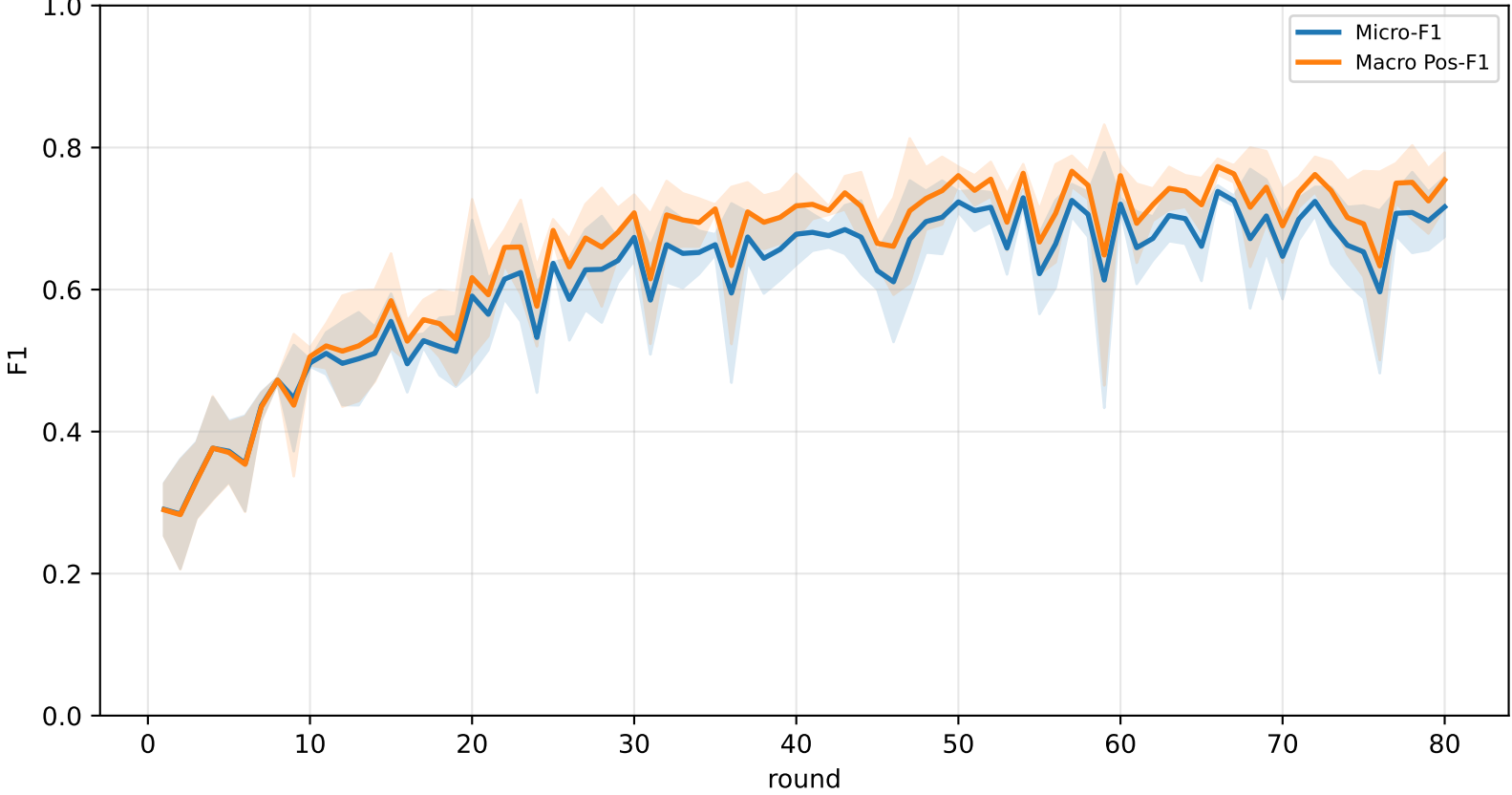
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.9% ± 2.7%	67.2% ± 2.6%	84.2% ± 3.3%	78.8% ± 3.0%	67.1% ± 2.5%	59.7% ± 2.9%	46.4% ± 2.7%
FedProx	66.5% ± 1.1%	69.0% ± 0.9%	85.2% ± 0.5%	73.8% ± 4.6%	66.4% ± 1.2%	64.4% ± 0.3%	55.4% ± 2.4%

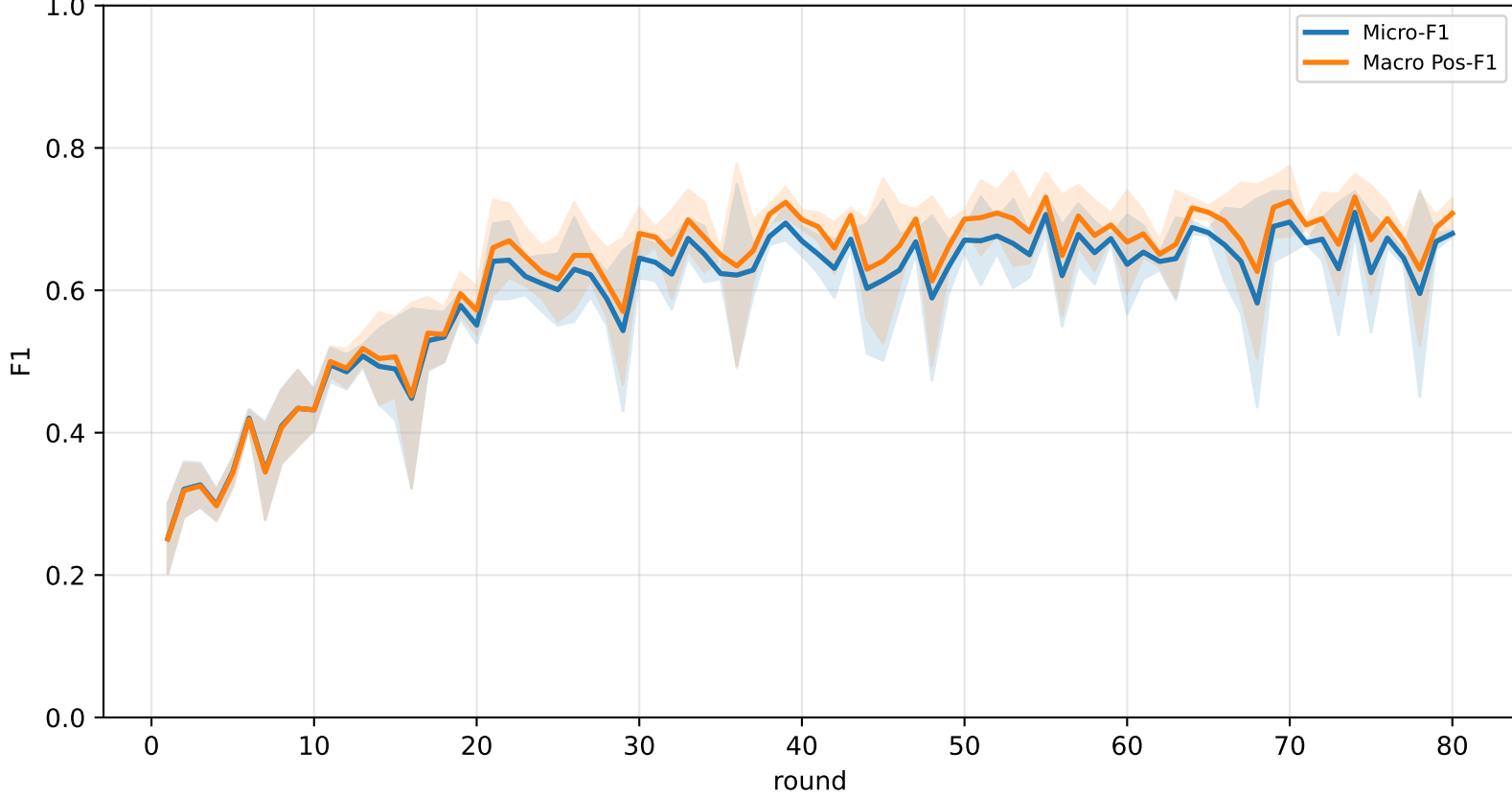
Validation loss comparison



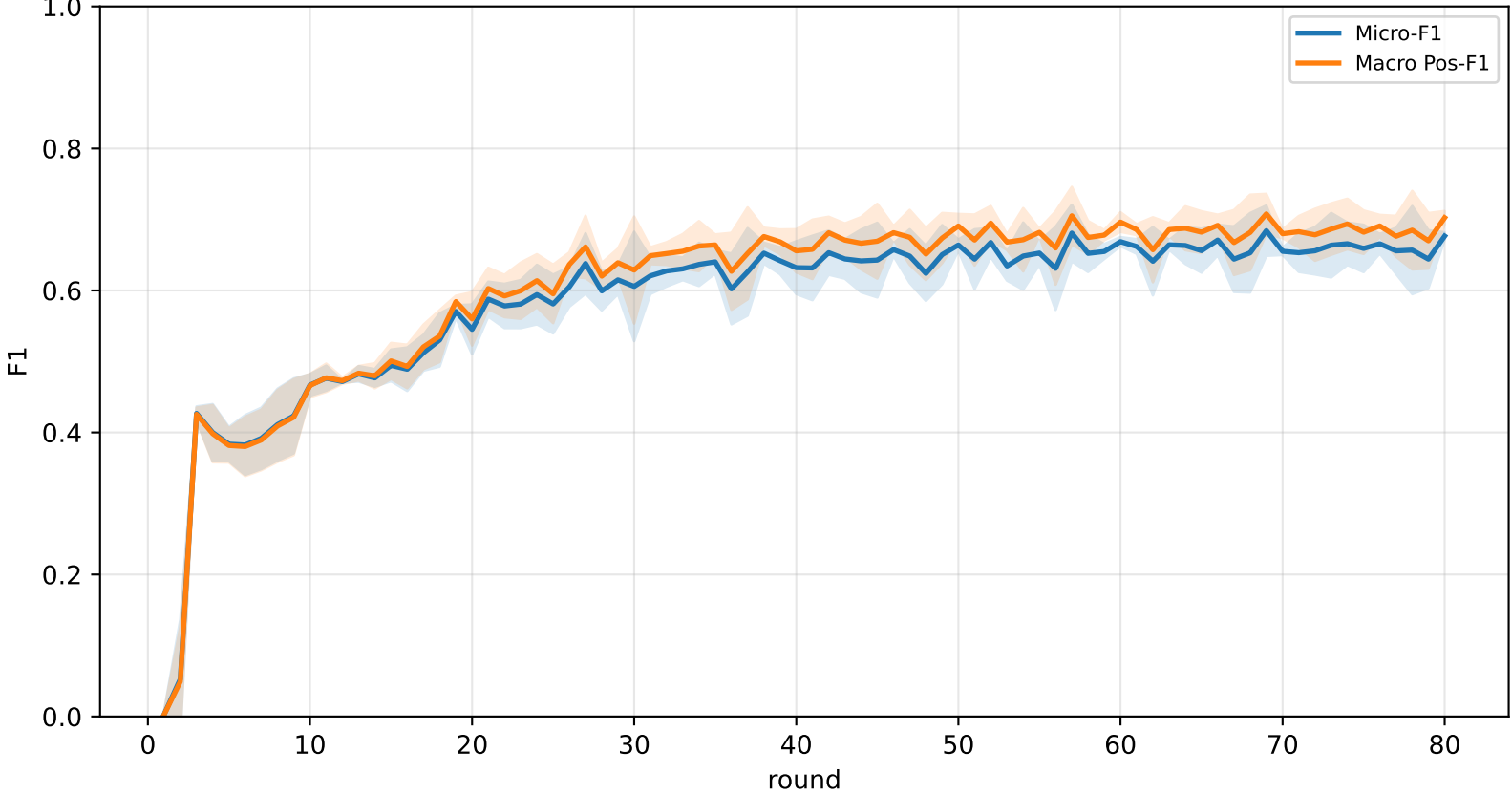
Pooled local centralized micro / macro F1



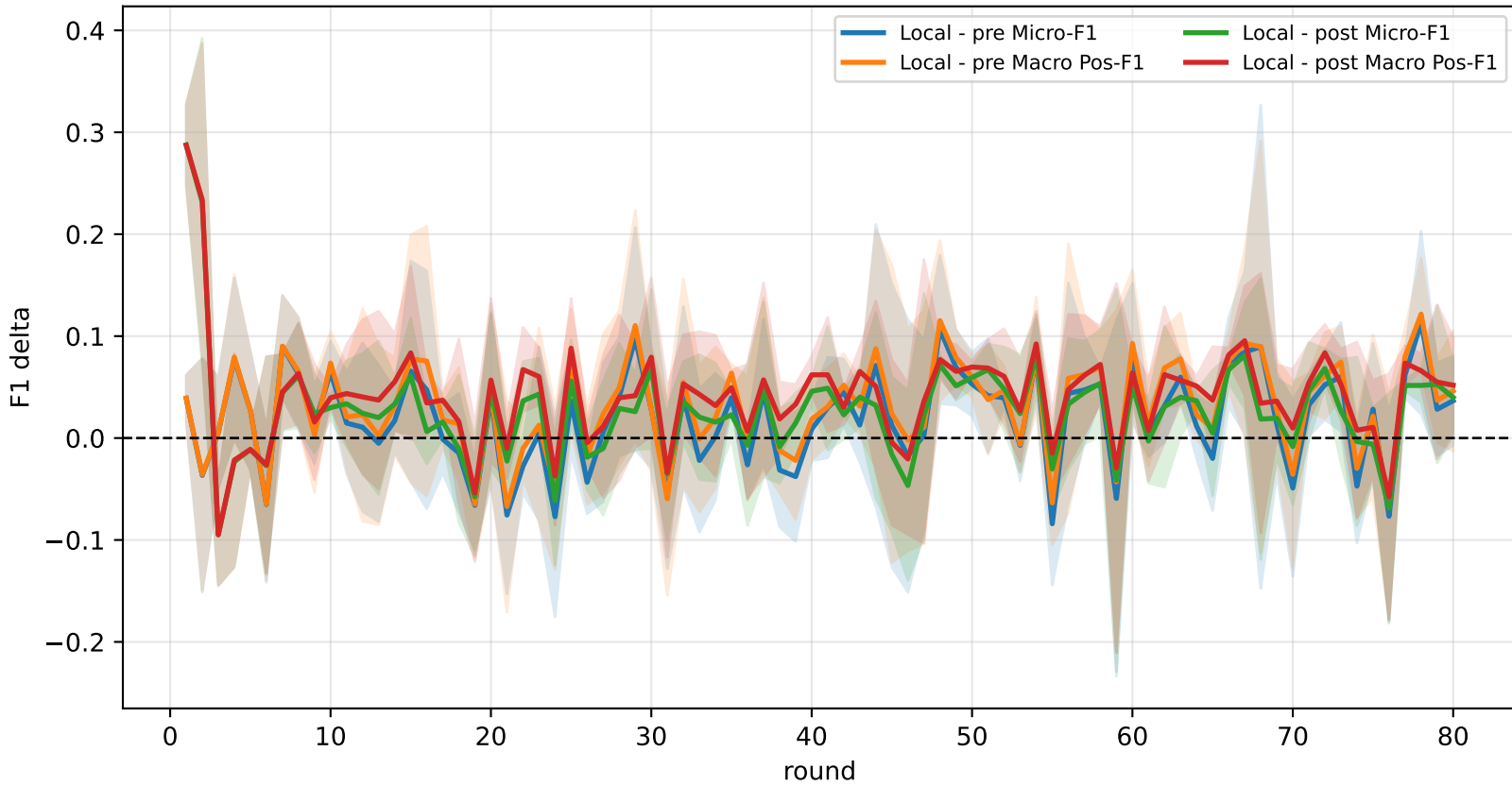
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

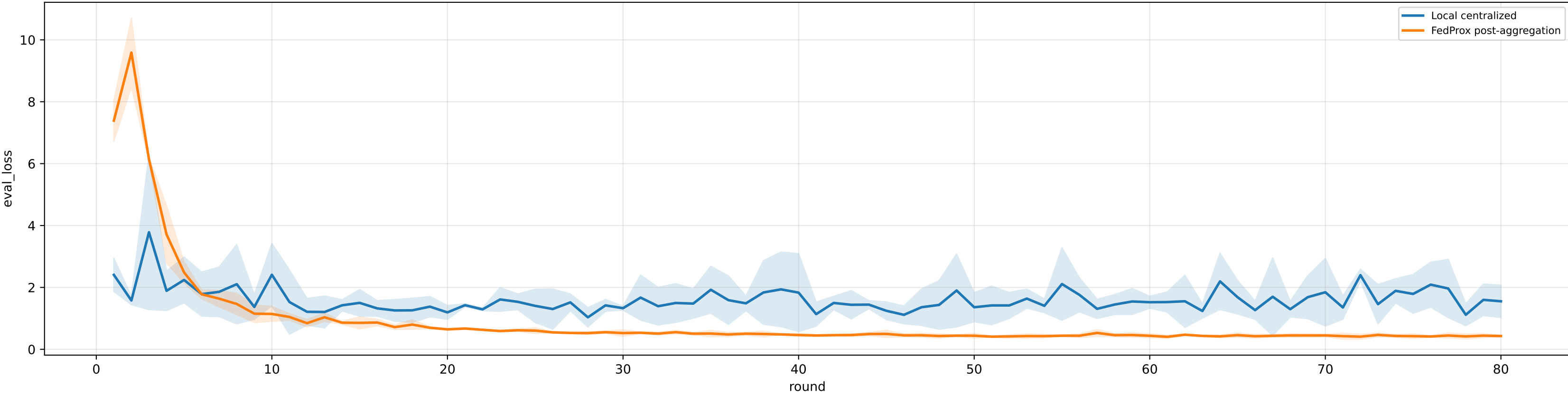


family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

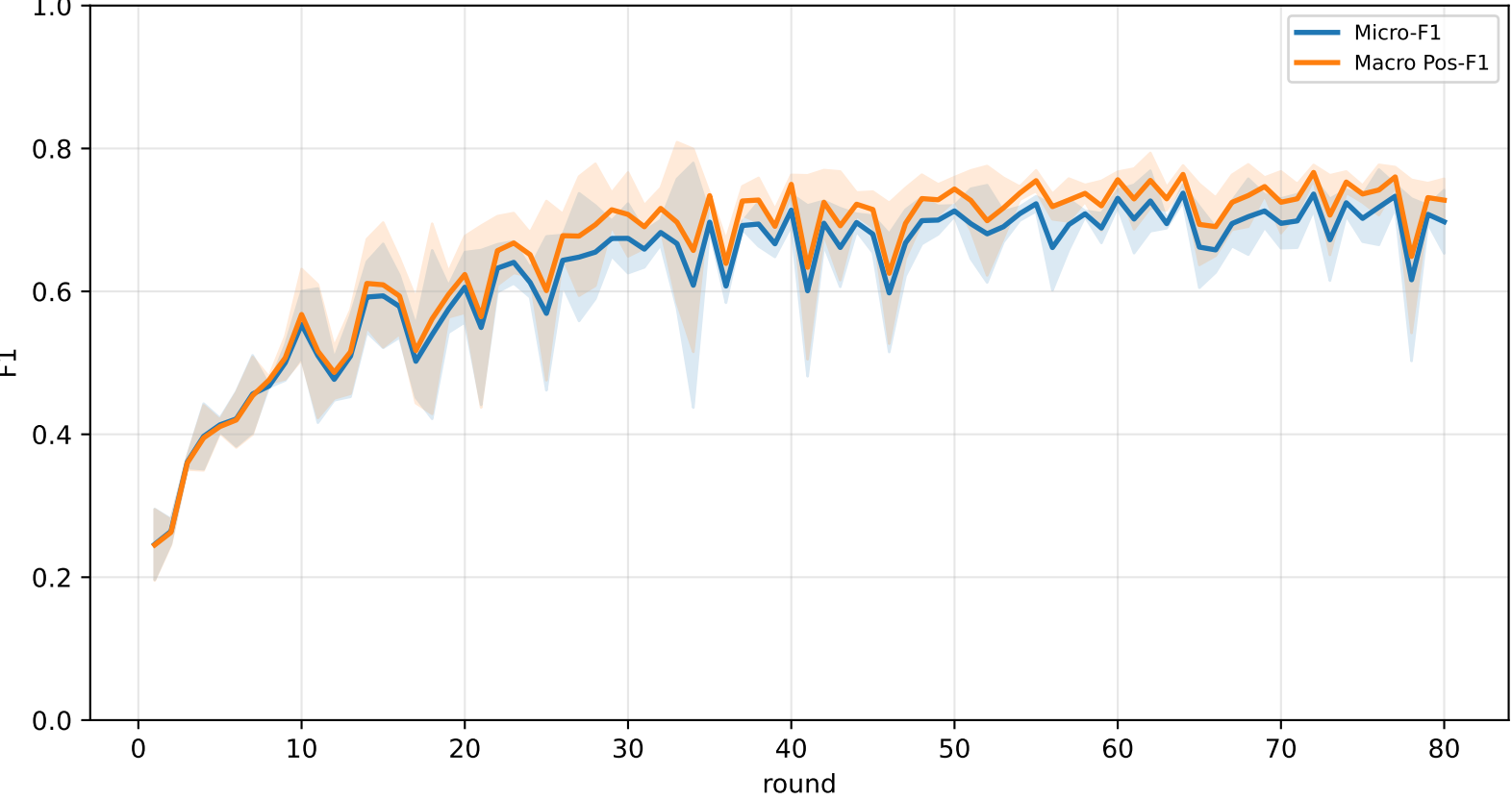
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	55.7% ± 8.8%	57.9% ± 10.2%	66.6% ± 17.9%	62.6% ± 13.2%	58.0% ± 7.4%	56.9% ± 12.7%	45.4% ± 3.0%
FedProx	66.5% ± 1.6%	69.0% ± 1.1%	84.3% ± 2.5%	74.9% ± 5.4%	66.7% ± 1.3%	63.9% ± 0.9%	55.3% ± 2.8%

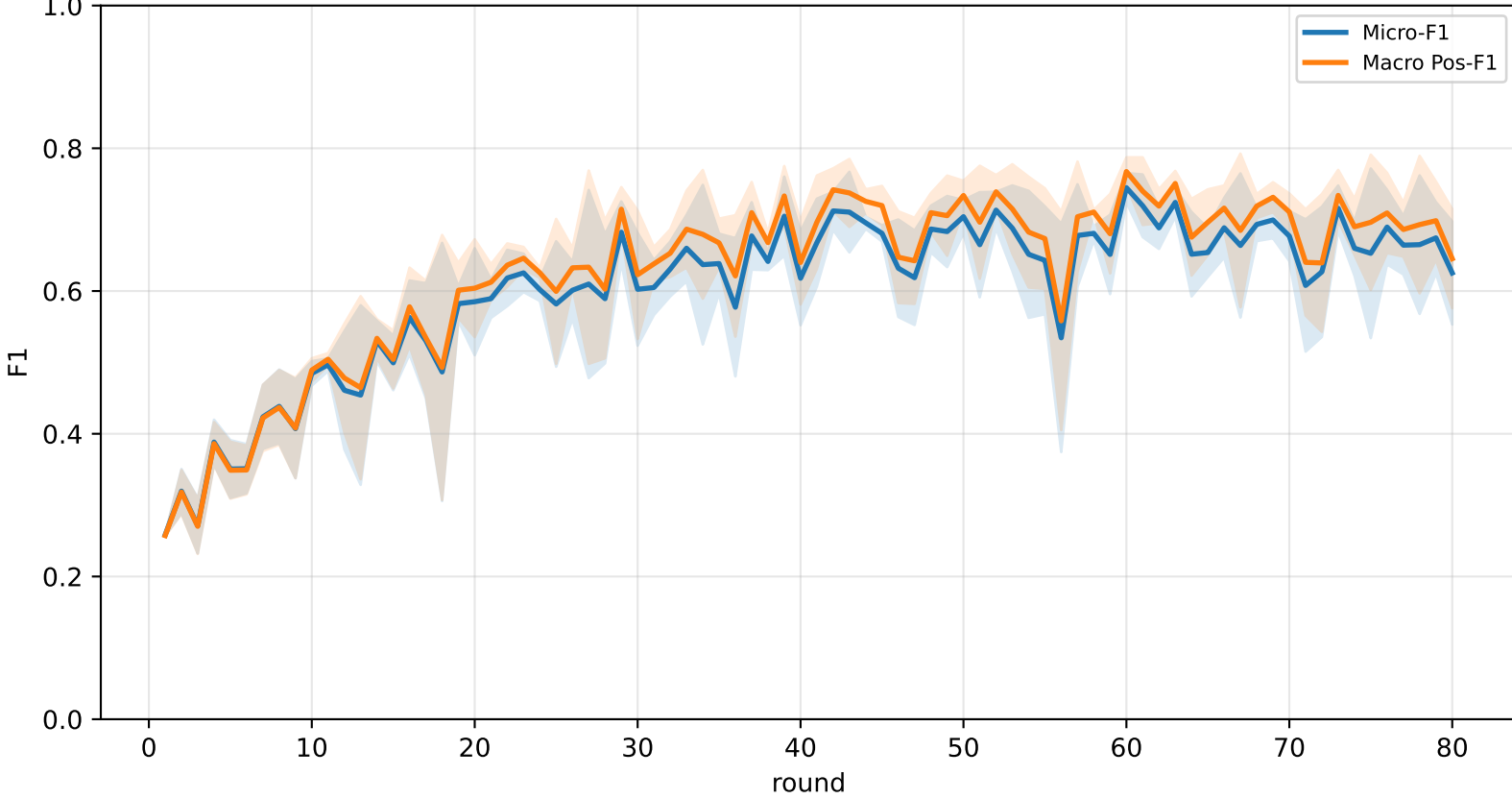
Validation loss comparison



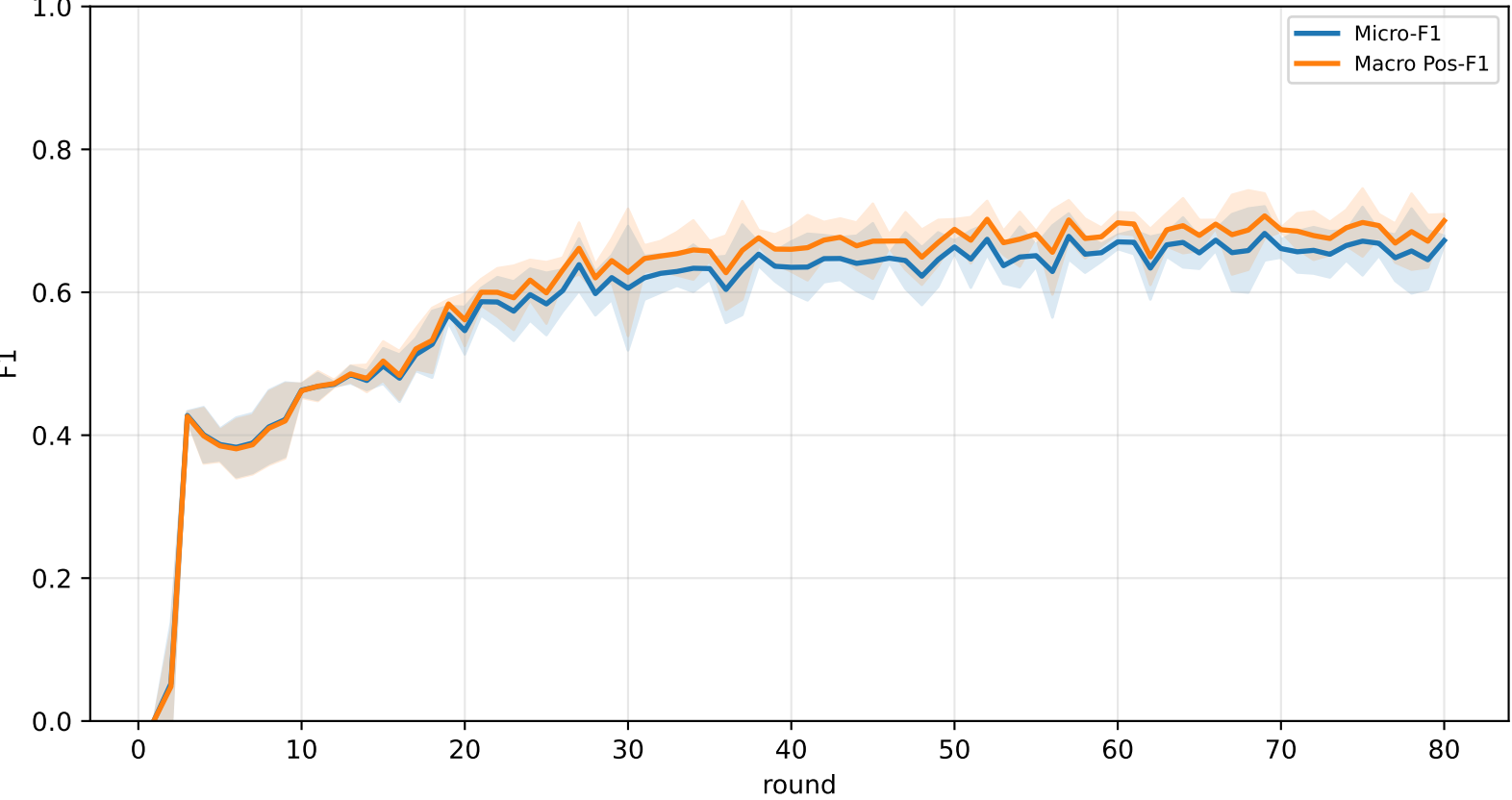
Pooled local centralized micro / macro F1



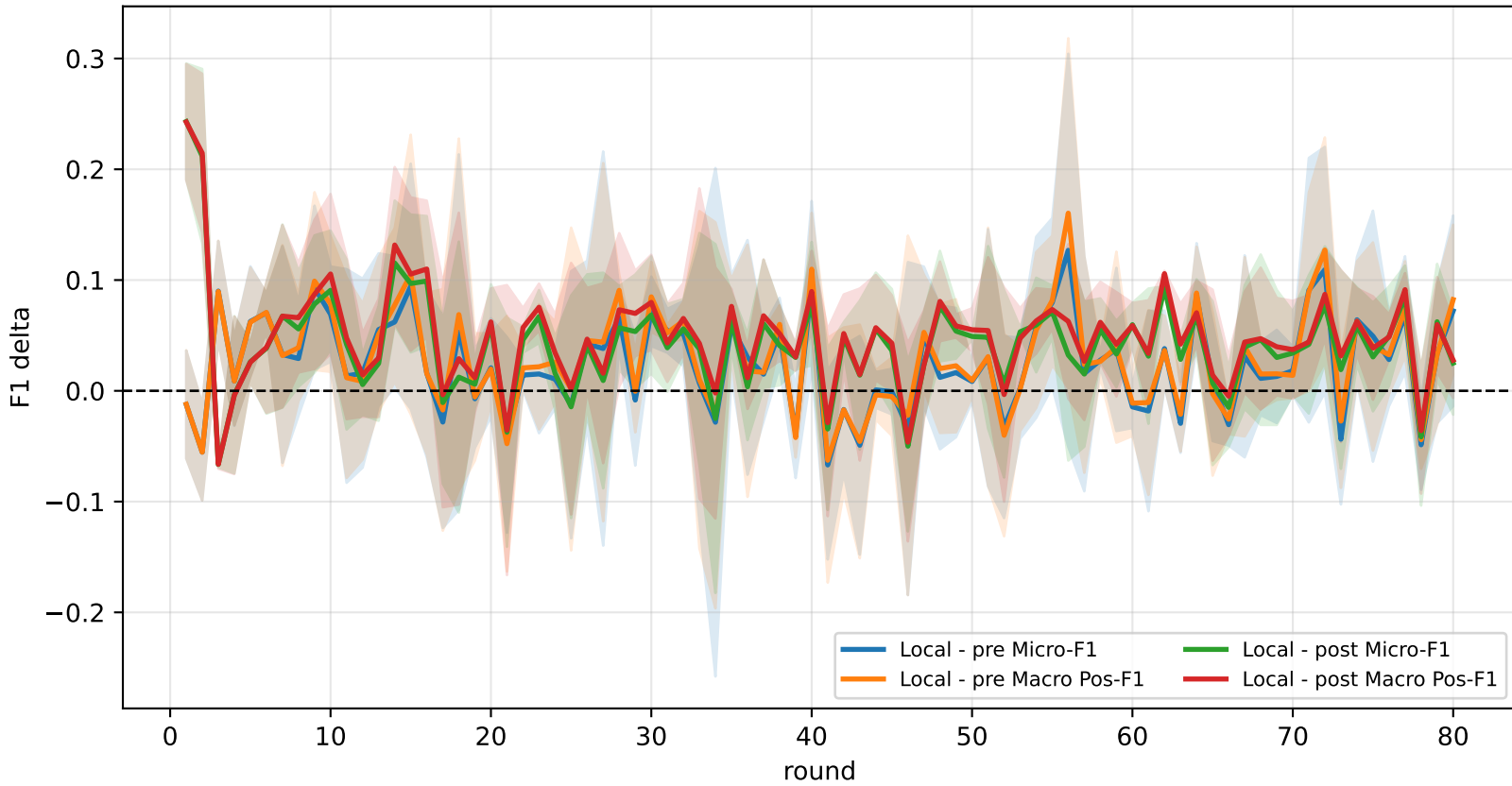
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

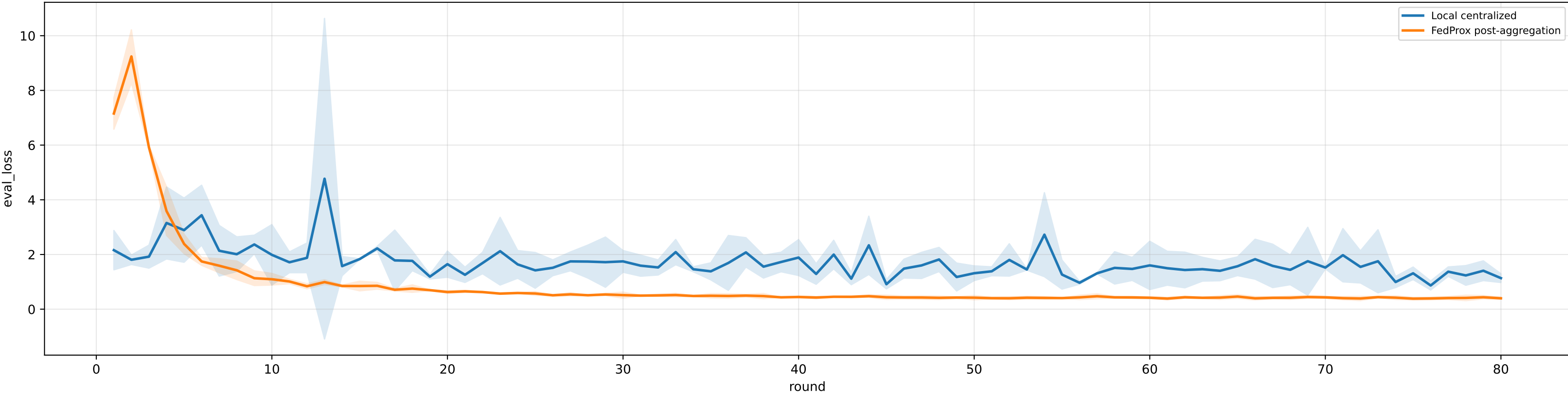


family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

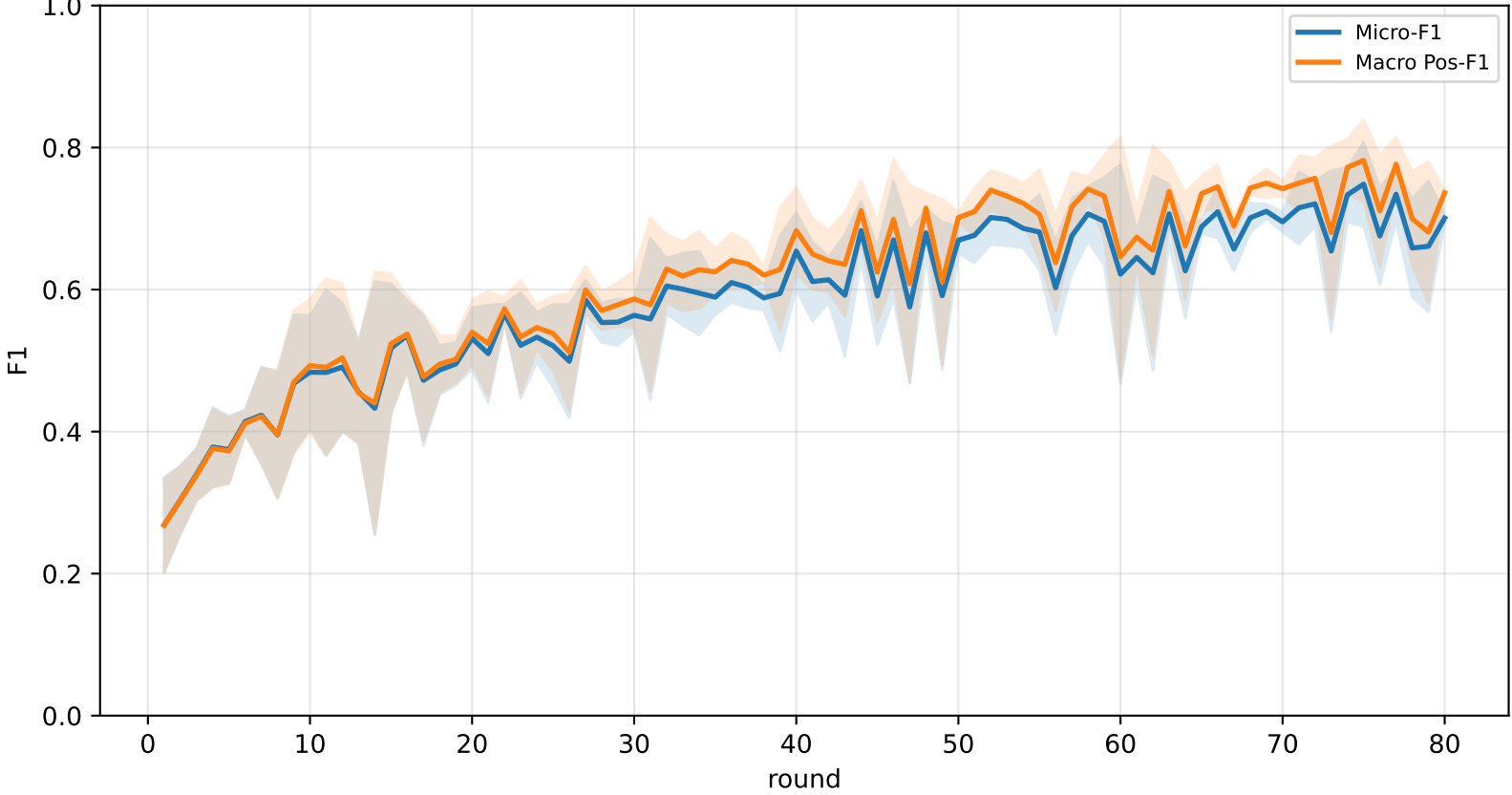
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	57.1% ± 7.9%	59.7% ± 8.6%	71.0% ± 8.9%	68.6% ± 10.4%	60.3% ± 10.1%	53.0% ± 10.3%	45.6% ± 4.2%
FedProx	66.2% ± 1.4%	68.6% ± 1.0%	83.7% ± 1.0%	74.3% ± 4.2%	66.1% ± 0.9%	64.1% ± 0.9%	54.9% ± 2.6%

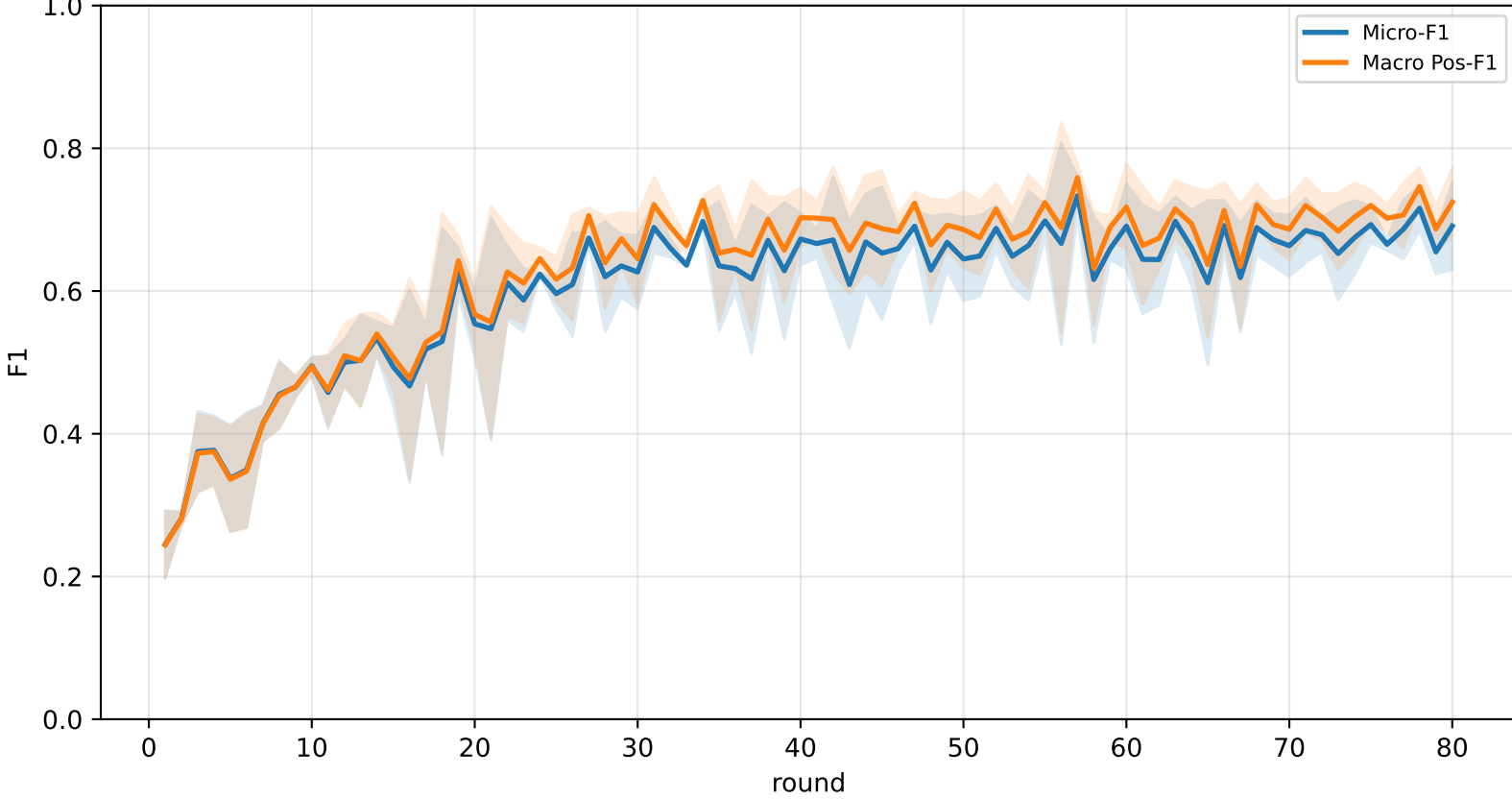
Validation loss comparison



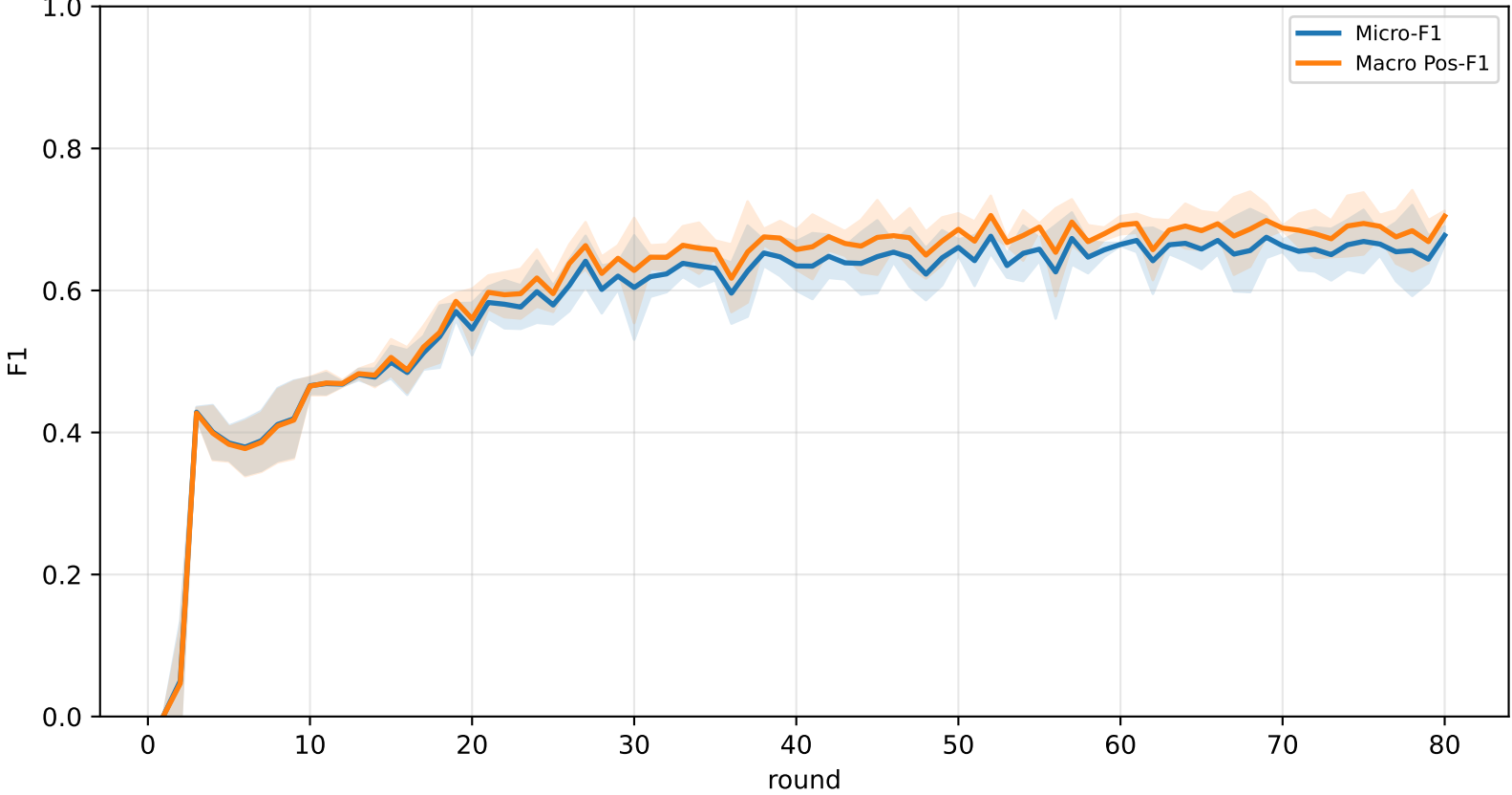
Pooled local centralized micro / macro F1



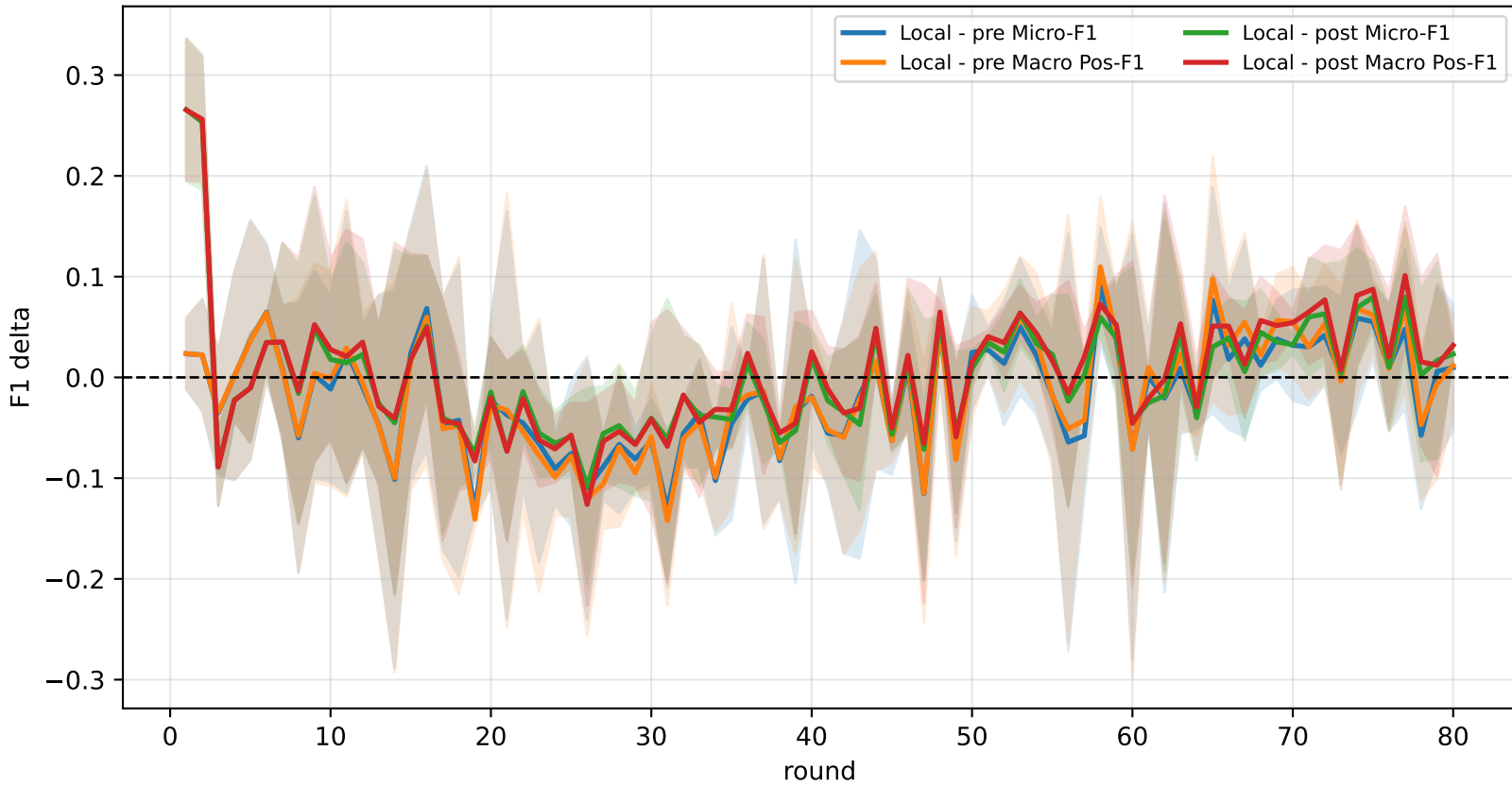
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

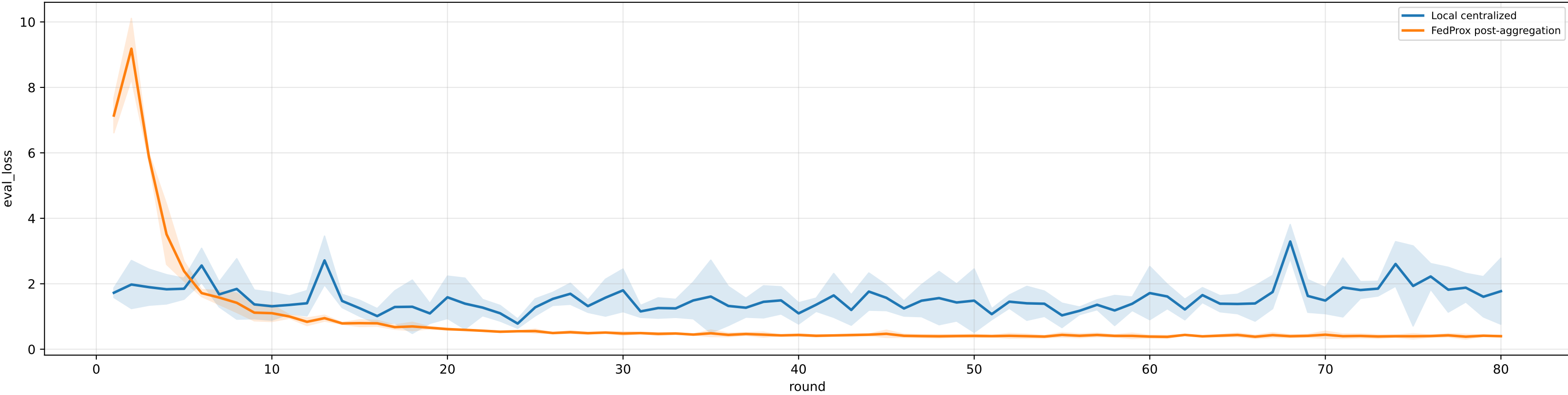


family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

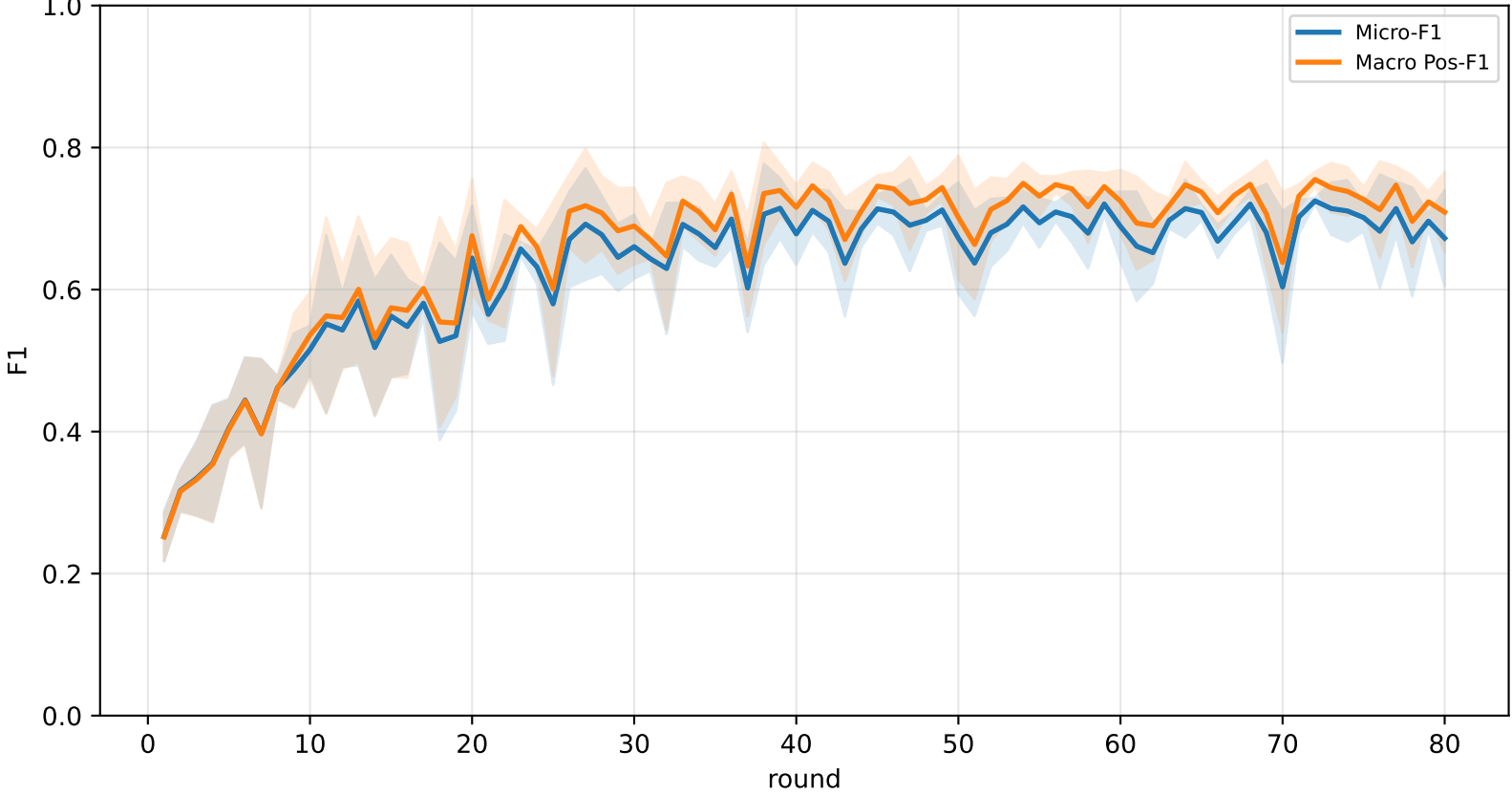
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	62.7% ± 2.4%	65.2% ± 2.6%	76.5% ± 5.8%	75.7% ± 4.0%	59.8% ± 2.9%	64.6% ± 3.0%	49.4% ± 2.2%
FedProx	65.9% ± 1.1%	68.2% ± 0.8%	82.6% ± 2.5%	73.4% ± 4.0%	66.3% ± 0.9%	63.6% ± 0.3%	55.2% ± 2.4%

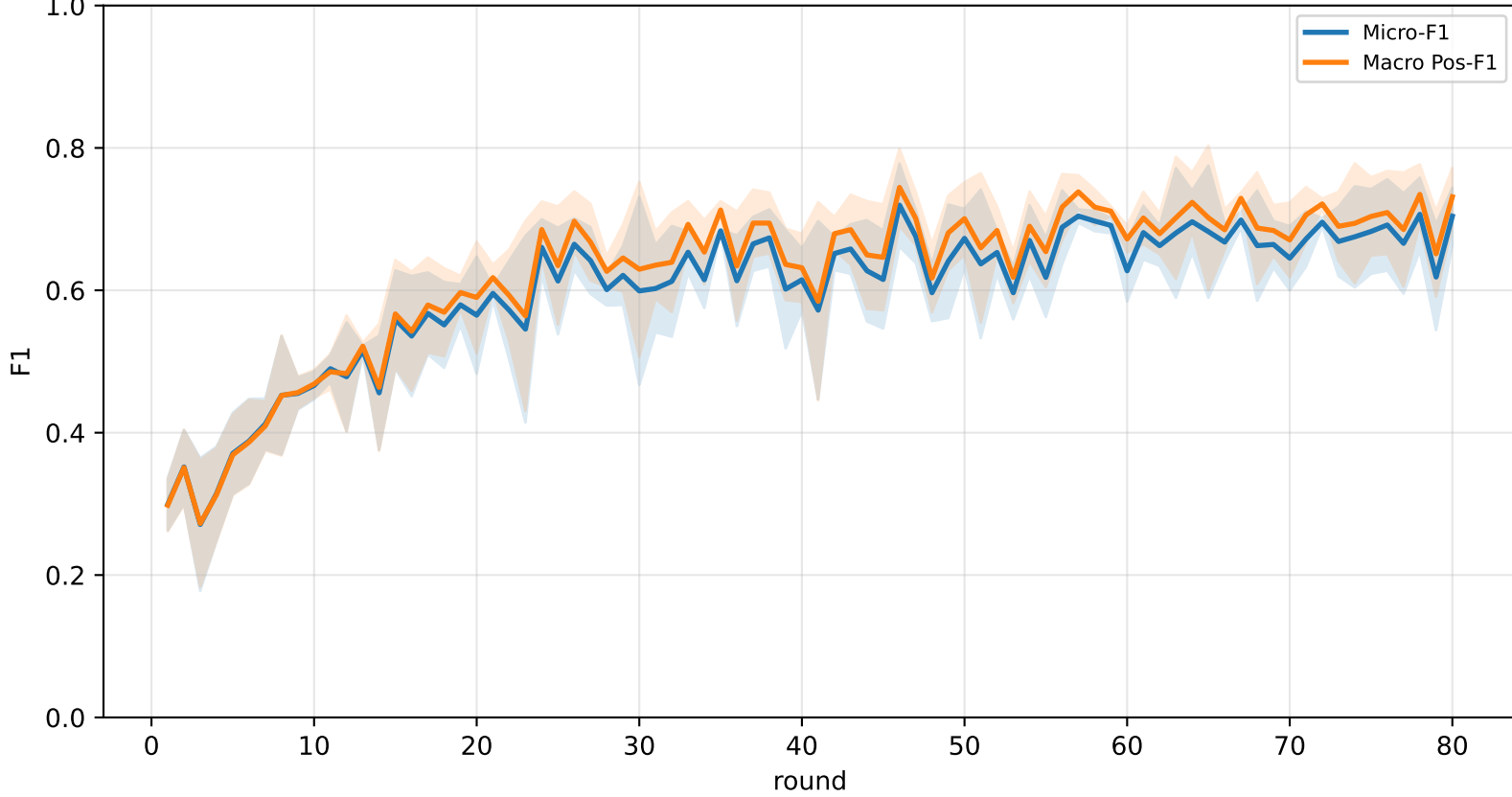
Validation loss comparison



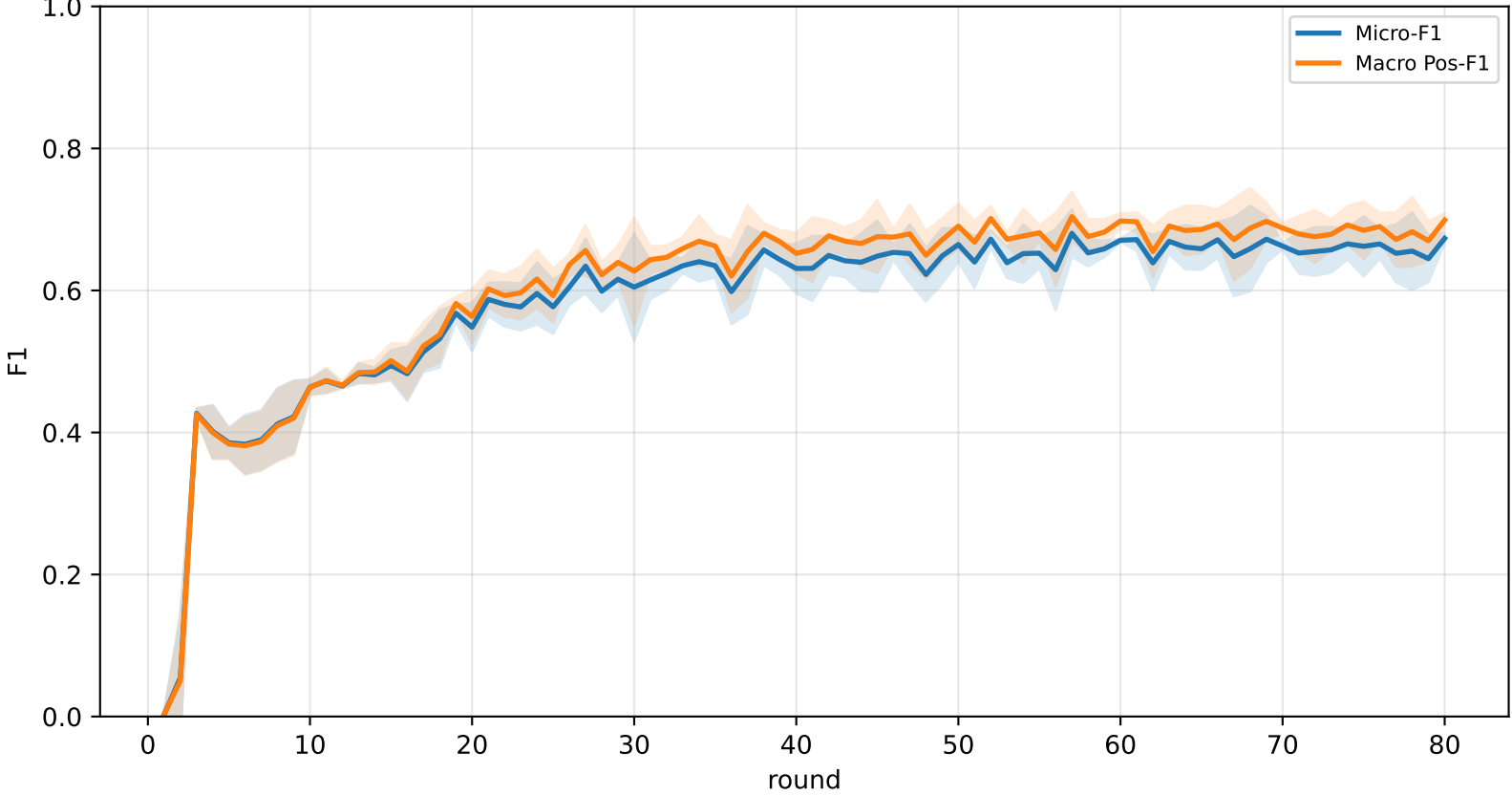
Pooled local centralized micro / macro F1



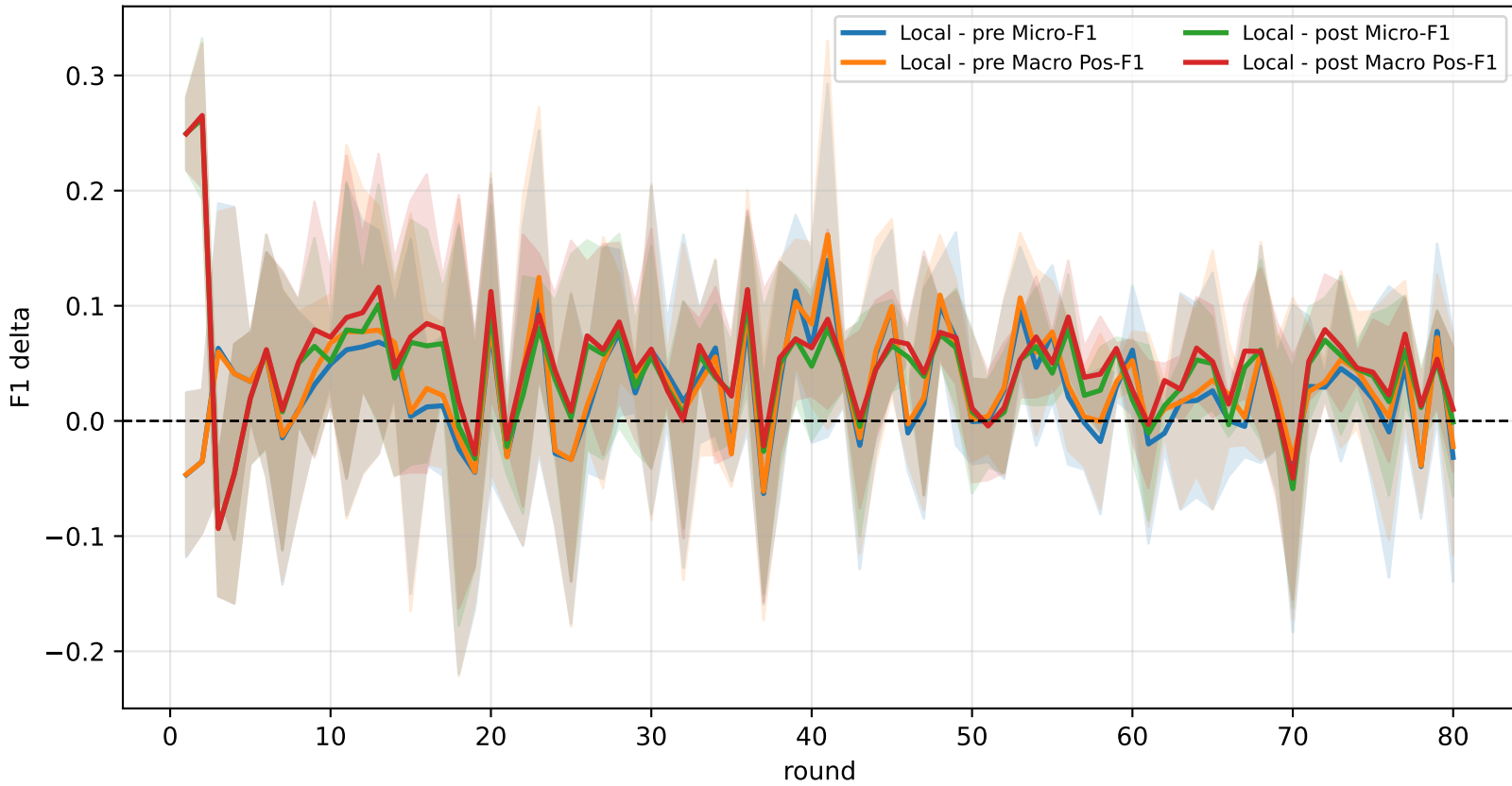
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

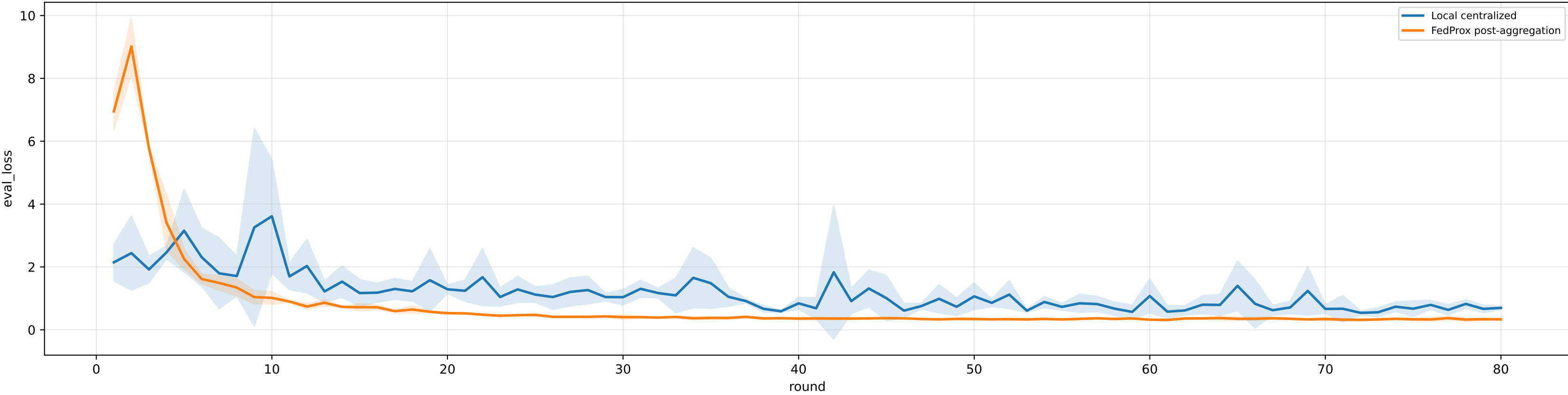


family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

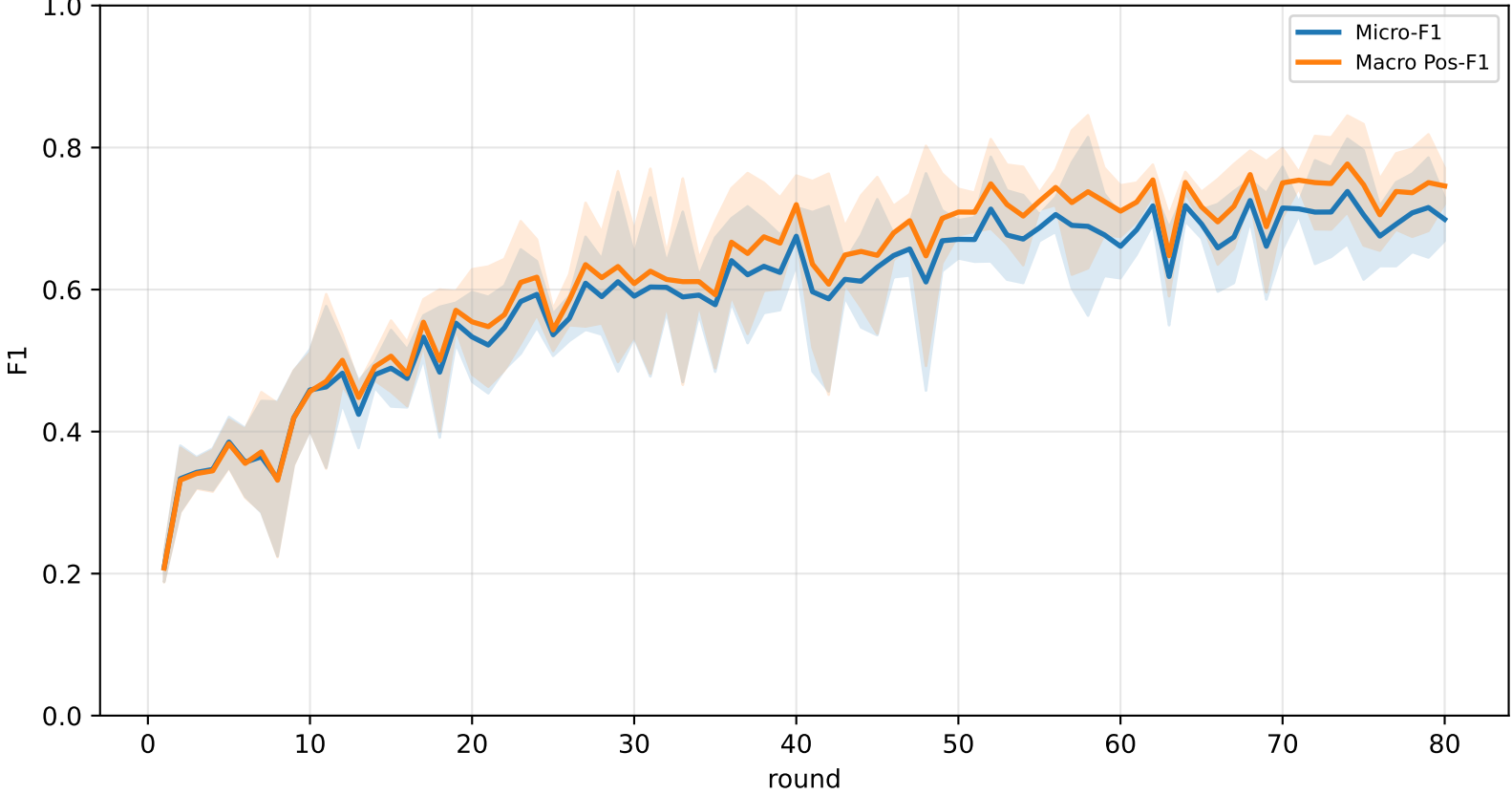
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	63.1% ± 3.5%	67.2% ± 3.7%	86.9% ± 6.6%	77.7% ± 9.8%	63.1% ± 2.3%	58.3% ± 3.3%	49.9% ± 3.9%
FedProx	66.6% ± 1.0%	69.1% ± 0.7%	84.1% ± 2.1%	74.0% ± 4.2%	66.8% ± 0.9%	65.3% ± 0.8%	55.1% ± 2.0%

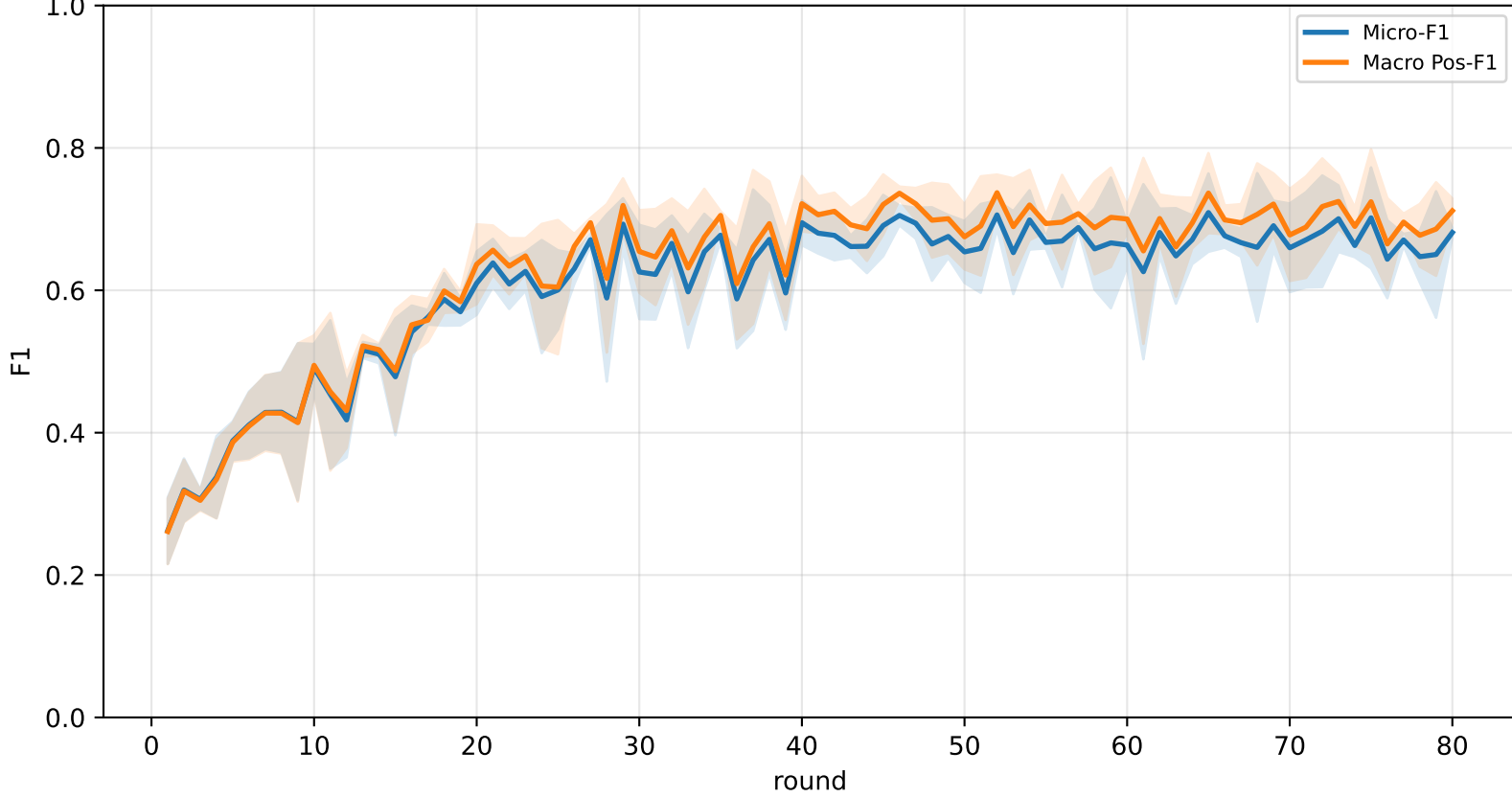
Validation loss comparison



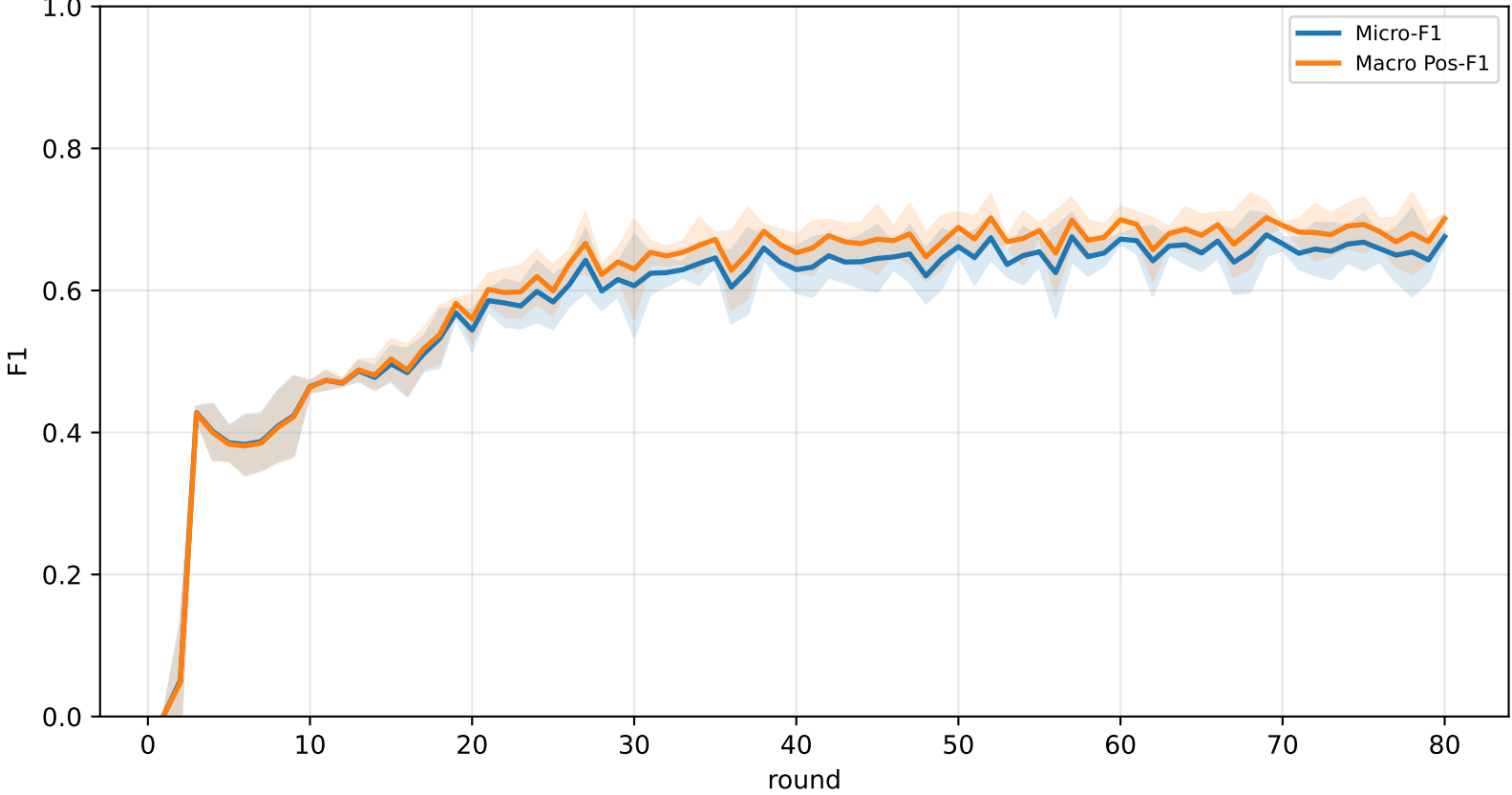
Pooled local centralized micro / macro F1



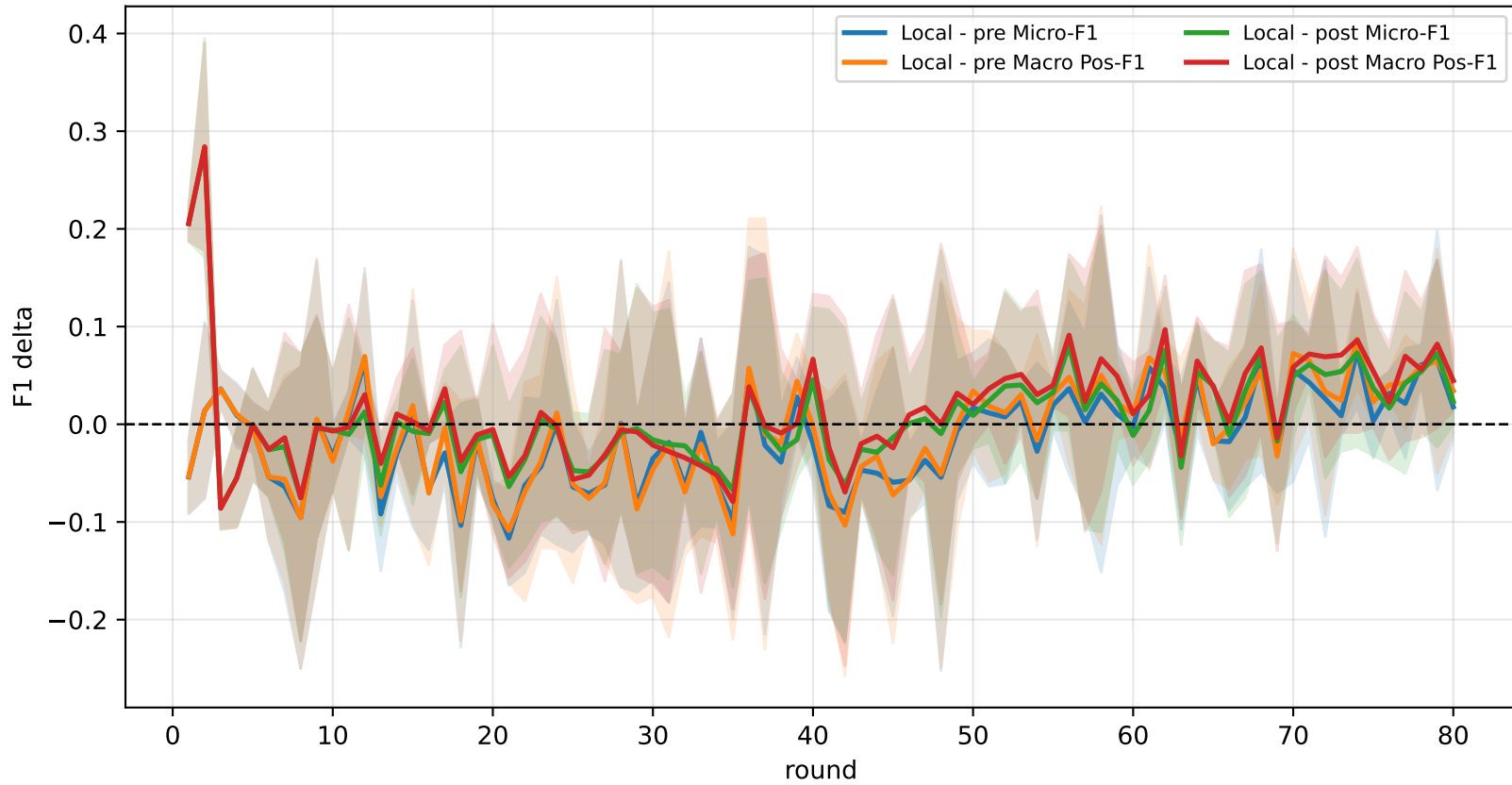
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)



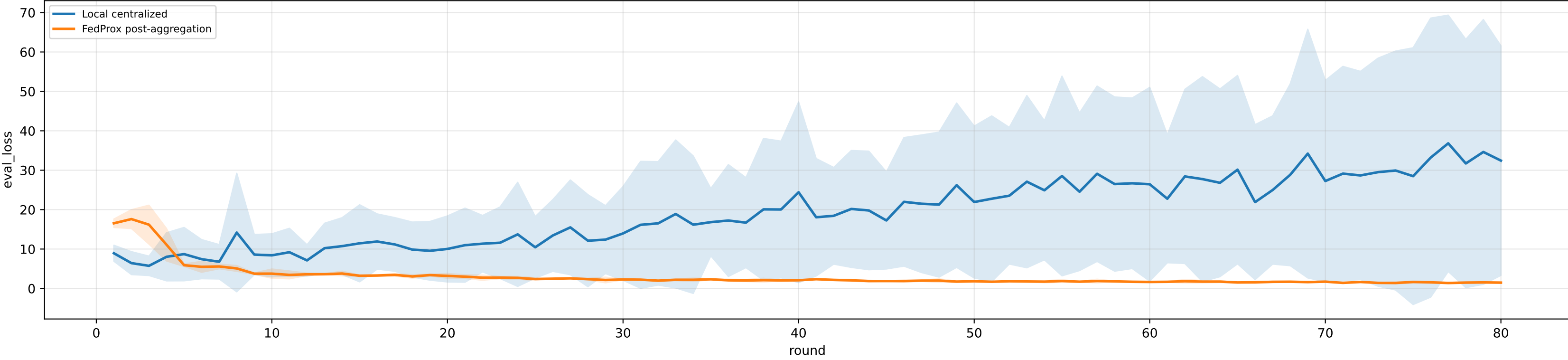
Controlled q-label heterogeneity: High | FedProx diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

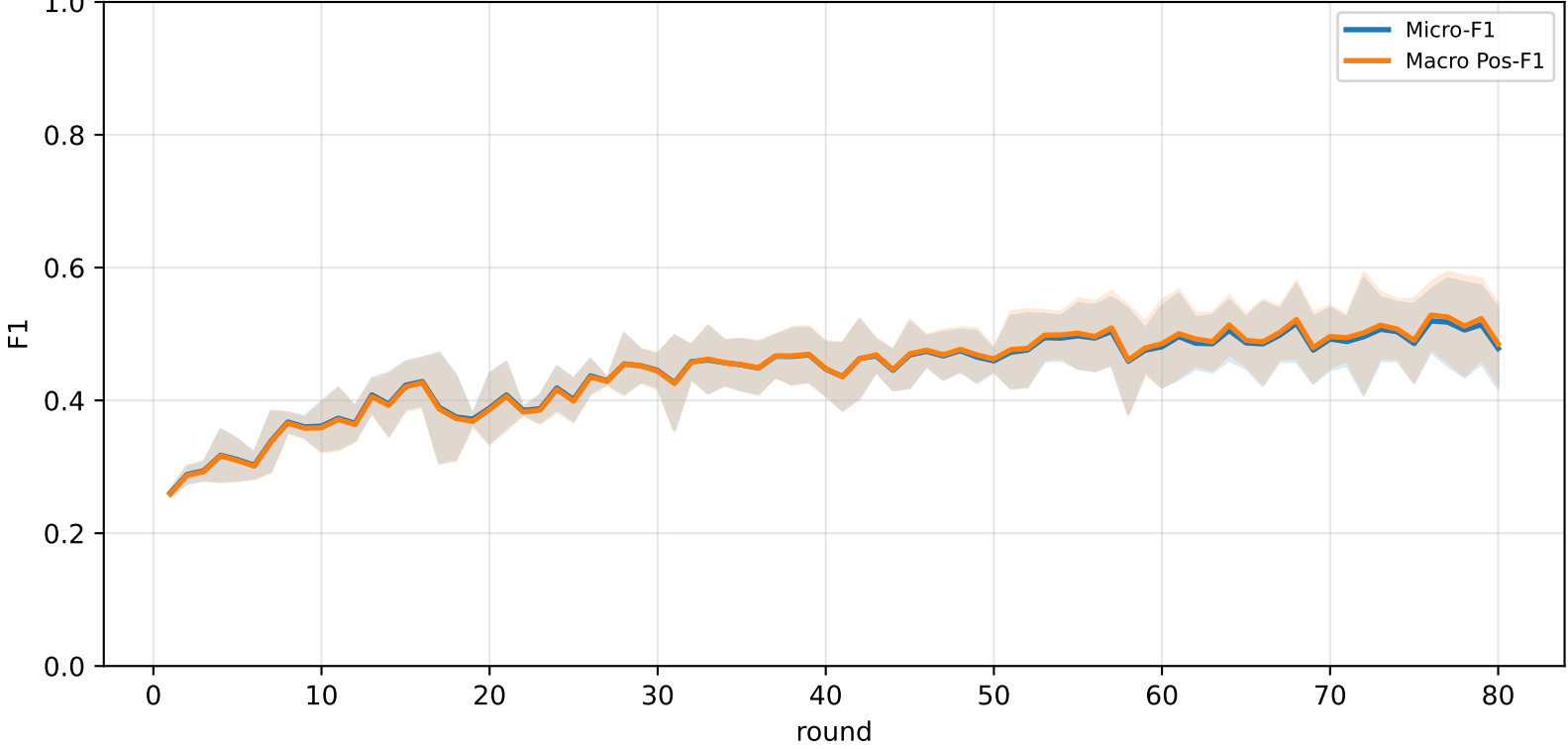
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	29.1% ± 5.1%	29.5% ± 5.3%	22.1% ± 2.8%	26.7% ± 3.4%	26.9% ± 3.1%	26.5% ± 3.3%	30.2% ± 3.0%
FedProx	52.8% ± 2.9%	53.2% ± 2.8%	53.5% ± 2.9%	55.1% ± 3.6%	54.7% ± 5.7%	53.5% ± 6.0%	49.1% ± 1.1%

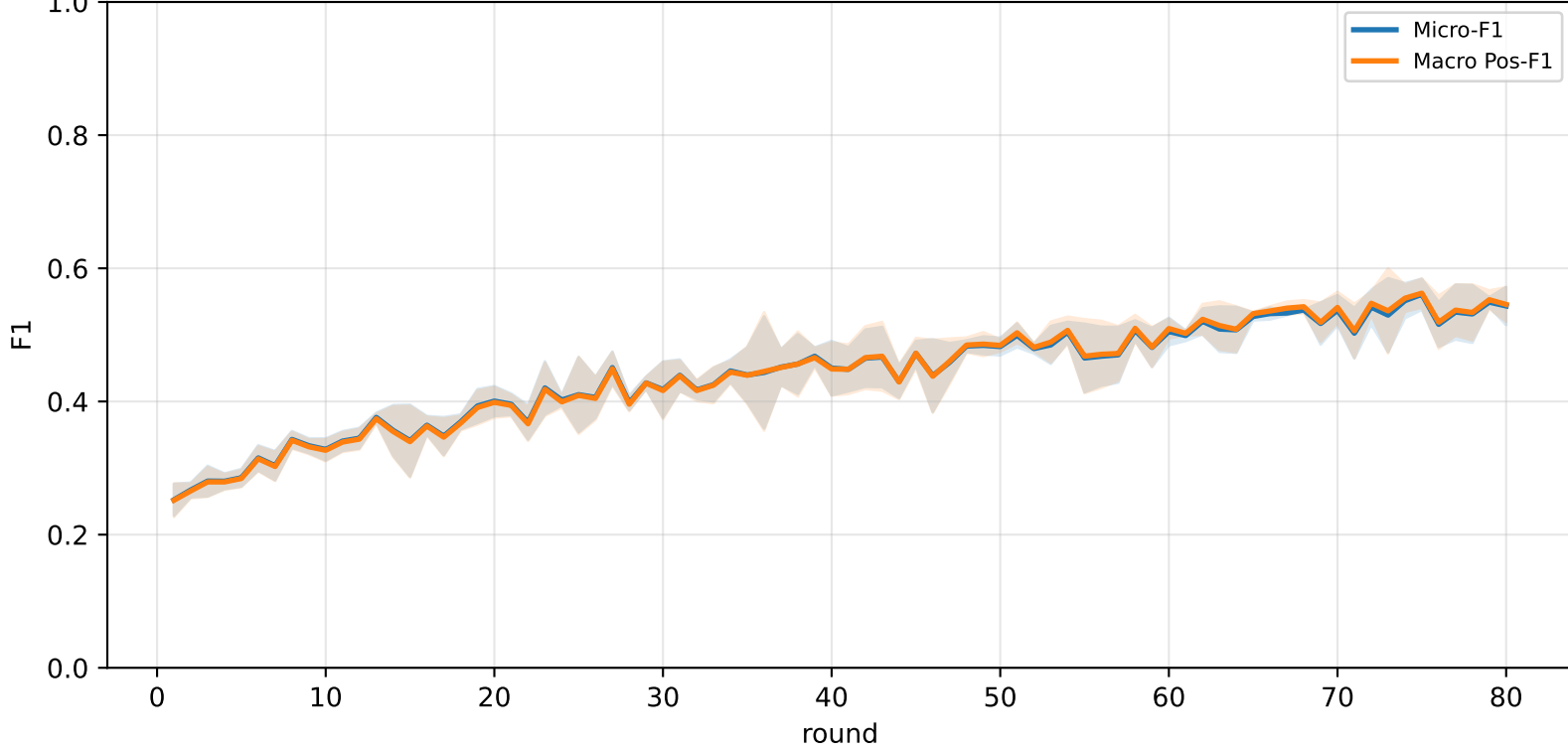
Validation loss comparison



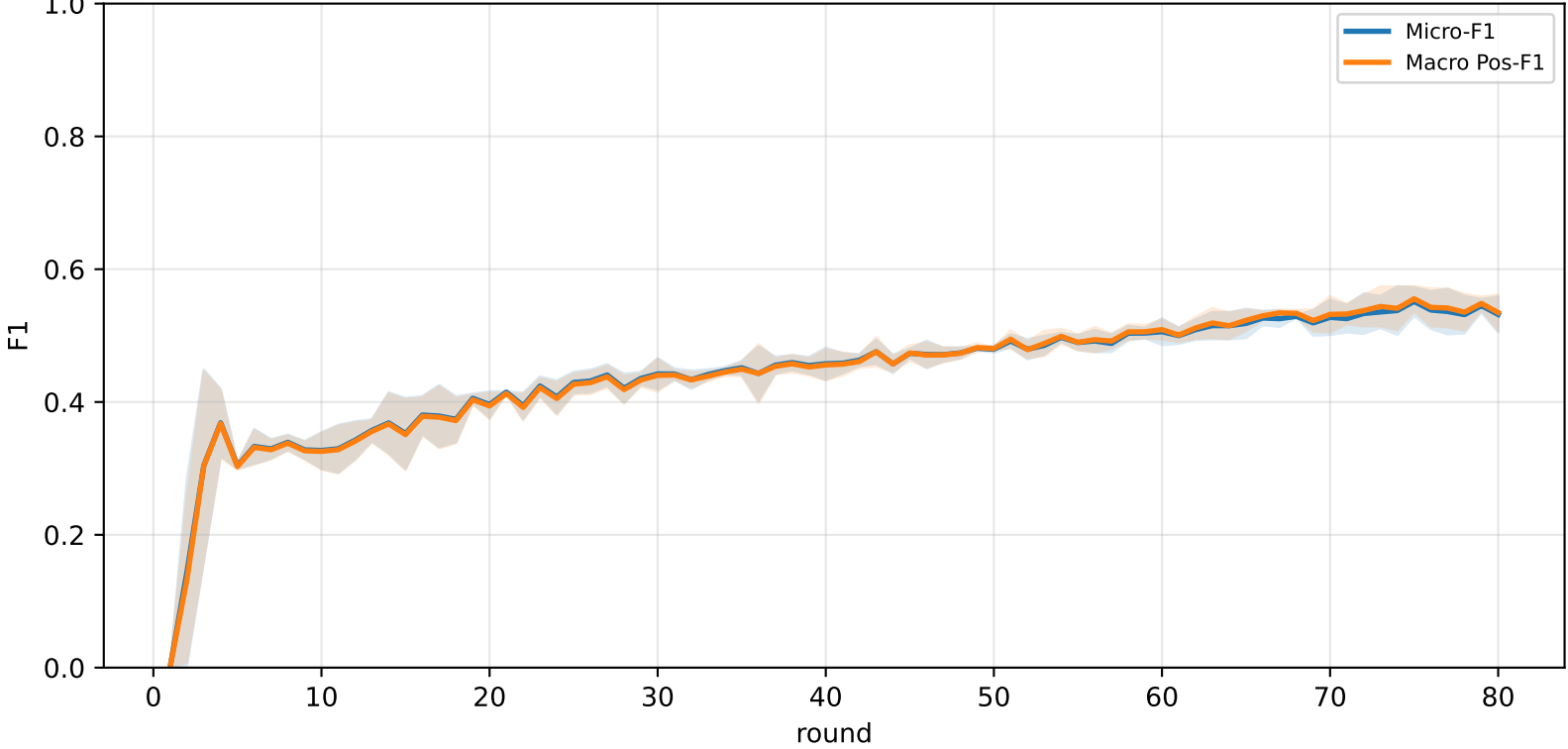
Pooled local centralized micro / macro F1



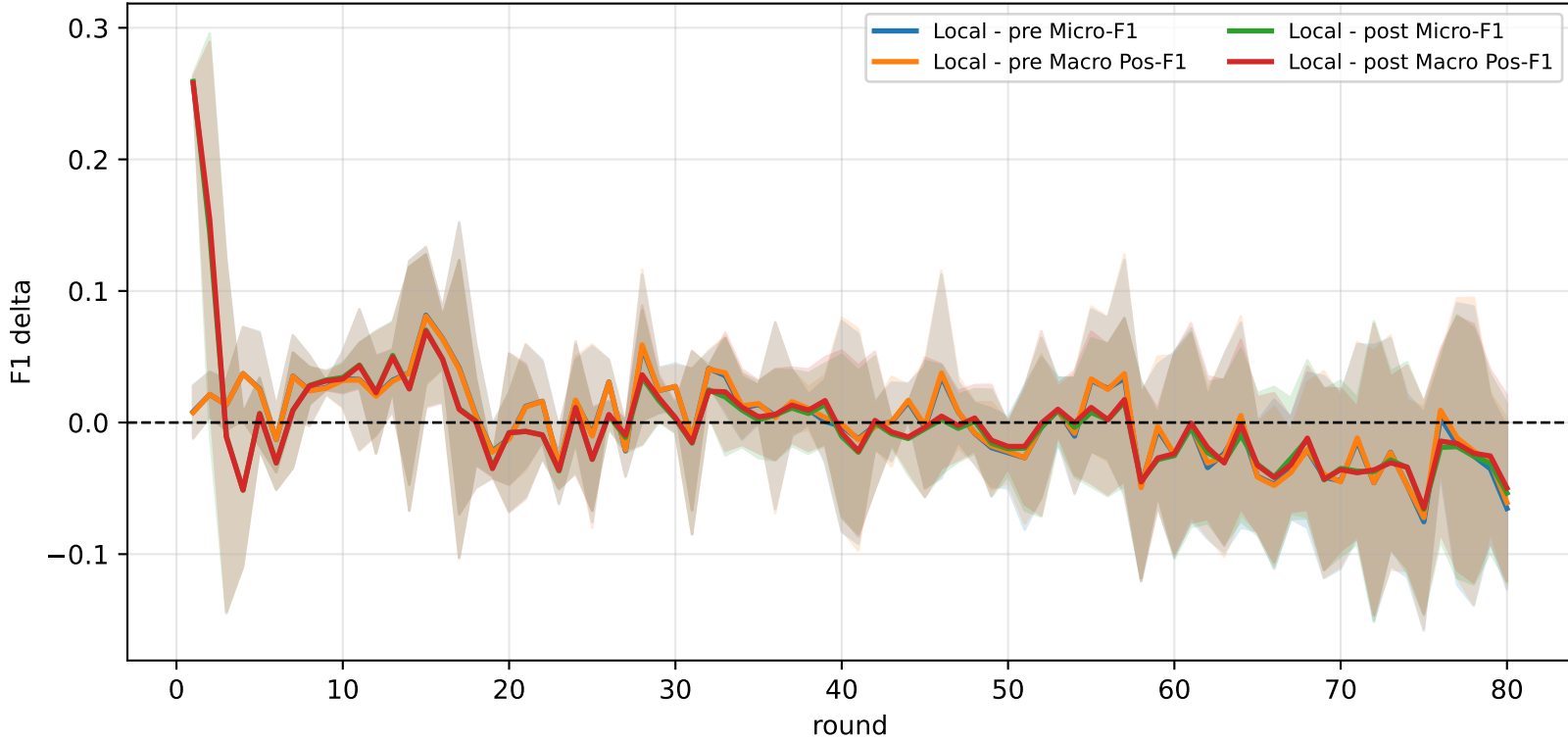
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

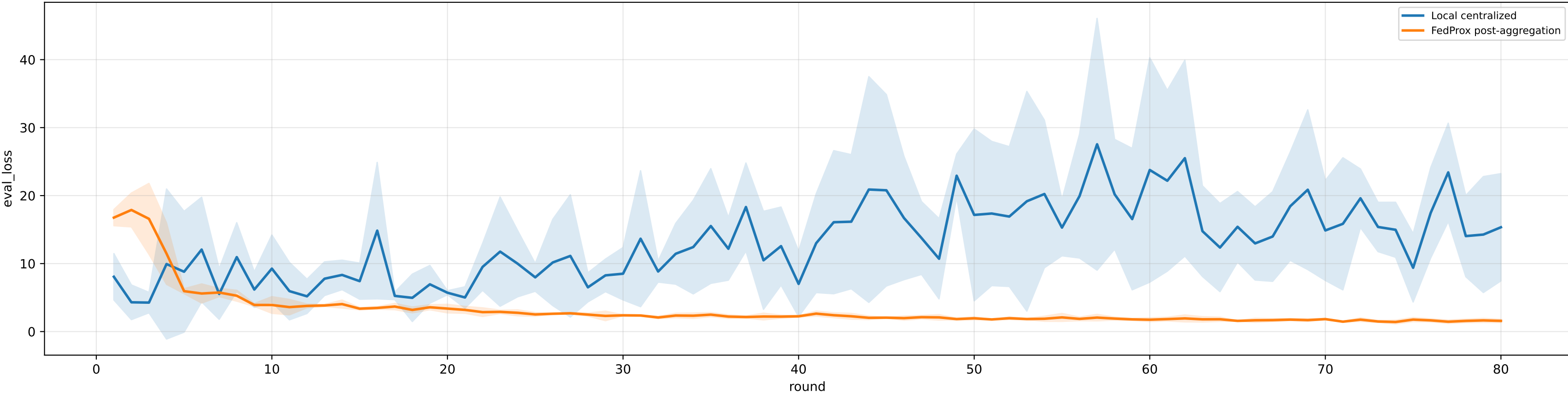


family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

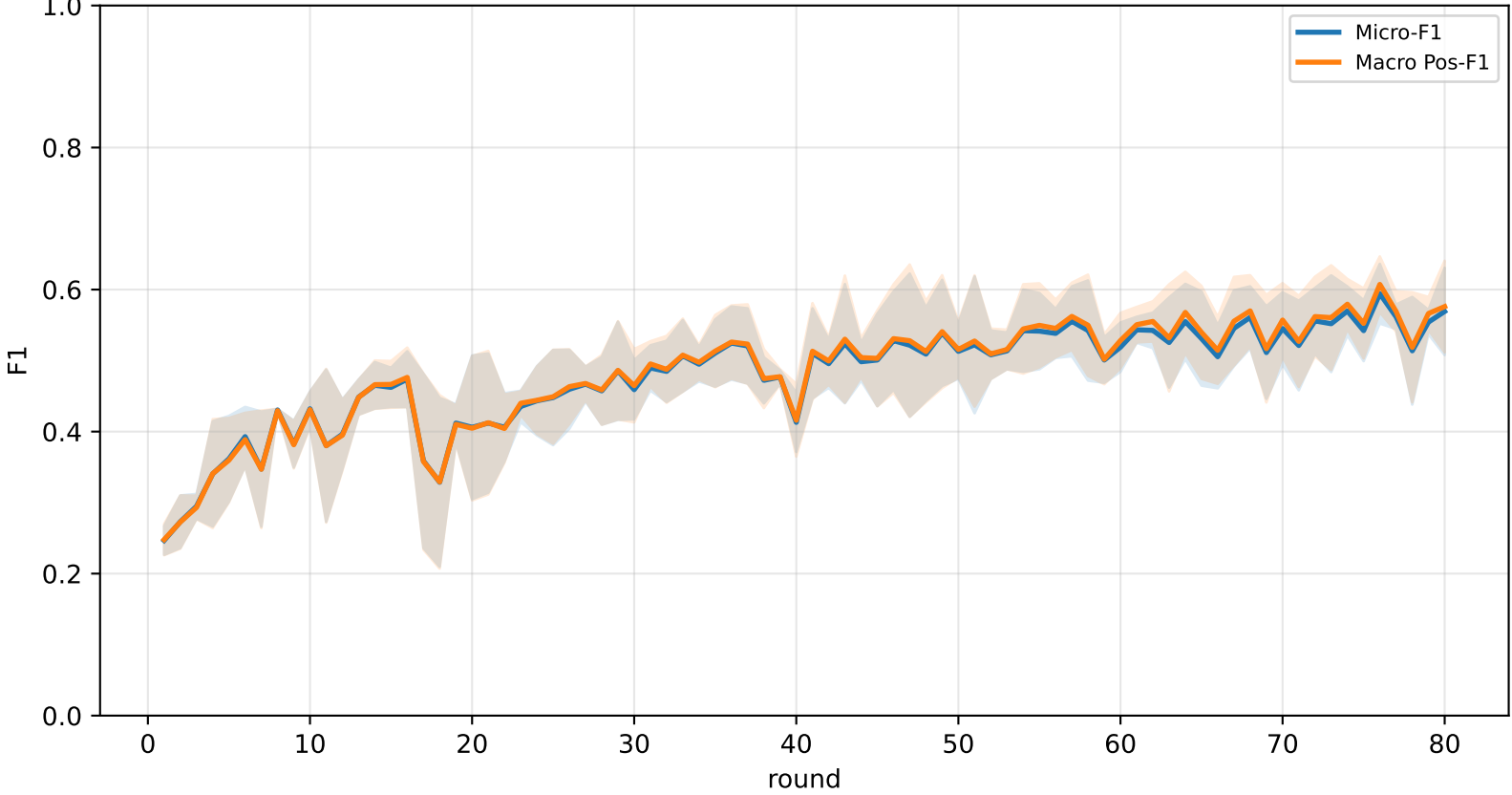
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	32.7% ± 12.1%	33.2% ± 12.8%	31.2% ± 16.3%	35.6% ± 15.5%	35.8% ± 15.4%	31.2% ± 10.3%	32.3% ± 6.7%
FedProx	52.9% ± 2.8%	53.3% ± 2.7%	53.6% ± 3.3%	55.0% ± 3.6%	54.9% ± 5.7%	53.7% ± 6.0%	49.4% ± 1.1%

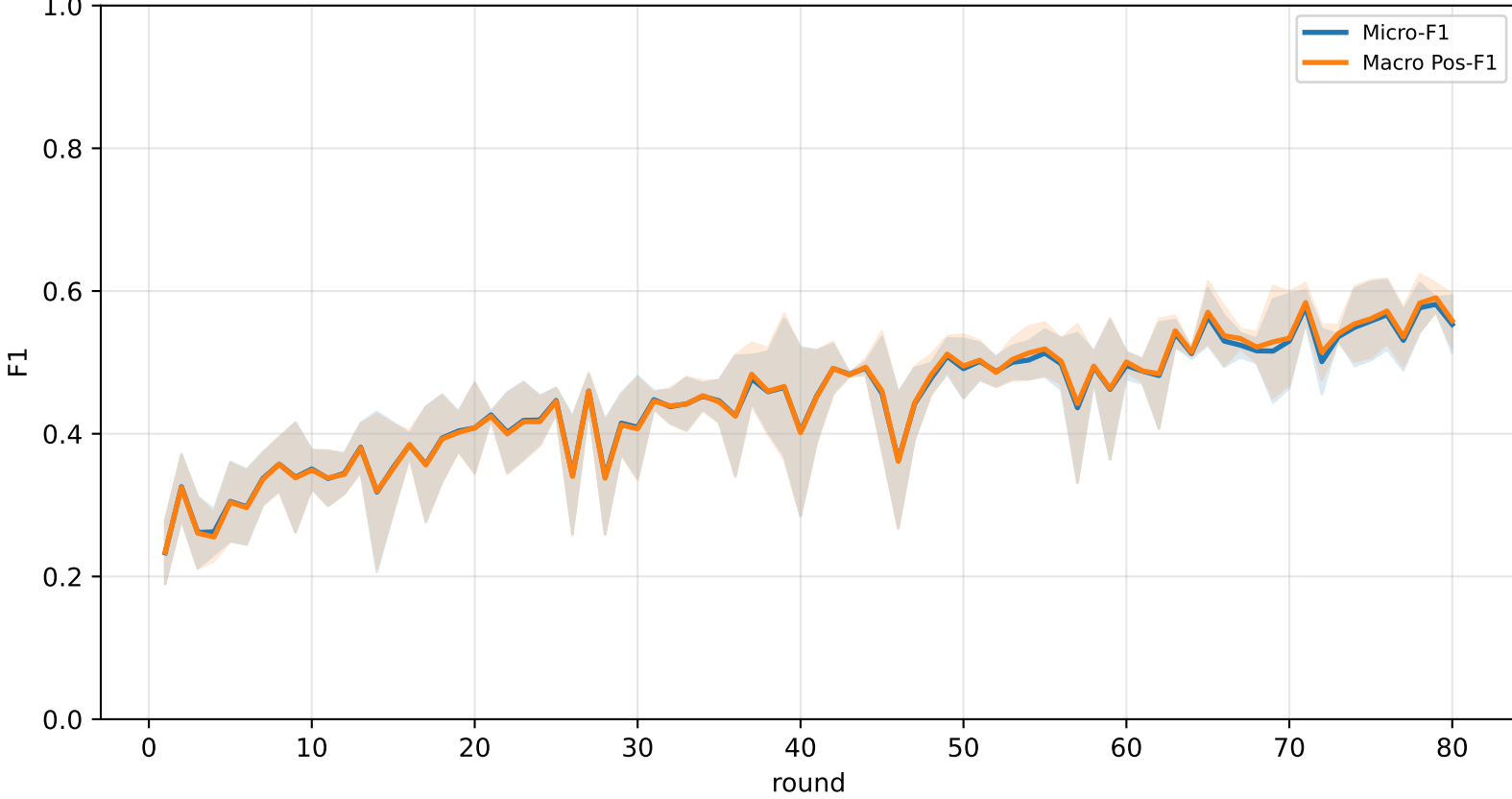
Validation loss comparison



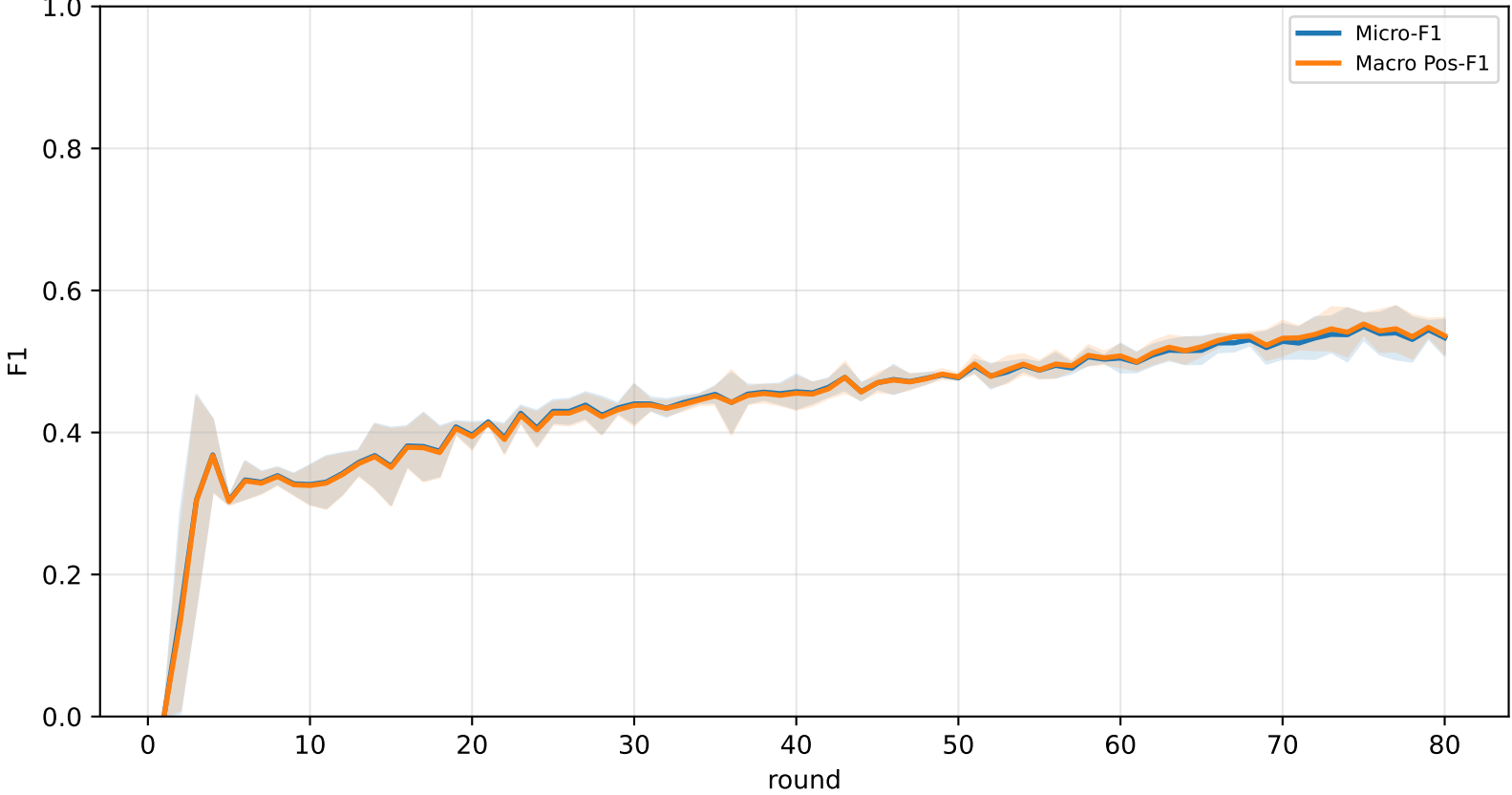
Pooled local centralized micro / macro F1



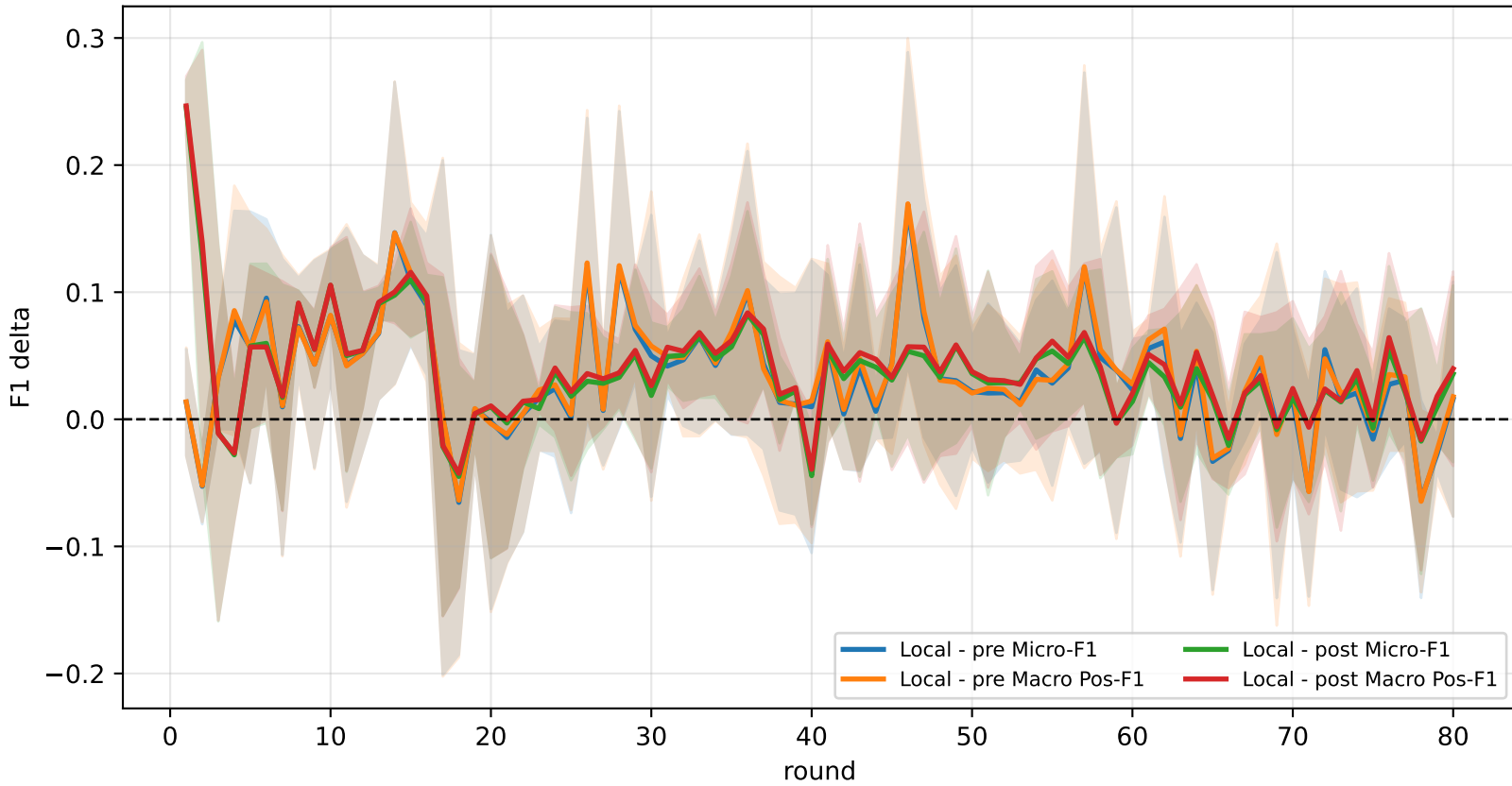
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

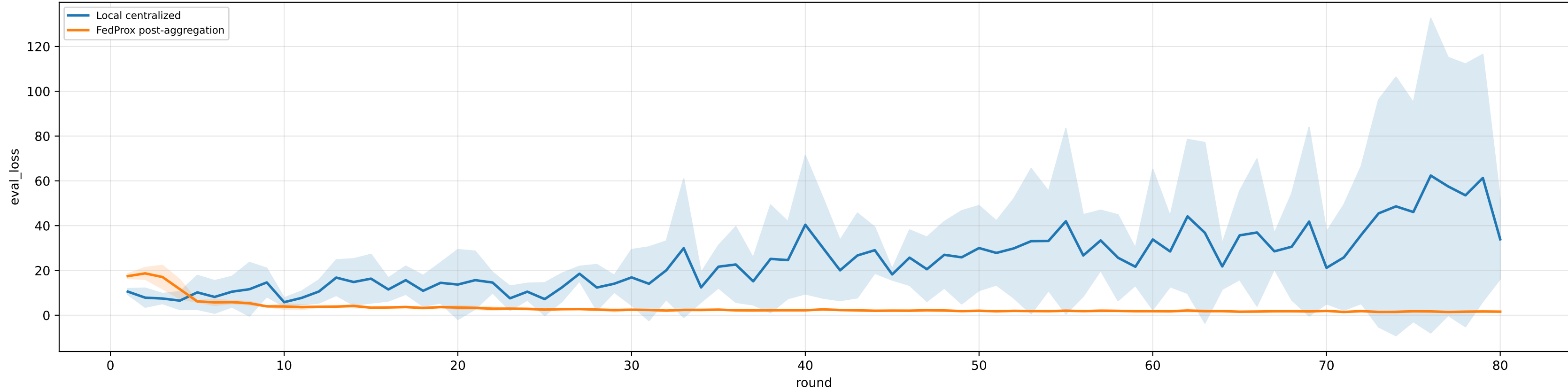


family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

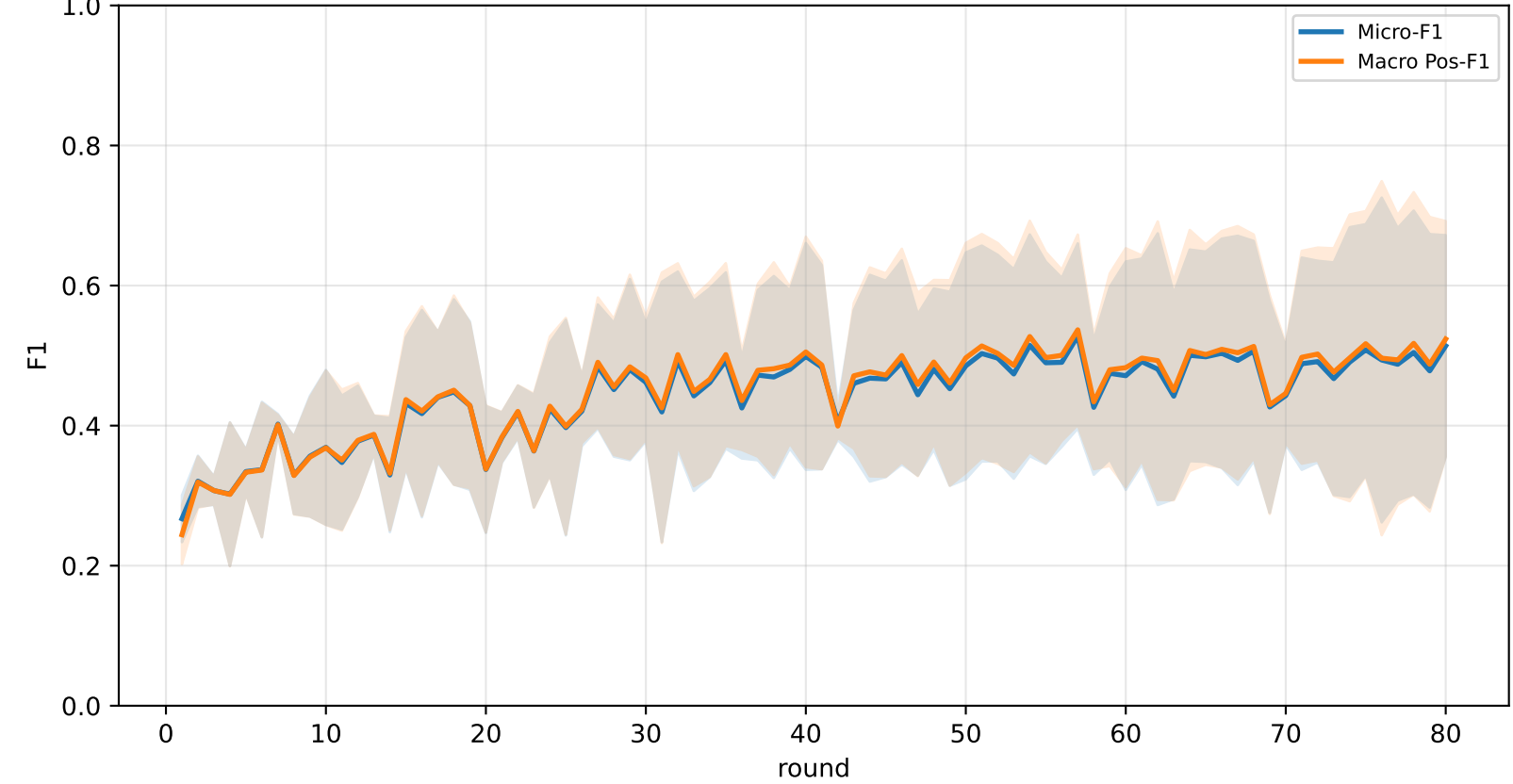
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	24.6% ± 3.5%	24.6% ± 3.5%	20.2% ± 3.0%	23.5% ± 4.2%	24.7% ± 3.6%	25.1% ± 3.4%	29.3% ± 3.3%
FedProx	52.8% ± 2.6%	53.2% ± 2.5%	53.4% ± 3.2%	55.1% ± 3.4%	54.8% ± 5.3%	53.4% ± 5.6%	49.2% ± 0.9%

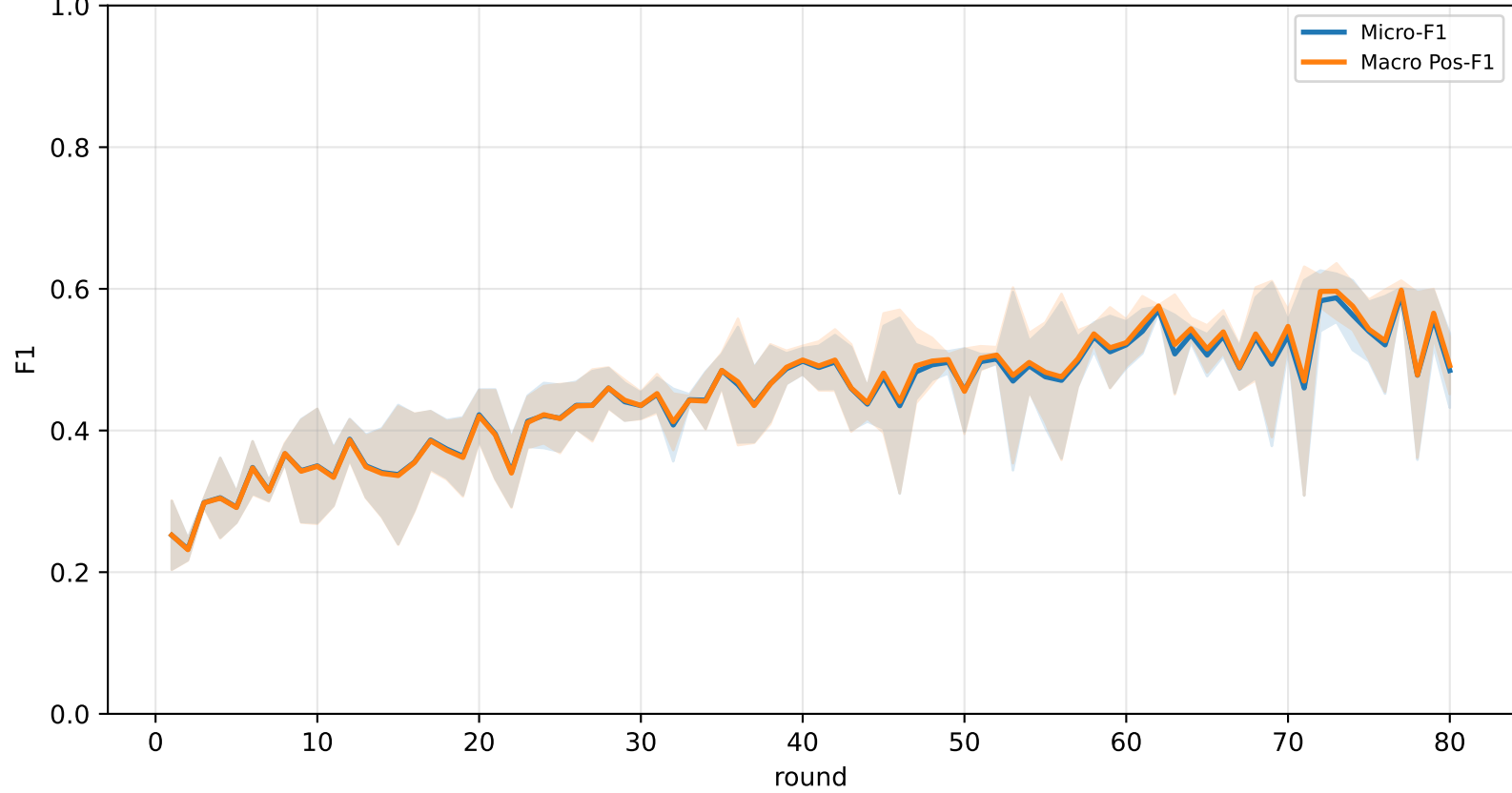
Validation loss comparison



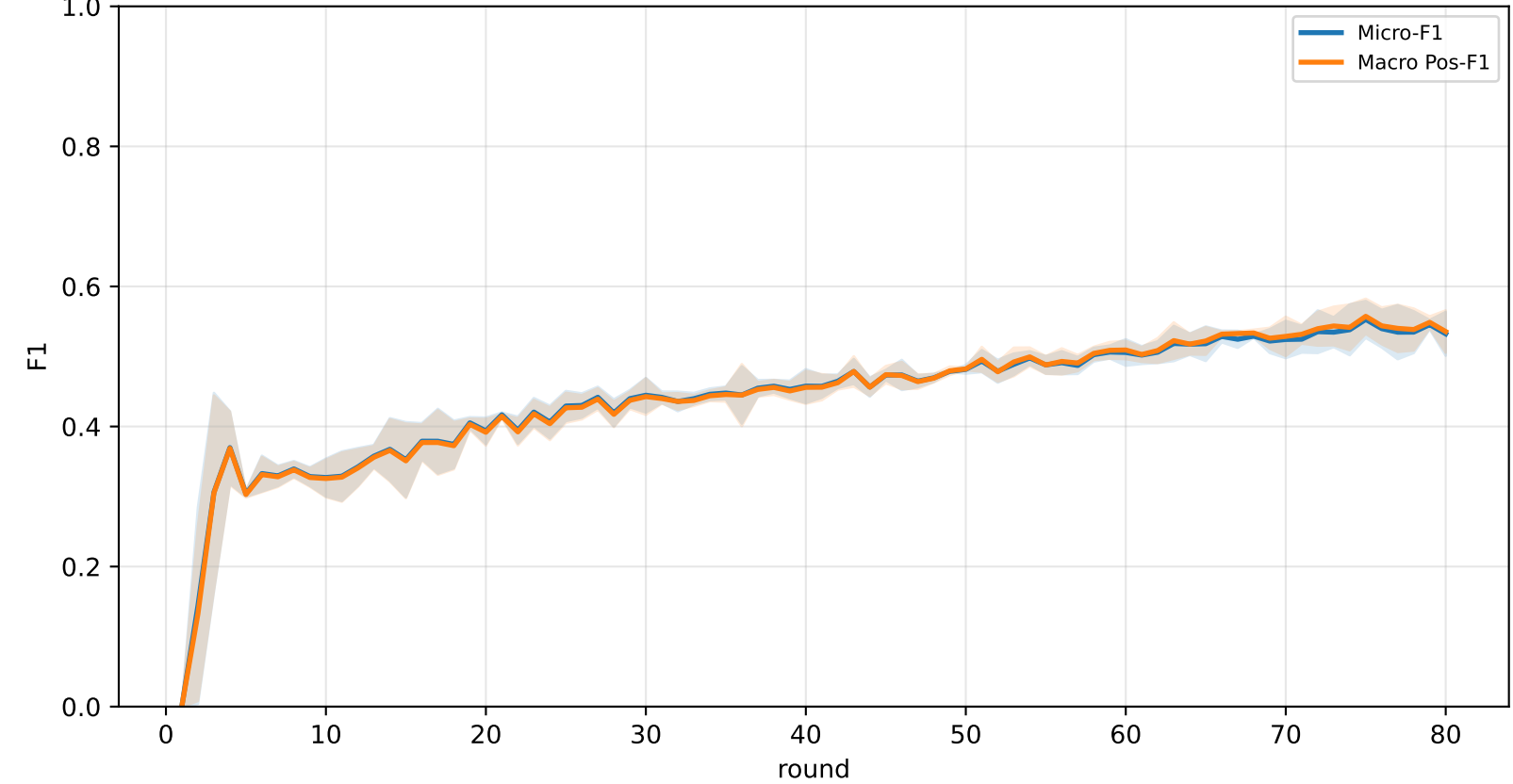
Pooled local centralized micro / macro F1



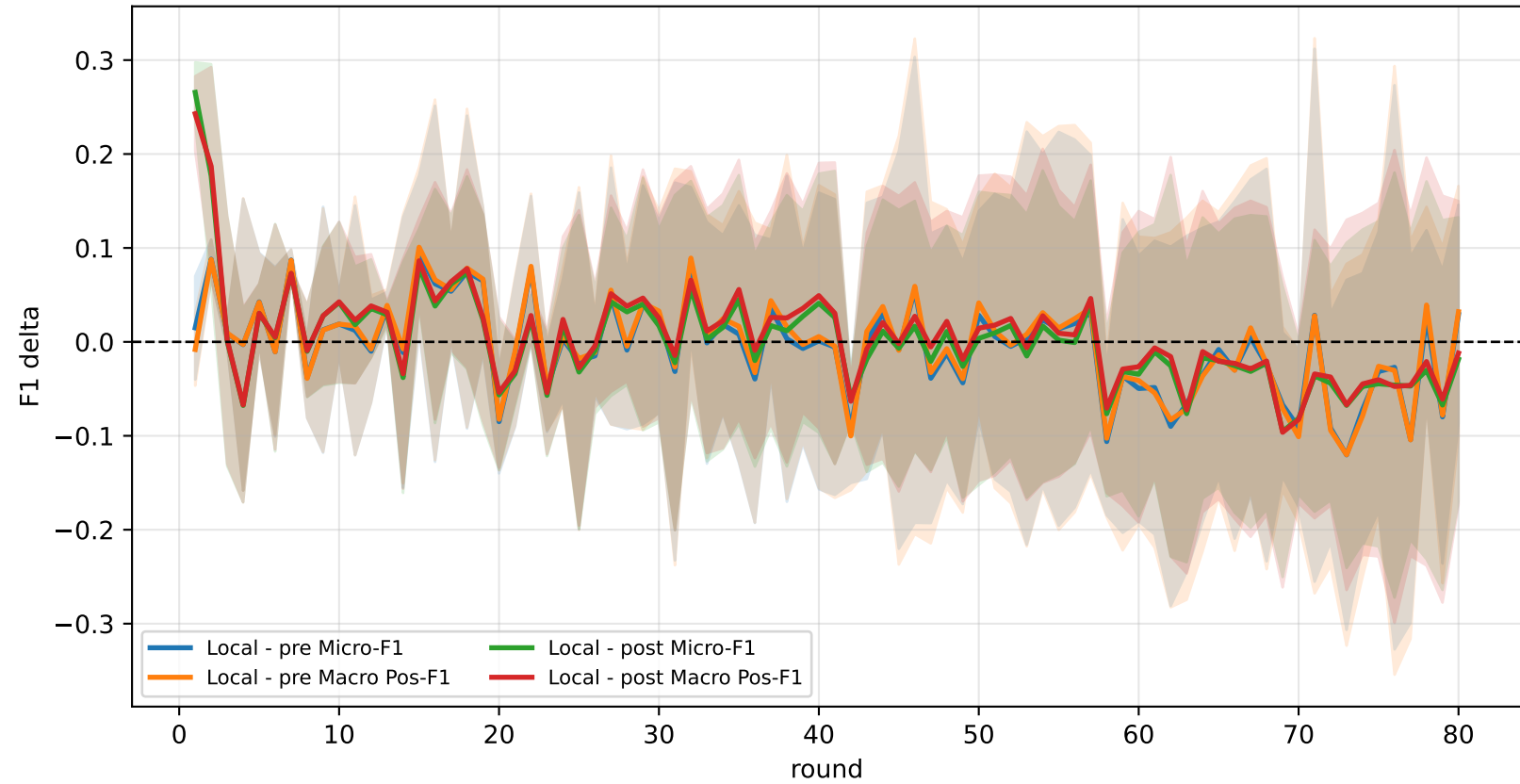
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

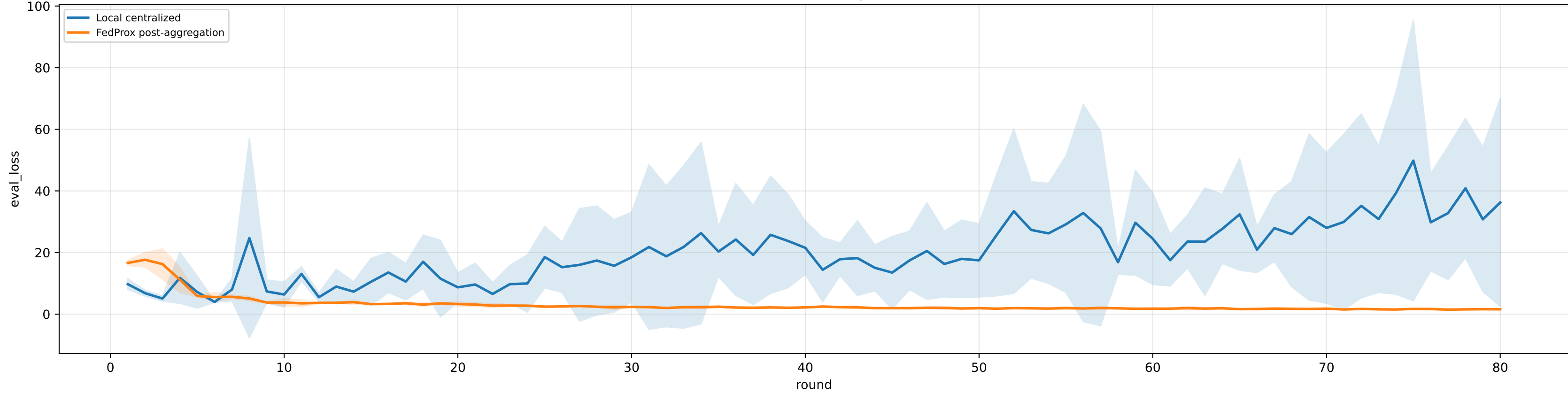


family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

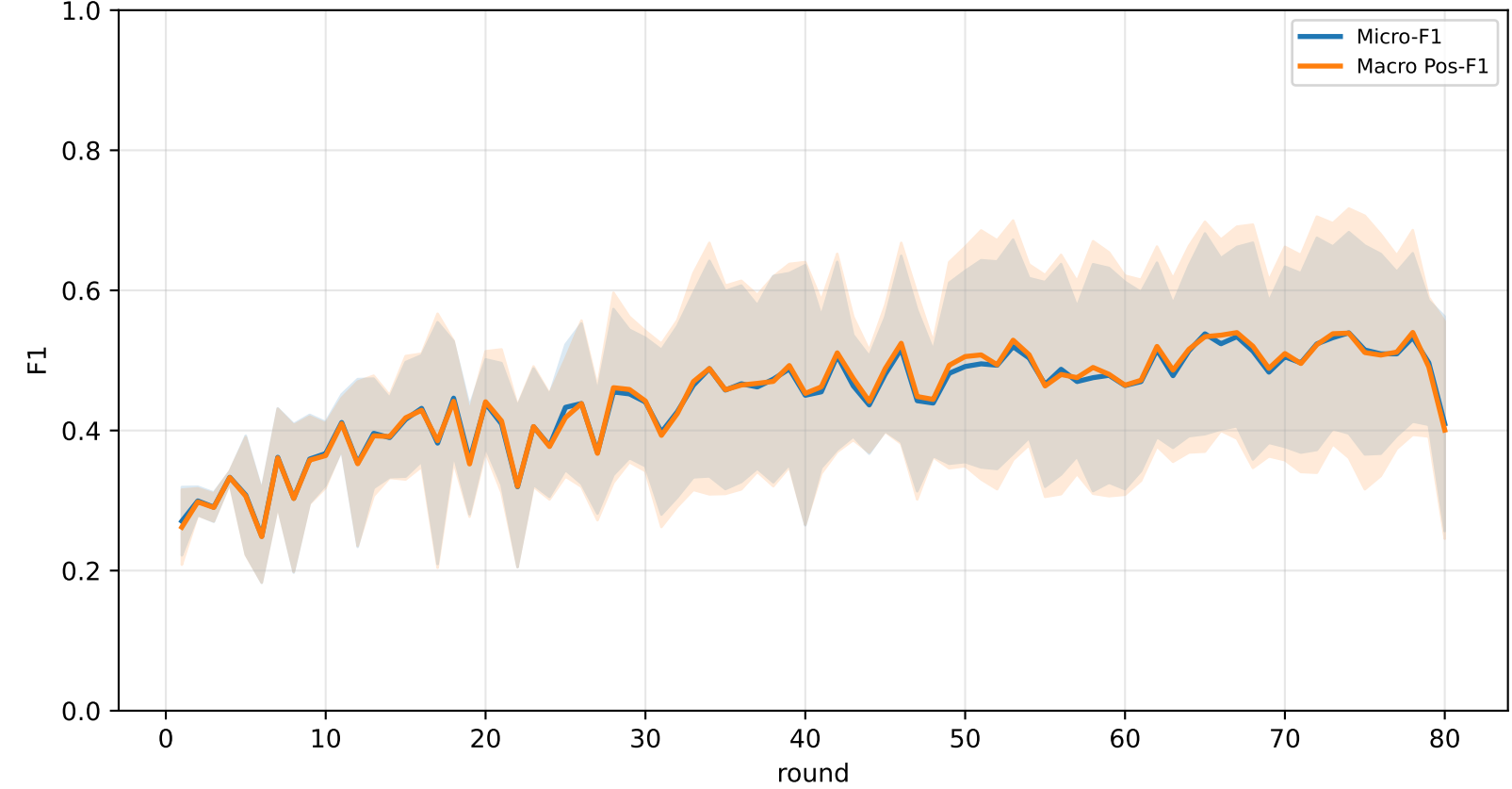
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	20.9% ± 2.3%	21.0% ± 2.5%	16.7% ± 1.9%	21.0% ± 1.6%	20.9% ± 2.8%	20.3% ± 1.5%	26.1% ± 4.9%
FedProx	53.0% ± 2.8%	53.4% ± 2.8%	53.9% ± 3.0%	55.3% ± 3.6%	55.0% ± 5.8%	53.7% ± 6.0%	49.4% ± 0.9%

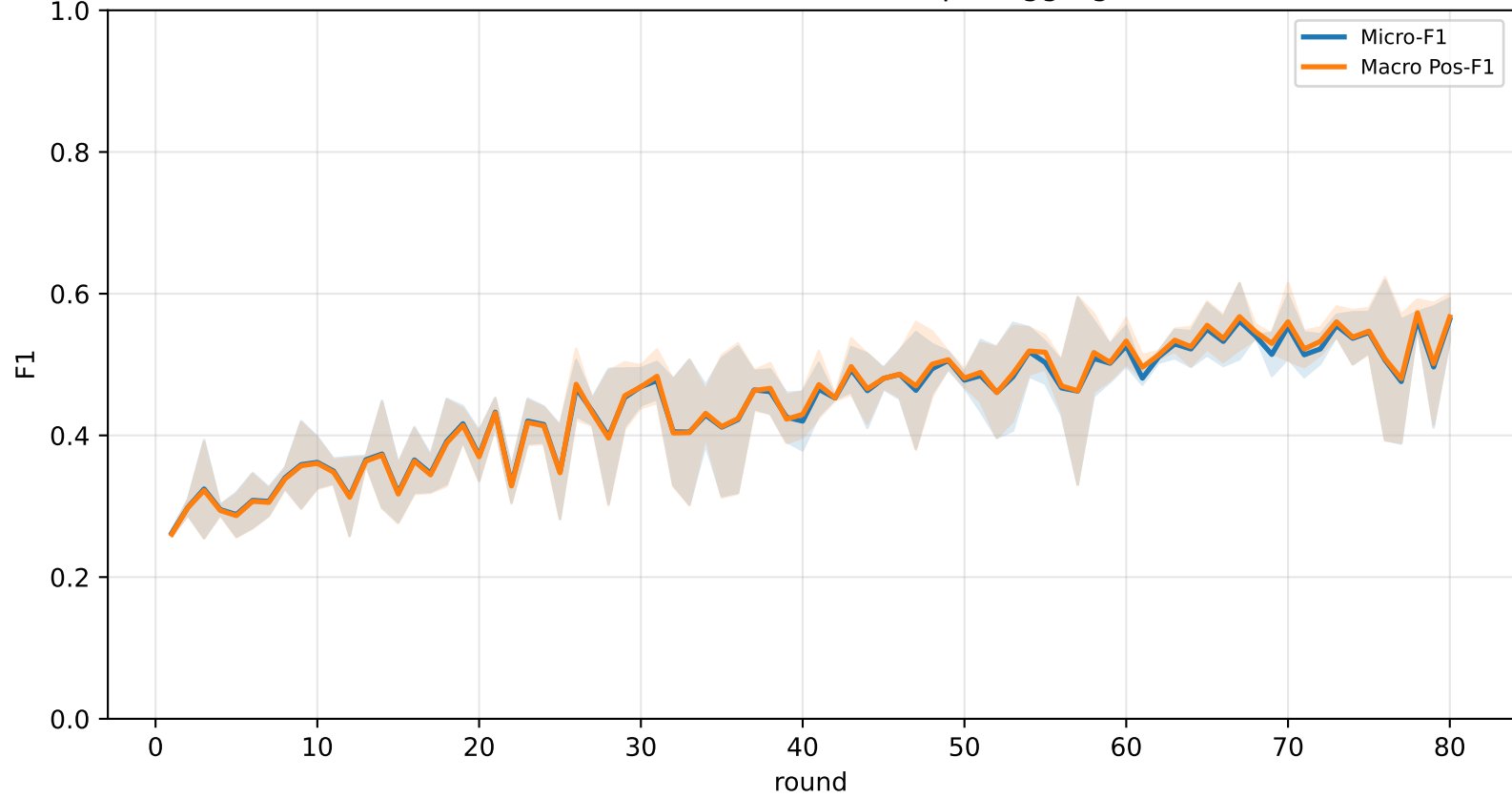
Validation loss comparison



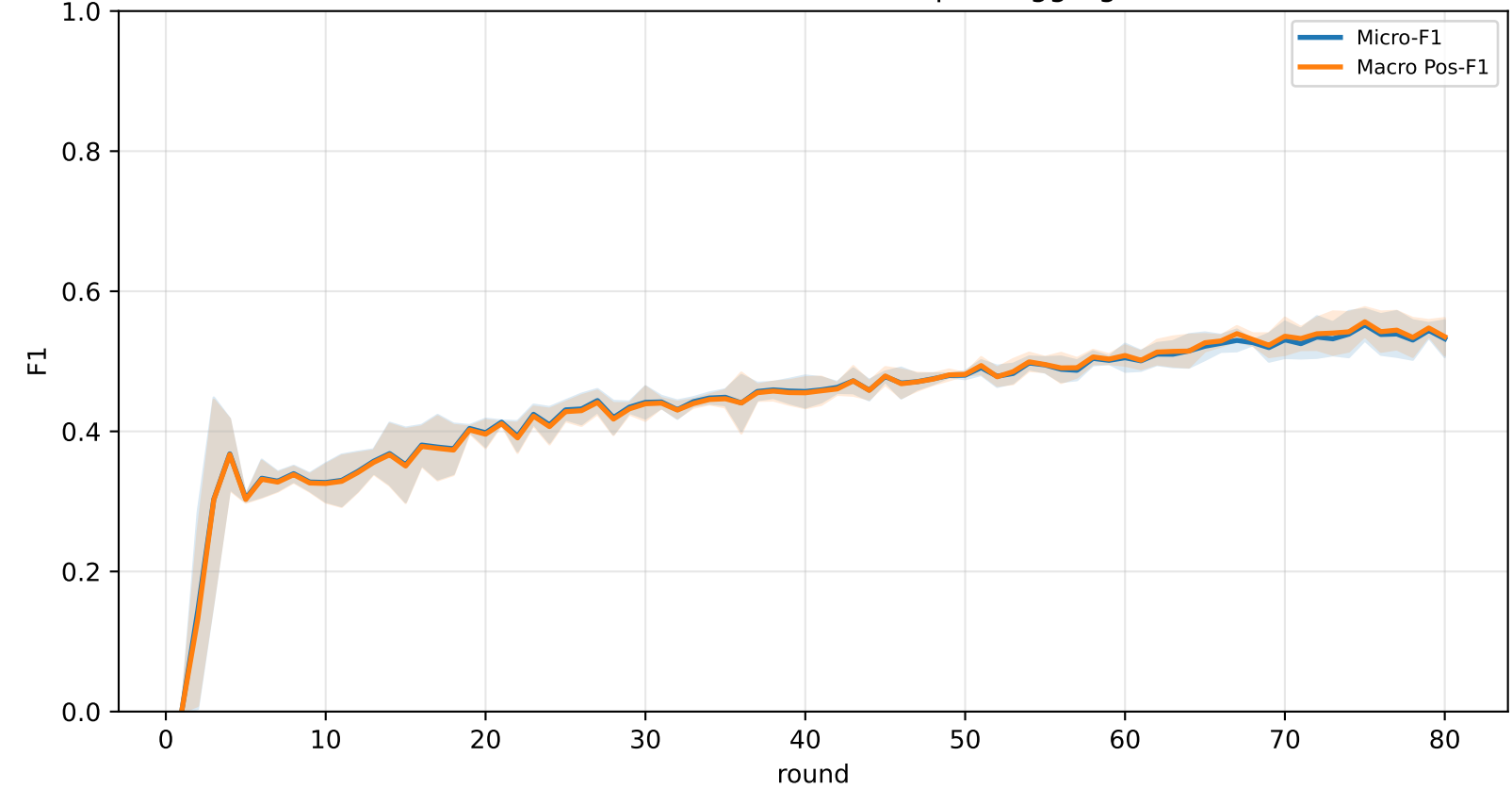
Pooled local centralized micro / macro F1



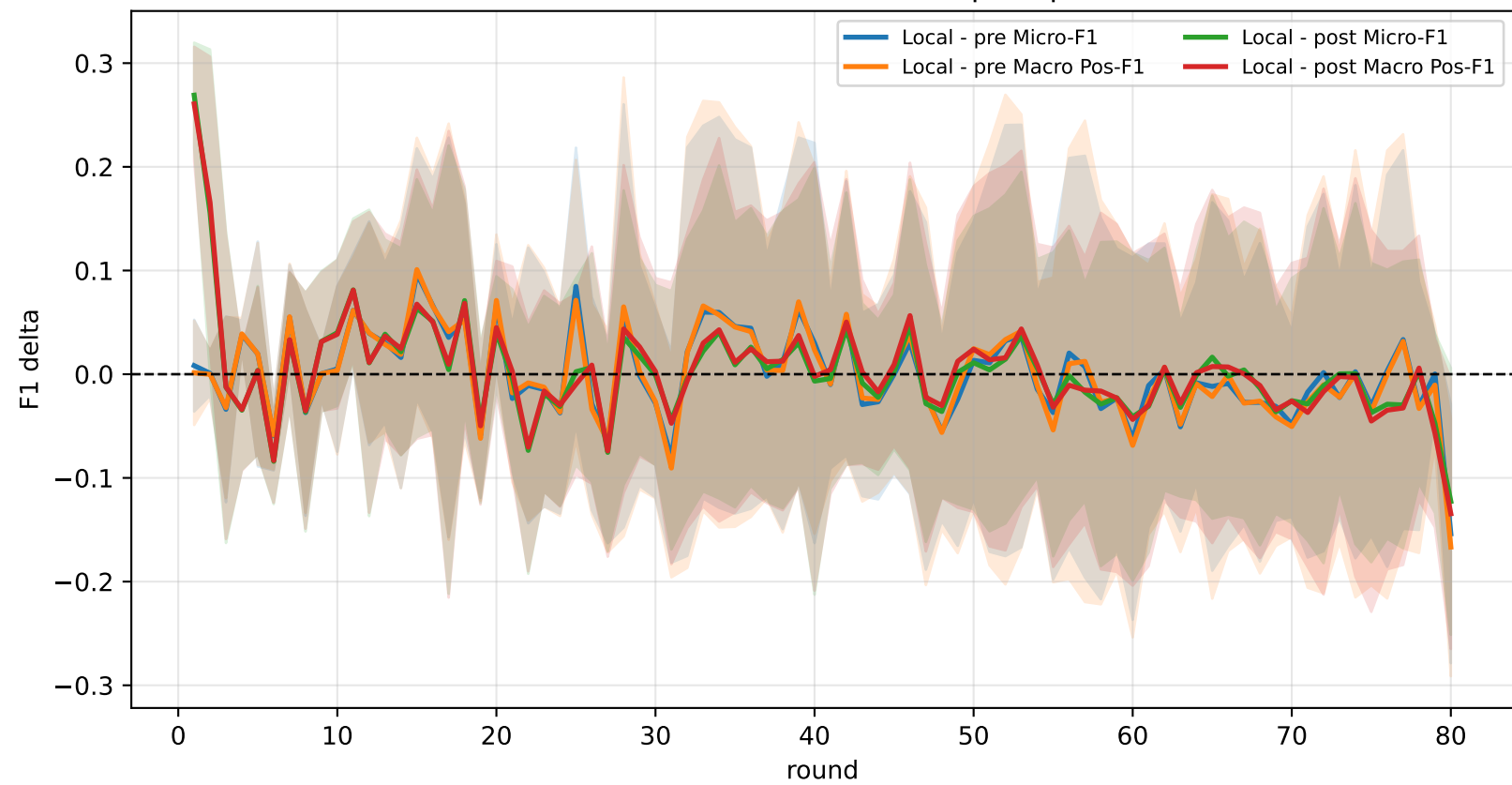
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

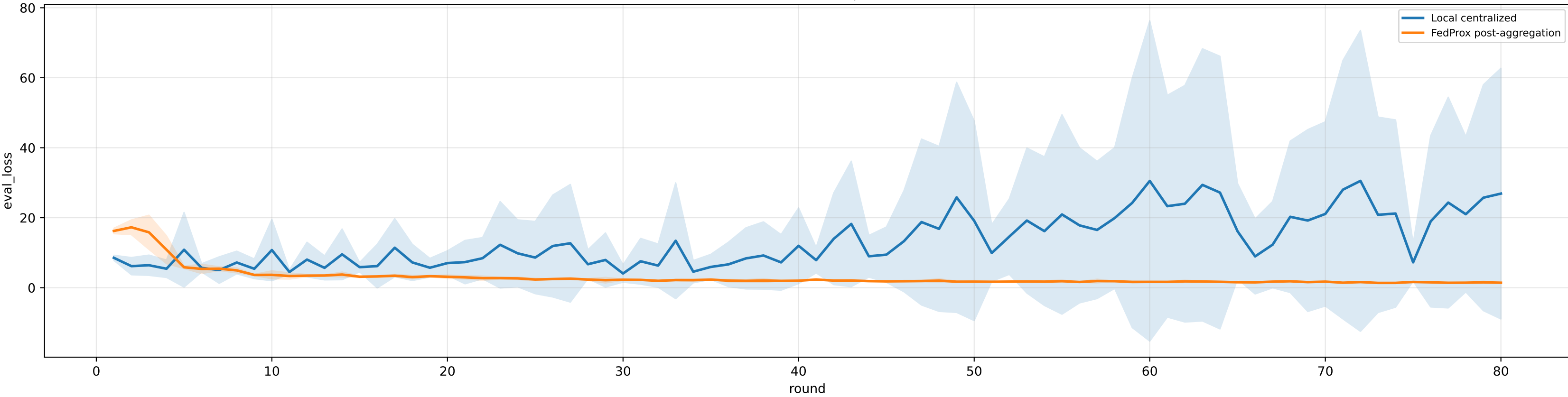


family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

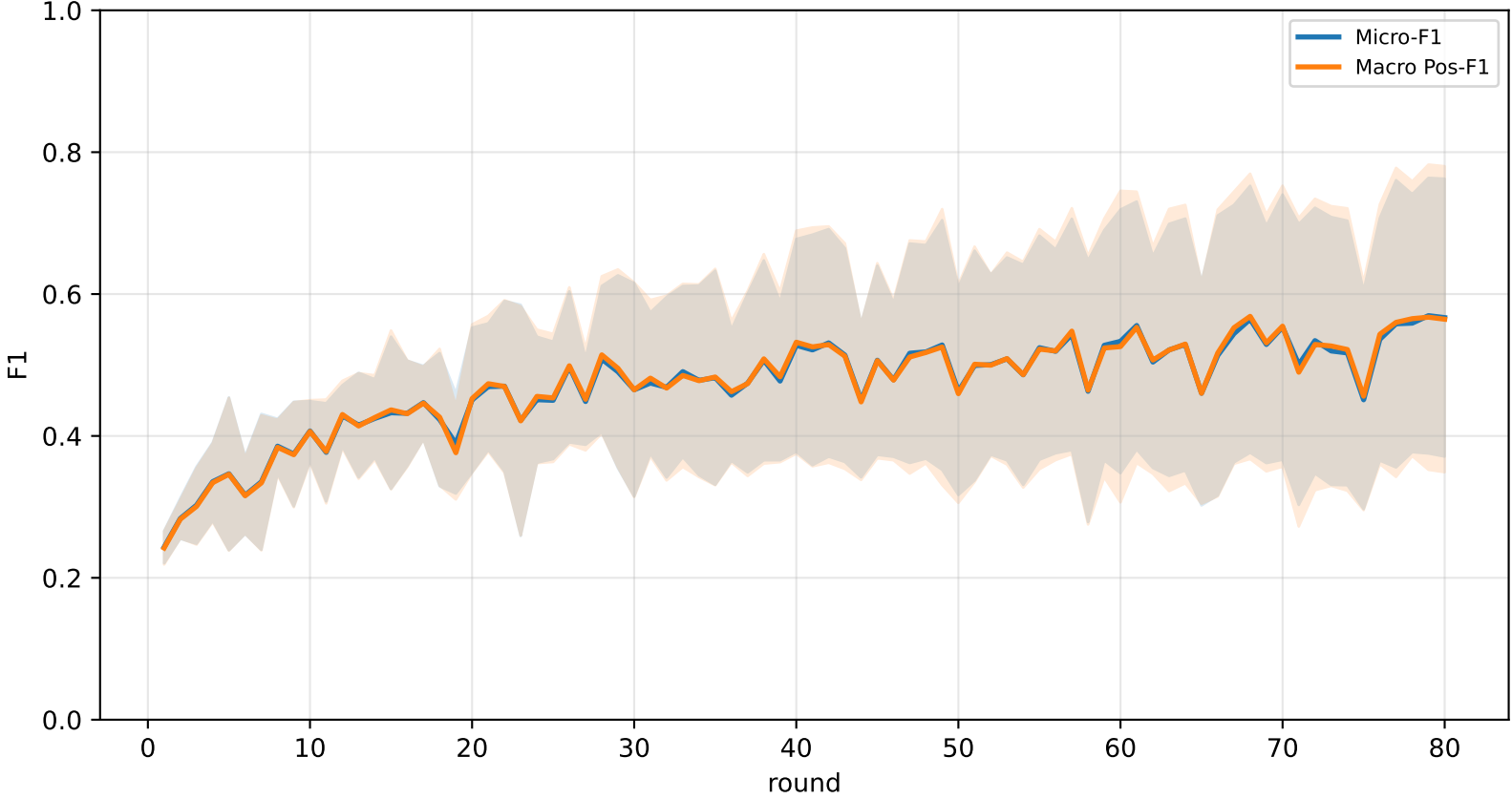
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	34.0% ± 13.0%	34.3% ± 13.3%	29.8% ± 14.7%	32.8% ± 11.9%	34.7% ± 10.6%	37.8% ± 19.0%	36.2% ± 10.7%
FedProx	52.5% ± 2.7%	52.9% ± 2.7%	53.1% ± 3.0%	55.0% ± 3.4%	54.5% ± 5.6%	53.2% ± 5.8%	48.7% ± 1.0%

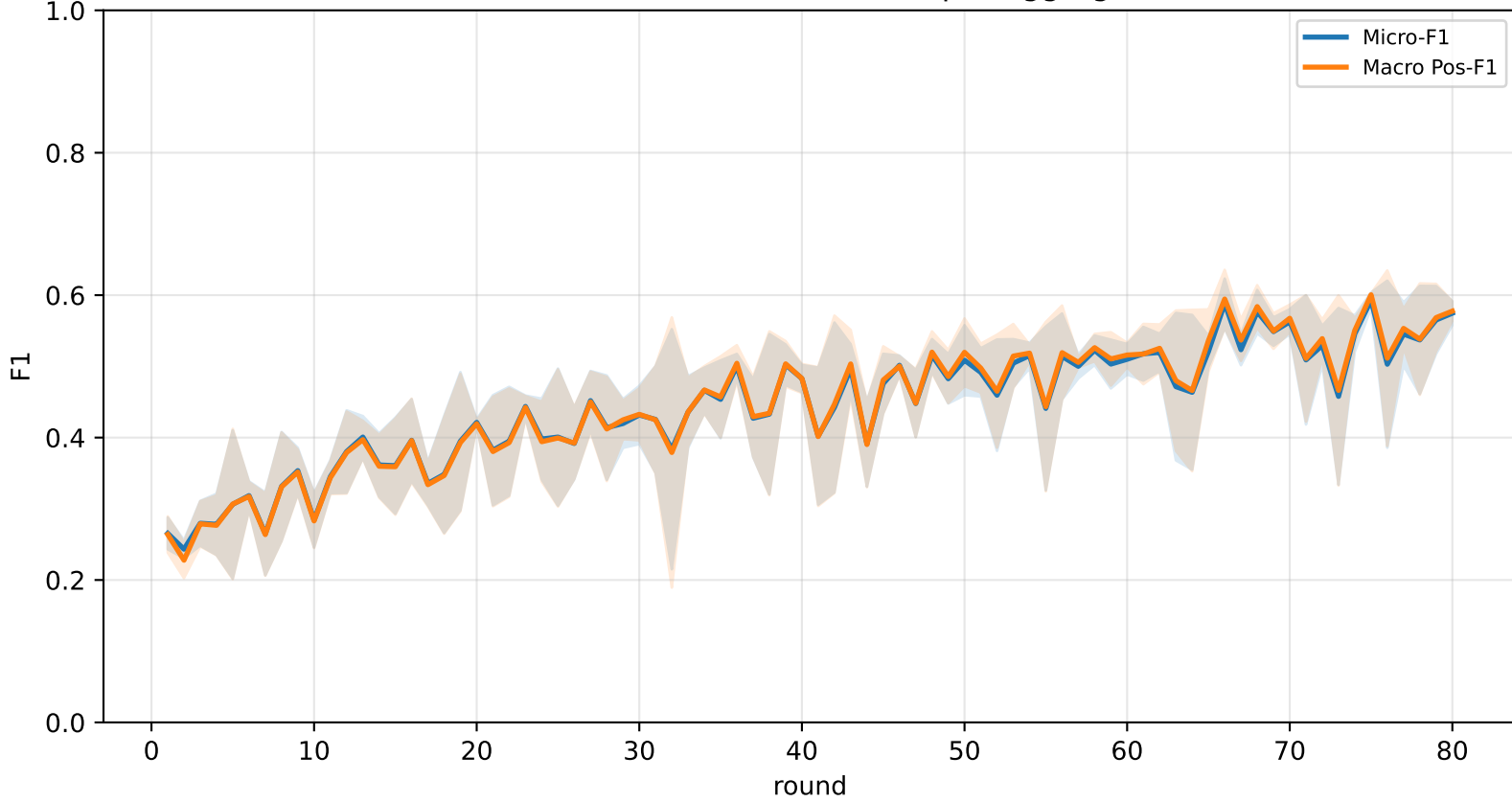
Validation loss comparison



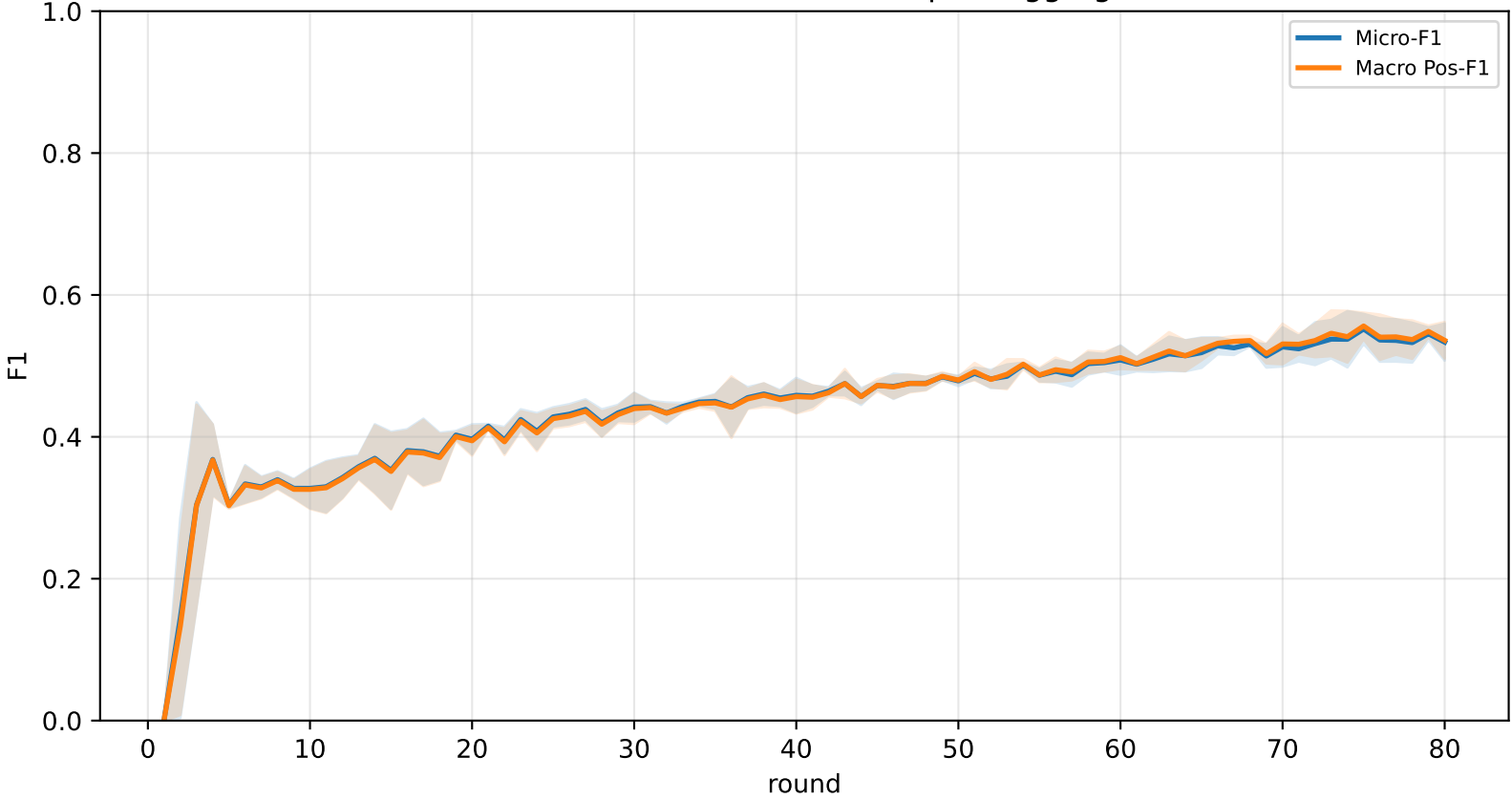
Pooled local centralized micro / macro F1



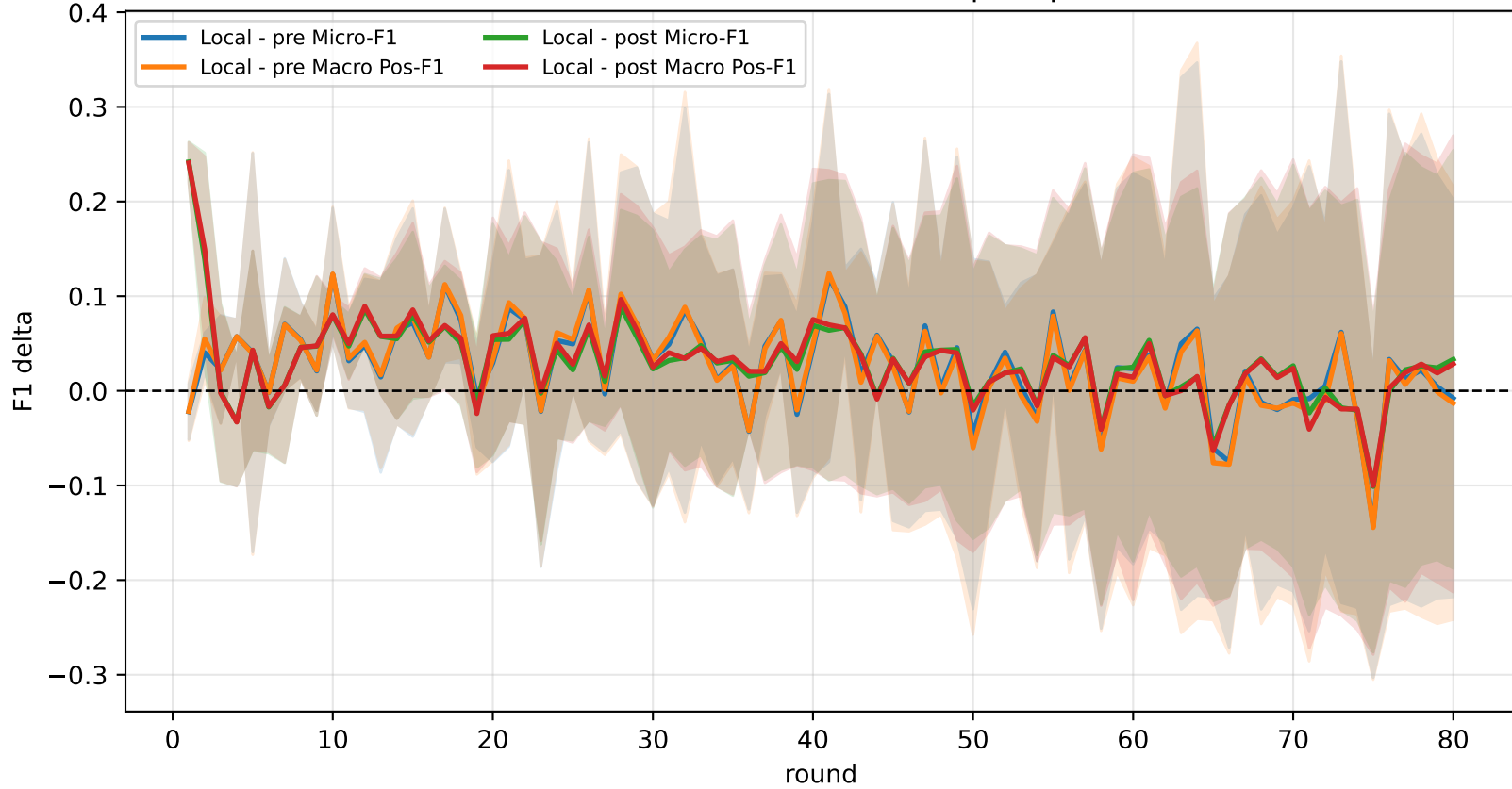
Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

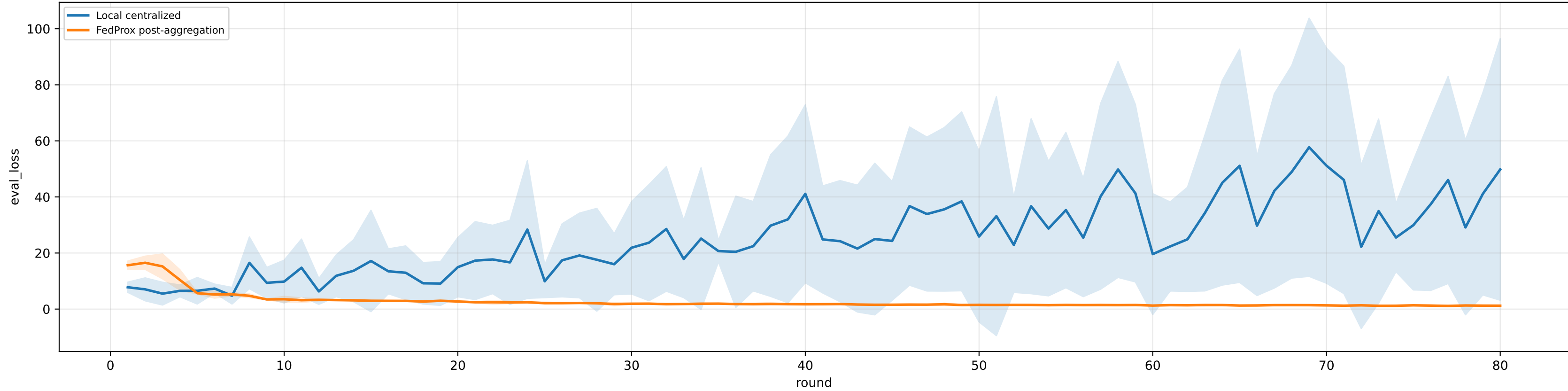


family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

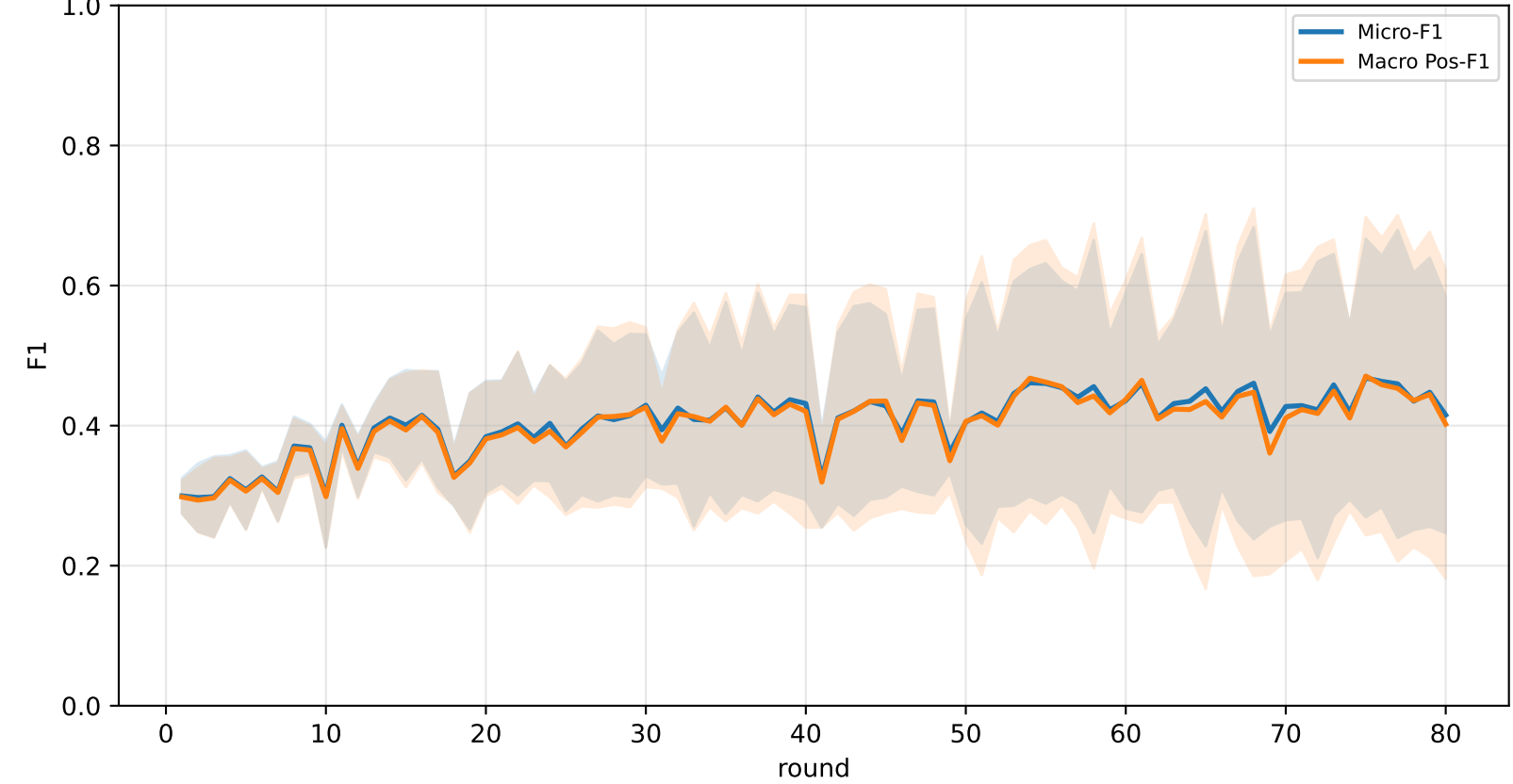
Best logged test performance (FedProx vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	33.5% ± 12.2%	34.3% ± 13.3%	36.4% ± 23.0%	36.9% ± 15.6%	32.4% ± 10.8%	32.2% ± 9.6%	33.5% ± 7.5%
FedProx	52.7% ± 3.3%	53.1% ± 3.3%	53.5% ± 2.4%	55.1% ± 4.0%	54.5% ± 6.3%	53.6% ± 6.4%	48.9% ± 1.5%

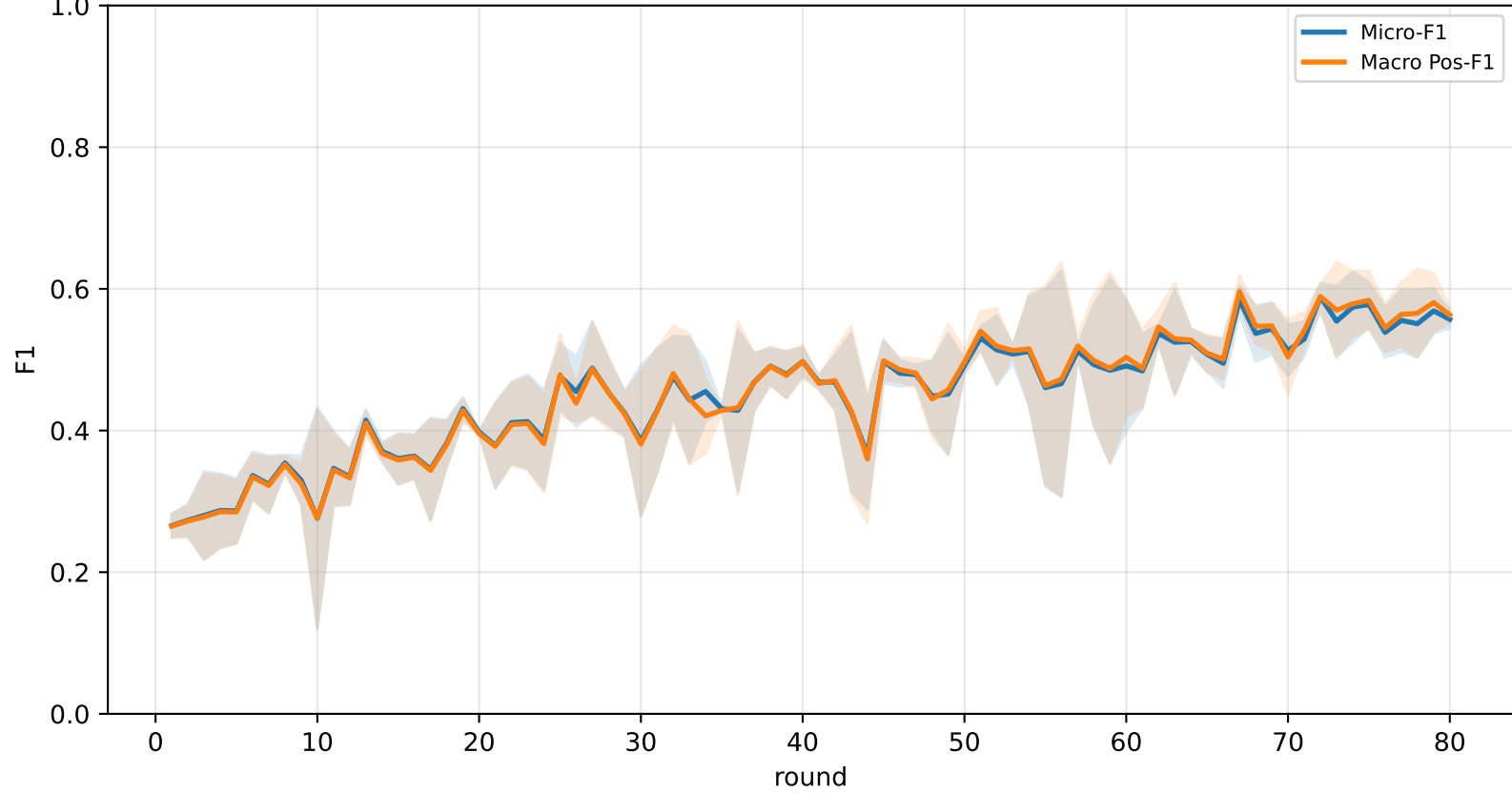
Validation loss comparison



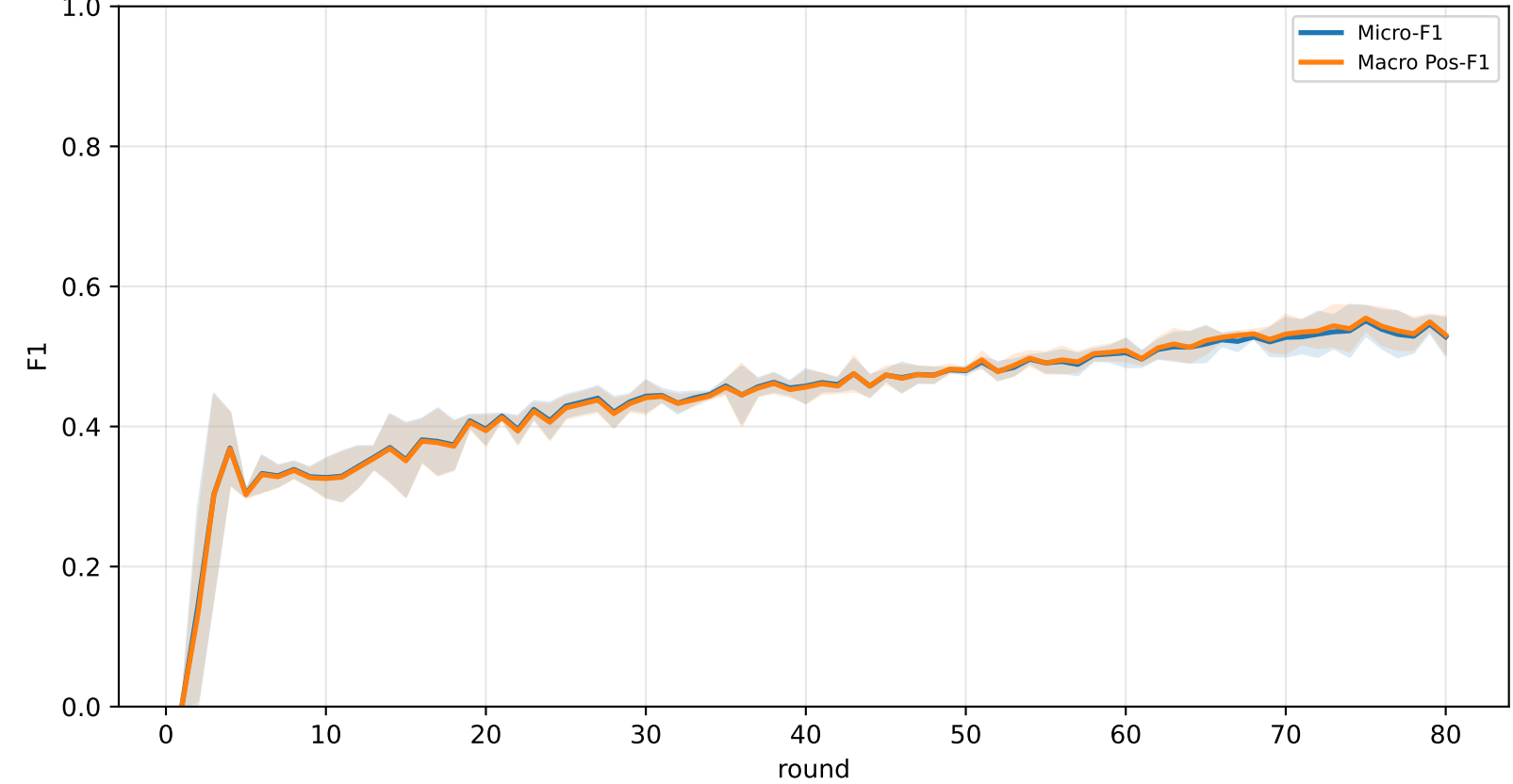
Pooled local centralized micro / macro F1



Pooled FedProx micro / macro F1 (pre-aggregation)



Pooled FedProx micro / macro F1 (post-aggregation)



Delta vs FedProx (local minus pre / post)

